

GRADE 4 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 5) • Forward 2 (Grade 6)
73 question sets • 1825 questions

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Interpret a multiplication equation as a comparison

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Compare the size of a product to the size of one factor	5.NF.B.5.a
FORWARD 2	Use ratio language to describe a relationship	6.RP.A.1

SET 1 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Which statement matches $24 = 6 \times 4$?

- A.** 24 is 6 more than 4.
- B.** 6 is 4 times as many as 24.
- C.** 24 is 4 more than 6.
- D.** 24 is 6 times as many as 4.

Write a comparison statement for $45 = 9 \times 5$.

Statement: _____

True or false?

$56 = 7 \times 8$ can mean that 56 is 8 times as many as 7.

A. True

B. False

True or false?

$18 = 3 \times 6$ means that 18 is 3 more than 6.

A. True

B. False

Which equation means "63 is 7 times as many as 9"?

A. $63 = 7 + 9$

B. $7 = 63 \times 9$

C. $63 = 7 \times 9$

D. $63 = 9 - 7$

SET 1 • 4.OA.A.1

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

What is the total in 4 groups of 6?

Total: _____

True or false?

5×3 can mean 5 groups with 3 objects in each group.

A. True

B. False

Which expression shows 3 groups of 7?

A. $7 - 3$

B. $3 + 7$

C. $7 \div 3$

D. 3×7

Write a multiplication equation for 6 groups of 2.

Equation: _____

True or false?

2×8 means 2 more than 8.

A. True

B. False

SET 1 • 4.OA.A.1

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 5$$

Product: _____

True or false?

$$7 \times 8 = 54$$

A. True

B. False

$$6 \times 4 = ?$$

A. 10

B. 20

C. 24

D. 28

Multiply.

$$8 \times 6$$

Product: _____

Multiply.

$$7 \times 9$$

Product: _____

SET 1 • 4.OA.A.1

FORWARD 1

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, how does 3×25 compare with 25?

- A.** It is 3 more.
- B.** It is smaller than 25.
- C.** It is equal to 25.
- D.** It is 3 times as large.

True or false?

$\frac{1}{2} \times 40$ is less than 40.

A. True

B. False

A rope is 12 feet long. A second rope is $2\frac{1}{2}$ times as long.

Without multiplying, is the second rope longer or shorter than the first rope? Explain.

Answer: _____

Explanation: _____

True or false?

5×18 is less than 18.

A. True

B. False

Which product is greater than 30?

A. $\frac{1}{2} \times 30$

B. 2×30

C. $\frac{3}{4} \times 30$

D. $\frac{1}{3} \times 30$

SET 1 • 4.OA.A.1

FORWARD 2

Use ratio language to describe a relationship • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A class has 12 girls and 4 boys.

Write the ratio of girls to boys.

Ratio: _____

True or false?

If there are 3 times as many cats as dogs, the ratio of cats to dogs is 3 : 1.

A. True

B. False

A bag has 5 red marbles for every 2 blue marbles.

Which statement describes the ratio?

- A.** There are 5 more red marbles than blue.
- B.** For every 5 red marbles there are 2 blue marbles.
- C.** There are 2 times as many red marbles as blue.
- D.** There are 7 red marbles.

There are 18 apples and 6 pears.

How many apples are there for every 1 pear?

Apples: _____

True or false?

The ratio 2 : 8 means there are 2 times as many of the first quantity as the second.

- A. True
- B. False

Write a comparison statement as a multiplication equation

MAIN	Grade 4 main question	4.OA.A.1
BACKWARD 1	Multiplication word problems within 100	3.OA.A.3
BACKWARD 2	Find the unknown number in a multiplication equation	3.OA.A.4
FORWARD 1	Write numerical expressions from words	5.OA.A.2
FORWARD 2	Write expressions that use a variable	6.EE.A.2.a

SET 2 • 4.OA.A.1

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Write an equation.

32 is 4 times as many as 8.

Equation: _____

Kai has 5 stickers. Lena has 3 times as many stickers as Kai.

Which equation gives Lena's stickers, s ?

A. $s = 3 \times 5$

B. $s = 5 - 3$

C. $s = 3 + 5$

D. $s = 5 \div 3$

Write an equation.

54 is 6 times as many as 9.

Equation: _____

True or false?

"40 is 8 times as many as 5" can be written as
 $40 = 8 + 5$.

A. True

B. False

A tree is 7 times as tall as a 2-meter bush.
Write an equation for the tree's height, t .

Equation: _____

SET 2 • 4.OA.A.1

BACKWARD 1

Multiplication word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 boxes with 8 crayons in each box.
How many crayons are there?

Crayons: _____

True or false?

5 bags with 9 apples in each bag hold 45 apples.

A. True

B. False

A garden has 4 rows with 7 plants in each row.

How many plants are there?

A. 21

B. 32

C. 28

D. 11

Each pack has 3 juice boxes.

How many juice boxes are in 9 packs?

Juice boxes: _____

True or false?

8 teams with 5 players each have 13 players in all.

- A.** True
- B.** False

SET 2 • 4.OA.A.1

BACKWARD 2

Find the unknown number in a multiplication equation • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 6 = 30$$

Unknown: _____

True or false?

If $4 \times n = 36$, then $n = 8$.

A. True

B. False

$$8 \times ? = 56$$

A. 8

B. 7

C. 6

D. 48

Find the unknown number.

$$3 \times ? = 27$$

Unknown: _____

True or false?

If $m \times 7 = 42$, then $m = 6$.

A. True

B. False

SET 2 • 4.OA.A.1

FORWARD 1

Write numerical expressions from words • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression.

3 times the sum of 8 and 4

Expression: _____

Which expression means "add 9 and 6, then multiply by 2"?

A. $9 \times 6 + 2$

B. $9 + 6 \times 2$

C. $(9 + 6) \times 2$

D. $2 + 9 \times 6$

True or false?

$5 \times (12 - 3)$ is 5 times as large as $12 - 3$.

A. True

B. False

Write an expression.

4 times as large as $25 + 15$

Expression: _____

True or false?

$2 \times (7 + 1)$ is 2 more than $7 + 1$.

A. True

B. False

SET 2 • 4.OA.A.1

FORWARD 2

Write expressions that use a variable • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for 5 times a number n .

Expression: _____

Which expression means "8 more than y "?

A. $y + 8$

B. $8 - y$

C. $y - 8$

D. $8y$

Maya has m marbles. Jo has 4 times as many marbles as Maya.

Write an expression for Jo's marbles.

Expression: _____

True or false?

"3 times the sum of x and 2" can be written as $3(x + 2)$.

A. True

B. False

True or false?

"7 less than k" can be written as $7 - k$.

A. True

B. False

Solve multiplicative comparison word problems by multiplying

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Multiplication word problems within 100	3.OA.A.3
BACKWARD 2	Multiply in either order	3.OA.B.5
FORWARD 1	Solve real-world problems by multiplying by a fraction or mixed nu...	5.NF.B.6
FORWARD 2	Solve unit rate problems	6.RP.A.3.b

SET 3 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A puppy weighs 6 pounds. Its mother weighs 8 times as much.

How much does the mother weigh?

Puppy

6

Mother

6	6	6	6	6	6	6	6
---	---	---	---	---	---	---	---

Weight: _____

Sam read 9 pages. Ana read 5 times as many pages as Sam.

How many pages did Ana read?

Pages: _____

A red ribbon is 7 inches long. A blue ribbon is 6 times as long.

How long is the blue ribbon?

A. 13 inches

B. 48 inches

C. 1 inch

D. 42 inches

A small tank holds 9 gallons. A large tank holds 7 times as much.

How much does the large tank hold?

Gallons: _____

True or false?

A bike costs \$80. A used car costs 100 times as much. The car costs \$800.

A. True

B. False

SET 3 • 4.OA.A.2

BACKWARD 1

Multiplication word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

7 vans each carry 8 students.

How many students are there in all?

Students: _____

4 children each have 9 crayons.

How many crayons are there in all?

Crayons: _____

True or false?

6 rows of 6 chairs make 36 chairs.

A. True

B. False

A book has 9 chapters with 9 pages in each chapter.

How many pages does it have?

A. 18

B. 72

C. 81

D. 90

True or false?

5 piles with 8 cards in each pile make 45 cards.

A. True

B. False

SET 3 • 4.OA.A.2

BACKWARD 2

Multiply in either order • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times 8 = 8 \times 6$$

A. True

B. False

Fill in the blank.

$$9 \times 4 = 4 \times \underline{\quad}$$

Blank: _____

Which is equal to 7×5 ?

A. 5×7

B. $7 + 5$

C. 75

D. $5 + 5 + 5$

$$3 \times 9 = 27.$$

What is 9×3 ?

Product: _____

True or false?

$$8 \times 2 = 2 + 8$$

A. True

B. False

SET 3 • 4.OA.A.2

FORWARD 1

Solve real-world problems by multiplying by a fraction or mixed number •
Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A pup weighs 6 pounds. An adult dog weighs $4\frac{1}{2}$ times as much.

How much does the adult weigh?

Weight: _____

A path is 12 miles long. Kim walked $\frac{2}{3}$ of the path.

How far did she walk?

Distance: _____

A red rope is 8 feet long. A blue rope is $1\frac{1}{4}$ times as long.

How long is the blue rope?

A. 10 feet

B. $9\frac{1}{4}$ feet

C. 2 feet

D. 6 feet

True or false?

A 10-cm plant grows until it is $2\frac{1}{2}$ times as tall. It is then 25 cm tall.

- A. True
- B. False

A recipe uses 3 cups of flour. Mia makes $\frac{1}{2}$ of the recipe.

How much flour does she use?

Flour: _____

SET 3 • 4.OA.A.2

FORWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

5 notebooks cost \$15.

At the same price, how much do 8 notebooks cost?

Cost: _____

True or false?

If 4 pounds of grapes cost \$10, then 1 pound costs \$2.50.

A. True

B. False

A car goes 180 miles in 3 hours at a steady speed.

How far does it go in 5 hours?

- A.** 240 miles
- B.** 360 miles
- C.** 300 miles
- D.** 185 miles

A printer prints 42 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

True or false?

If 3 tickets cost \$27, then 7 tickets cost \$56.

- A. True
- B. False

Solve multiplicative comparison word problems by dividing

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find the unknown number in a multiplication or division equation	3.OA.A.4
FORWARD 1	Solve division word problems with two-digit divisors	5.NBT.B.6
FORWARD 2	Solve one-step equations of the form $px = q$	6.EE.B.7

SET 4 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A dog weighs 48 pounds. That is 6 times as much as a cat weighs.

How much does the cat weigh?

Weight: _____

Leo has 72 cards. That is 9 times as many cards as Ty has.

How many cards does Ty have?

Cards: _____

A building is 60 feet tall. That is 5 times as tall as a tree.
How tall is the tree?

- A.** 300 feet
- B.** 55 feet
- C.** 65 feet
- D.** 12 feet

A jacket costs \$56. A hat costs \$8.

The jacket costs how many times as much as the hat?

Times as much: _____

True or false?

Mo ran 36 laps. That is 4 times as many laps as Ed ran. Ed ran 9 laps.

- A. True
- B. False

SET 4 • 4.OA.A.2

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $42 \div 6$ by thinking $6 \times ? = 42$.

Quotient: _____

True or false?

$54 \div 9 = 6$ because $9 \times 6 = 54$.

A. True

B. False

Which fact helps you find $63 \div 7$?

A. $9 + 7 = 16$

B. $63 - 7 = 56$

C. $7 \times 9 = 63$

D. $7 \times 7 = 49$

Divide.

$$32 \div 8$$

Quotient: _____

True or false?

$48 \div 6 = 9$ because $6 \times 9 = 48$.

A. True

B. False

SET 4 • 4.OA.A.2

BACKWARD 2

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$5 \times ? = 40$$

Unknown: _____

True or false?

If $27 \div ? = 3$, the unknown number is 9.

A. True

B. False

$$? \times 9 = 72$$

A. 63

B. 9

C. 81

D. 8

Find the unknown number.

$$? \div 4 = 6$$

Unknown: _____

True or false?

If $6 \times ? = 36$, the unknown number is 5.

A. True

B. False

SET 4 • 4.OA.A.2

FORWARD 1

Solve division word problems with two-digit divisors • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

A truck carries 384 boxes. That is 12 times as many boxes as a van carries.

How many boxes does the van carry?

Boxes: _____

A school has 864 students in 24 classes of the same size.

How many students are in each class?

A. 38

B. 36

C. 34

D. 840

True or false?

$$1,050 \div 15 = 70$$

A. True

B. False

Divide.

$$2,016 \div 36$$

Quotient: _____

True or false?

A farm has 630 trees in rows of 18 trees. There are 45 rows.

A. True

B. False

SET 4 • 4.OA.A.2

FORWARD 2

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for c.

$$6c = 48$$

$$c = \underline{\hspace{10cm}}$$

True or false?

The solution of $9x = 72$ is $x = 8$.

A. True

B. False

Solve for n.

$$5n = 65$$

A. 325

B. 60

C. 70

D. 13

A dog weighs 48 pounds, which is 6 times the weight c of a cat.

Write and solve an equation for c .

Equation: _____

$c =$ _____

True or false?

The solution of $4y = 36$ is $y = 32$.

A. True

B. False

Tell multiplicative comparison from additive comparison

MAIN	Grade 4 main question	4.OA.A.2
BACKWARD 1	Compare quantities by adding or subtracting	2.OA.A.1
BACKWARD 2	Multiplication facts within 100	3.OA.C.7
FORWARD 1	Interpret numerical expressions without evaluating them	5.OA.A.2
FORWARD 2	Tell an additive relationship from a multiplicative one in tables a...	6.EE.C.9

SET 5 • 4.OA.A.2

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

Ana has 4 shells. Ben has 12 shells.

Which statement is true?

- A.** Ben has 3 times as many shells as Ana.
- B.** Ben has 3 more shells than Ana.
- C.** Ana has 8 times as many shells as Ben.
- D.** Ben has 4 more shells than Ana.

The red team scored 24 points. The blue team scored 6 points.

How many more points did the red team score? How many times as many points?

How many more: _____

How many times as many: _____

True or false?

"8 more than 5" is 40.

A. True

B. False

Which question is answered by multiplying?

A. A cat is 3 years old. A dog is 4 times as old. How old is the dog?

B. A cat ate 4 treats and then 3 more. How many did it eat?

C. A cat is 3 years old. A dog is 4 years older. How old is the dog?

D. A cat is 7 years old. A dog is 3 years old. How much older is the cat?

True or false?

A pencil is 6 inches long. A ruler that is 2 times as long is 12 inches, but a crayon that is 2 inches longer than the pencil is 8 inches.

- A.** True
- B.** False

SET 5 • 4.OA.A.2

BACKWARD 1

Compare quantities by adding or subtracting • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Jo has 15 stamps. Ty has 9 stamps.

How many more stamps does Jo have?

Stamps: _____

True or false?

Mia has 8 beads. That is 5 more beads than Kim has. Kim has 3 beads.

A. True

B. False

A red rope is 34 cm long. A green rope is 18 cm long.
How much longer is the red rope?

- A.** 14 cm
- B.** 16 cm
- C.** 52 cm
- D.** 26 cm

A tower has 27 blocks. Another tower has 8 more blocks.

How many blocks does it have?

Blocks: _____

True or false?

Sam has 20 cards. Lee has 6 fewer cards. Lee has 26 cards.

A. True

B. False

SET 5 • 4.OA.A.2

BACKWARD 2

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$4 \times 9$$

Product: _____

True or false?

$$6 \times 7 = 42$$

A. True

B. False

$$3 \times 8 = ?$$

A. 21

B. 24

C. 11

D. 32

Multiply.

$$5 \times 6$$

Product: _____

True or false?

$$8 \times 8 = 16$$

A. True

B. False

SET 5 • 4.OA.A.2

FORWARD 1

Interpret numerical expressions without evaluating them • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$6 \times (25 + 9)$ is 6 times as large as $25 + 9$.

A. True

B. False

How does $4 \times (17 - 3)$ compare with $17 - 3$?

- A.** It is 4 times as large.
- B.** It is the same.
- C.** It is 4 less.
- D.** It is 4 more.

True or false?

$(48 + 6) + 3$ is 3 times as large as $48 + 6$.

A. True

B. False

Without calculating, how does $10 \times (325 + 19)$ compare with $325 + 19$?

Answer: _____

Which expression is 2 more than 9×7 ?

A. $9 \times (7 + 2)$

B. $2 \times 9 \times 7$

C. $9 \times 7 + 2$

D. $(9 + 2) \times 7$

SET 5 • 4.OA.A.2

FORWARD 2

Tell an additive relationship from a multiplicative one in tables and equations •
Grade 6 • Expressions & Equations • 6.EE.C.9

Which equation shows that y is always 4 times x ?

A. $y = 4x$

B. $y = x + 4$

C. $y = x - 4$

D. $x = 4y$

True or false?

The table fits the equation $y = 5x$.

x	1	2	3
y	5	6	7

A. True

B. False

Write an equation for y in terms of x .

x	2	4	6
y	6	12	18

Equation: _____

Write an equation for y in terms of x .

x	2	4	6
y	5	7	9

Equation: _____

True or false?

In $y = x + 6$, y is always 6 more than x .

A. True

B. False

Solve multistep word problems and write an equation with a letter for the unknown

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 2	Subtract within 1000	3.NBT.A.2
BACKWARD 3	One-step equal-groups problems with a symbol for the unknown	3.OA.A.3
FORWARD 1	Evaluate expressions with parentheses, brackets, and braces	5.OA.A.1
FORWARD 2	Use variables to write expressions for real-world problems	6.EE.B.6

SET 6 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A school bought 8 packs of 24 pencils. It gave away 57 pencils.

Write an equation with a letter for the number of pencils left. Then solve it.

Equation: _____

Pencils left: _____

Ty has \$250. He buys 3 games that cost \$45 each.

Write an equation with a letter for the money left. Then solve it.

Equation: _____

Money left: _____

A farmer has 6 crates with 35 apples in each crate. She sells 98 apples.

Which equation gives the number of apples left, a ?

A. $a = 6 \times 35 - 98$

B. $a = 6 + 35 - 98$

C. $a = 98 - 6 \times 35$

D. $a = 6 \times (35 - 98)$

Tickets cost \$9 for adults and \$6 for children. A group buys 4 adult tickets and 7 child tickets.

Write an equation with a letter for the total cost. Then solve it.

Equation: _____

Total cost: _____

True or false?

A baker makes 128 rolls and puts them in bags of 8. She sells 11 bags. The equation $b = 128 \div 8 - 11$ gives the bags left, and $b = 5$.

- A.** True
- B.** False

SET 6 • 4.OA.A.3

BACKWARD 2

Subtract within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Subtract.

$$725 - 257$$

Difference: _____

Subtract.

$$803 - 469$$

Difference: _____

True or false?

$$825 - 475 = 350$$

A. True

B. False

$$612 - 238 = ?$$

A. 484

B. 374

C. 386

D. 850

True or false?

$$900 - 345 = 665$$

A. True

B. False

SET 6 • 4.OA.A.3

BACKWARD 3

One-step equal-groups problems with a symbol for the unknown • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.3

There are 8 packs with 6 pencils in each pack.

Write an equation with a symbol for the total number of pencils. Then find the total.

Equation: _____

Pencils: _____

True or false?

Ann buys 3 packs with 8 beads in each pack. The equation $3 \times 8 = b$ gives the number of beads she buys, and $b = 24$.

- A.** True
- B.** False

There are 7 rows with 6 chairs in each row.

Which equation gives the number of chairs, c ?

A. $7 - 6 = c$

B. $7 + 6 = c$

C. $7 \div 6 = c$

D. $7 \times 6 = c$

Leo has 3 boxes with 9 markers in each box.
How many markers does he have?

Markers: _____

True or false?

6 bags with 9 oranges in each bag hold 15 oranges.

A. True

B. False

SET 6 • 4.OA.A.3

FORWARD 1

Evaluate expressions with parentheses, brackets, and braces • Grade 5 •
Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$8 \times 24 - 57$$

Value: _____

Evaluate.

$$250 - 3 \times 45$$

Value: _____

Evaluate.

$$(36 + 12) \div 6$$

A. 38

B. 8

C. 12

D. 42

True or false?

$$4 \times (9 + 6) = 42$$

A. True

B. False

Evaluate.

$$[40 - (5 \times 6)] \times 3$$

Value: _____

SET 6 • 4.OA.A.3

FORWARD 2

Use variables to write expressions for real-world problems • Grade 6 •
Expressions & Equations • 6.EE.B.6

Pens cost \$3 each.

Write an expression for the cost of p pens.

Expression: _____

Ty has 40 stickers and gives away s stickers.

Which expression shows how many he has left?

A. $40 - s$

B. $s - 40$

C. $40s$

D. $40 + s$

A movie ticket costs \$9 and a popcorn costs \$4.

Write an expression for the cost of t tickets and one popcorn.

Expression: _____

True or false?

If n is the number of packs with 12 cards in each pack, then $12n$ is the total number of cards.

A. True

B. False

True or false?

The expression $5 + b$ means 5 times b .

A. True

B. False

Interpret remainders in word problems

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Quotients as shares or as groups	3.OA.A.2
BACKWARD 2	Division facts within 100	3.OA.C.7
FORWARD 1	Interpret a fraction as division to write a remainder as a fraction	5.NF.B.3
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 7 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

29 students ride in cars that hold 4 students each.
How many cars are needed?

Cars: _____

A baker has 50 muffins. She puts 6 muffins in each box.
How many boxes can she fill completely?

Boxes: _____

Ella shares 35 stickers equally among 4 friends.

How many stickers are left over?

A. 8

B. 3

C. 4

D. 1

A rope 46 feet long is cut into 5-foot pieces.

How many full pieces are there, and how much rope is left over?

Full pieces: _____

Left over: _____

True or false?

70 people need tables that seat 8 people each.
They need 8 tables.

A. True

B. False

SET 7 • 4.OA.A.3

BACKWARD 1

Quotients as shares or as groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

24 pencils are shared equally by 6 students.

How many pencils does each student get?

Pencils: _____

True or false?

$30 \div 5$ can mean 30 objects put into groups of 5.

A. True

B. False

Which question does $18 \div 3$ answer?

- A.** What is 3 more than 18?
- B.** How many groups of 3 are in 18?
- C.** What is 18 take away 3?
- D.** How many are in 18 groups of 3?

40 crayons are put into boxes of 8.

How many boxes are filled?

Boxes: _____

True or false?

$12 \div 4$ means 12 groups of 4.

A. True

B. False

SET 7 • 4.OA.A.3

BACKWARD 2

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$45 \div 5 = 9$$

A. True

B. False

$$64 \div 8 = ?$$

A. 56

B. 7

C. 9

D. 8

Divide.

$$36 \div 4$$

Quotient: _____

True or false?

$$81 \div 9 = 8$$

A. True

B. False

SET 7 • 4.OA.A.3

FORWARD 1

Interpret a fraction as division to write a remainder as a fraction • Grade 5 •
Number & Operations–Fractions • 5.NF.B.3

3 friends share 7 pizzas equally.

How much pizza does each friend get? Write a mixed number.

Pizza: _____

True or false?

When 9 feet of ribbon is cut into 4 equal pieces, each piece is $2\frac{1}{4}$ feet long.

A. True

B. False

5 sandwiches are shared equally by 2 people.

How much does each person get?

- A.** $\frac{2}{5}$ sandwich
- B.** 2 sandwiches
- C.** $2\frac{1}{2}$ sandwiches
- D.** 3 sandwiches

Write $17 \div 5$ as a mixed number.

Mixed number: _____

True or false?

$$11 \div 4 = 2\frac{1}{4}$$

A. True

B. False

SET 7 • 4.OA.A.3

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$4,368 \div 52$$

Quotient: _____

Divide.

$$7,695 \div 81$$

Quotient: _____

Divide.

$$9,072 \div 36$$

A. 25

B. 252

C. 242

D. 262

Divide. Write the quotient and the remainder.

$$1,000 \div 23$$

Quotient: _____

Remainder: _____

True or false?

$$3,240 \div 45 = 72$$

A. True

B. False

Solve multistep word problems in which a remainder must be interpreted

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Division word problems within 100	3.OA.A.3
BACKWARD 2	Subtract within 1000	3.NBT.A.2
FORWARD 1	Interpret the quotient in multistep problems with two-digit divisors	5.NBT.B.6
FORWARD 2	Divide multi-digit numbers to solve multistep problems — nearest...	6.NS.B.2

SET 65 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

32 students are going on a trip, but 3 students stay home. The rest ride in cars that hold 4 students each.

Write an equation with a letter for the number of students who ride. Then find how many cars are needed.

Equation: _____

Cars needed: _____

A baker makes 4 trays of 15 muffins. She packs all the muffins in boxes that hold 8 muffins each.

How many boxes can she fill completely? How many muffins are left over?

Full boxes: _____

Left over: _____

A class has \$100. It buys a game for \$30, then spends the rest on puzzles that cost \$9 each.

How many puzzles can it buy?

A. 8

B. 11

C. 9

D. 7

Ms. Ruiz has 3 bags with 18 apples in each bag. She puts all the apples in baskets that hold 5 apples each.

Write an equation with a letter for the number of apples. Then find the fewest baskets she needs.

Equation: _____

Baskets: _____

True or false?

Ty has 50 feet of ribbon. He uses 8 feet for a bow and cuts the rest into 5-foot pieces. He gets 8 full pieces, and 2 feet are left over.

- A.** True
- B.** False

SET 65 • 4.OA.A.3

BACKWARD 1

Division word problems within 100 • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.3

28 students ride in 4 cars with the same number of students in each car.

How many students ride in each car?

Students: _____

True or false?

48 muffins packed 8 to a box fill 6 boxes.

A. True

B. False

A coach puts 45 balls into 5 bins with the same number in each bin.

How many balls are in each bin?

A. 40

B. 8

C. 9

D. 50

Ty cuts 36 feet of rope into 4-foot pieces.
How many pieces does he get?

Pieces: _____

True or false?

63 apples shared equally by 9 baskets puts 8 apples in each basket.

- A. True
- B. False

SET 65 • 4.OA.A.3

BACKWARD 2

Subtract within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Subtract.

$$100 - 28$$

Difference: _____

True or false?

$$250 - 76 = 174$$

A. True

B. False

$$512 - 145 = ?$$

A. 433

B. 657

C. 377

D. 367

Subtract.

$$403 - 167$$

Difference: _____

True or false?

$$600 - 254 = 356$$

A. True

B. False

SET 65 • 4.OA.A.3

FORWARD 1

Interpret the quotient in multistep problems with two-digit divisors • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.6

A school has 412 students, and 20 are absent. The rest ride buses that hold 48 students each.

How many buses are needed?

Buses: _____

True or false?

A farm packs 4 crates of 96 eggs into cartons of 12. It fills exactly 32 cartons.

A. True

B. False

A club has \$500. It spends \$86 on snacks and the rest on tickets that cost \$25 each.

How many tickets can it buy?

A. 414

B. 16

C. 17

D. 20

A printer makes 3 boxes of 250 cards. The cards are tied in stacks of 35.

How many full stacks are there? How many cards are left over?

Full stacks: _____

Left over: _____

True or false?

A hall has 340 seats, and 15 are closed. The open seats are split into rows of 26. There are 13 full rows.

A. True

B. False

FORWARD 2

Divide multi-digit numbers to solve multistep problems • Grade 6 • The Number System • 6.NS.B.2

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A theater has 2,450 seats, and 90 are closed for repairs. The open seats are sold in blocks of 24.

Use the standard algorithm to find how many full blocks can be sold. Show your work.

Full blocks: _____

A factory makes 6 batches of 1,250 bolts. It packs them in boxes that hold 144 bolts each.

How many boxes are needed to pack every bolt?

Boxes: _____

A charity has \$5,000. It spends \$310 on shipping and the rest on blankets that cost \$45 each.

How many blankets can it buy?

A. 104

B. 105

C. 111

D. 4,690

1,000 runners sign up for a race, and 64 do not start.
The starters run in waves of 36.

How many waves are needed so every starter runs?

Waves: _____

True or false?

A school orders 3 cases of 1,200 pencils and shares them equally among 28 classes. Each class gets 128 pencils, and 16 pencils are left over.

A. True

B. False

Use rounding and mental math to check whether an answer is reasonable

MAIN	Grade 4 main question	4.OA.A.3
BACKWARD 1	Round whole numbers to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Check answers with rounding	3.OA.D.8
FORWARD 1	Round decimals to estimate	5.NBT.A.4
FORWARD 2	Use estimates to check computations with multi-digit decimals —...	6.NS.B.3

SET 8 • 4.OA.A.3

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Round each number to the nearest hundred. Then estimate the sum.

$$512 + 289$$

Rounded numbers: _____

Estimate: _____

Mia says $48 \times 6 = 188$.

Round 48 to the nearest ten and estimate 48×6 . Is her answer reasonable?

Estimate: _____

Reasonable? _____

Round each number to the nearest ten. Which is the estimate of $73 + 28 + 41$?

A. 150

B. 142

C. 140

D. 130

A store had 1,215 shirts and sold 487 of them.

Round each number to the nearest hundred to estimate how many shirts are left.

Estimate: _____

True or false?

Rounding each number to the nearest hundred,
 $395 + 612$ is about 900.

- A. True
- B. False

SET 8 • 4.OA.A.3

BACKWARD 1

Round whole numbers to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1

Round 67 to the nearest ten.

Rounded: _____

True or false?

348 rounded to the nearest hundred is 400.

A. True

B. False

Round 452 to the nearest ten.

A. 400

B. 460

C. 500

D. 450

Round 815 to the nearest hundred.

Rounded: _____

True or false?

75 rounded to the nearest ten is 80.

A. True

B. False

SET 8 • 4.OA.A.3

BACKWARD 2

Check answers with rounding • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

Rounding each number to the nearest ten,
 $38 + 51$ is about 90.

- A. True
- B. False

Leo says $297 + 102 = 499$. Rounding to the nearest hundred gives $300 + 100 = 400$.

What does the estimate show?

- A.** An estimate cannot check a sum.
- B.** His answer is too big.
- C.** His answer is correct.
- D.** His answer is too small.

Round each number to the nearest ten. Then estimate.

$$62 - 29$$

Estimate: _____

True or false?

Rounding each number to the nearest hundred,
 $710 - 290$ is about 600.

A. True

B. False

Round each number to the nearest ten. Then estimate.

$$47 + 33$$

Estimate: _____

SET 8 • 4.OA.A.3

FORWARD 1

Round decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round each number to the nearest whole number. Then estimate.

$$4.8 + 7.3$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $9.6 - 3.2$ is about 7.

A. True

B. False

Round 6.47 to the nearest tenth.

A. 6.4

B. 6

C. 6.5

D. 7

Round each number to the nearest whole number. Then estimate.

$$2.9 \times 4.1$$

Estimate: _____

True or false?

Rounding each number to the nearest whole number, $12.5 + 6.5$ is about 18.

A. True

B. False

FORWARD 2

Use estimates to check computations with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Round each number to the nearest whole number. Use your estimate to decide whether $18.75 + 6.4 = 25.15$ is reasonable.

Estimate: _____

Reasonable? _____

True or false?

Rounding each number to the nearest whole number, 3.9×5.2 is about 20, so a product of 202.8 is not reasonable.

A. True

B. False

Ann computed $45.6 \div 3.8 = 1.2$. Rounding each number to the nearest whole number gives $46 \div 4$, which equals 11.5.

What does this show?

- | | | | |
|------------------------------------|--|--|----------------------------------|
| A. Her answer is too large. | B. The estimate cannot be used. | C. Her answer is far too small. | D. Her answer is correct. |
|------------------------------------|--|--|----------------------------------|

Multiply.

$$12.4 \times 3.5$$

Product: _____

True or false?

Rounding each number to the nearest whole number, 7.1×2.9 is about 21, so a product of 2.059 is reasonable.

- A.** True
- B.** False

Find all factor pairs of a whole number from 1 to 100

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Solve a division problem by finding an unknown factor	3.OA.B.6
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Use multiples to find a common denominator	5.NF.A.1
FORWARD 2	Find the greatest common factor	6.NS.B.4

SET 9 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 24.

Factor pairs: _____

List all the factor pairs of 45.

Factor pairs: _____

Which is NOT a factor pair of 60?

A. 7×9

B. 5×12

C. 6×10

D. 4×15

True or false?

36 has exactly 5 factor pairs.

A. True

B. False

How many factors does 18 have? List them.

How many: _____

Factors: _____

SET 9 • 4.OA.B.4

BACKWARD 1

Solve a division problem by finding an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $24 \div 4$ by finding the unknown factor.

$$4 \times ? = 24$$

Quotient: _____

True or false?

$32 \div 8 = 4$ because $8 \times 4 = 32$.

A. True

B. False

Which unknown-factor equation helps you find $45 \div 5$?

A. $5 \times ? = 45$

B. $5 - ? = 45$

C. $45 + ? = 5$

D. $45 \times 5 = ?$

Write the unknown-factor equation for $28 \div 7$. Then find the quotient.

Equation: _____

Quotient: _____

True or false?

$48 \div 6 = 7$ because $6 \times 7 = 48$.

A. True

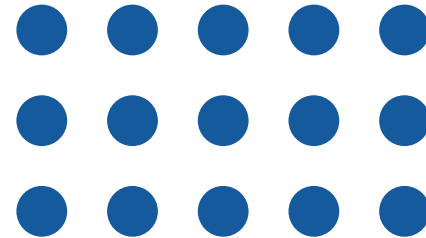
B. False

SET 9 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many dots are in the array?



Dots: _____

True or false?

An array with 4 rows of 5 dots has 20 dots.

- A. True
- B. False

Which addition equation matches an array with 2 rows of 4?

A. $4 - 2 = 2$

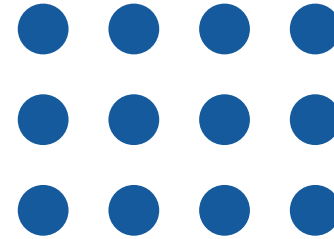
B. $2 + 2 = 4$

C. $4 + 4 = 8$

D. $2 + 4 = 6$

Write an addition equation for the array.

Equation: _____



True or false?

An array with 5 rows of 3 dots has 8 dots.

- A. True
- B. False

SET 9 • 4.OA.B.4

FORWARD 1

Use multiples to find a common denominator • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Find the least common denominator of $\frac{1}{4}$ and $\frac{1}{6}$.

Denominator: _____

True or false?

$\frac{2}{3}$ and $\frac{3}{4}$ can both be written with a denominator of 12.

- A.** True
- B.** False

Which is a common denominator of $\frac{1}{3}$ and $\frac{2}{5}$?

- A.** 3
- B.** 15
- C.** 10
- D.** 8

Rewrite $\frac{1}{2}$ and $\frac{1}{3}$ as fractions with a common denominator.

Fractions: _____

True or false?

A common denominator of $\frac{1}{4}$ and $\frac{1}{10}$ is 14.

A. True

B. False

SET 9 • 4.OA.B.4

FORWARD 2

Find the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

Find the greatest common factor of 18 and 45.

GCF: _____

What is the greatest common factor of 16 and 40?

- A.** 2
- B.** 16
- C.** 4
- D.** 8

True or false?

The greatest common factor of 14 and 35 is 7.

A. True

B. False

True or false?

The greatest common factor of 20 and 30 is 5.

A. True

B. False

A whole number is a multiple of each of its factors; find multiples

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Skip count and find patterns in the multiplication table	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Generate two patterns and compare their terms	5.OA.B.3
FORWARD 2	Find the least common multiple	6.NS.B.4

SET 10 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first six positive multiples of 7.

Multiples: _____

True or false?

Because 6 is a factor of 42, 42 is a multiple of 6.

A. True

B. False

Which number is a multiple of 9?

A. 39

B. 19

C. 54

D. 92

Is 84 a multiple of 6? Explain.

Answer: _____

Explanation: _____

True or false?

70 is a multiple of 4.

A. True

B. False

SET 10 • 4.OA.B.4

BACKWARD 1

Skip count and find patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

Skip count by 6.

6, 12, 18, __, __

Next two: _____

True or false?

Every number in the 5s row of the multiplication table ends in 0 or 5.

- A.** True
- B.** False

Which number is in the 4s row of the multiplication table?

- A.** 18
- B.** 30
- C.** 28
- D.** 26

What comes next?

9, 18, 27, 36, ____

Next: _____

True or false?

Every number in the 3s row of the multiplication table is odd.

- A.** True
- B.** False

SET 10 • 4.OA.B.4

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

16 is even.

A. True

B. False

Is 13 odd or even?

Answer: _____

Which number is even?

- A.** 7
- B.** 18
- C.** 15
- D.** 11

Write 14 as the sum of two equal addends.

Sum: _____

True or false?

An even number of objects cannot be put into two equal groups.

- A.** True
- B.** False

SET 10 • 4.OA.B.4

FORWARD 1

Generate two patterns and compare their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12.

Write the first five terms of each pattern.

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12. Each term of B is 3 times the matching term of A.

A. True

B. False

Pattern A: 0, 5, 10, 15. Pattern B: 0, 10, 20, 30.

Which ordered pair (A, B) comes from the patterns?

A. (15, 30)

B. (15, 20)

C. (30, 15)

D. (10, 30)

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 10.

What is the 5th term of Pattern B?

Term: _____

True or false?

Each term of Pattern B is 4 times the matching term of Pattern A. When A is 9, B is 13.

A. True

B. False

SET 10 • 4.OA.B.4

FORWARD 2

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

Find the least common multiple of 5 and 7.

LCM: _____

What is the least common multiple of 4 and 10?

- A.** 20
- B.** 2
- C.** 14
- D.** 40

True or false?

The least common multiple of 9 and 12 is 36.

A. True

B. False

True or false?

The least common multiple of 3 and 6 is 18.

A. True

B. False

Decide whether a whole number from 1 to 100 is prime or composite

MAIN	Grade 4 main question	4.OA.B.4
BACKWARD 1	Find the unknown number in a multiplication or division equation	3.OA.A.4
BACKWARD 2	Arrays with up to 5 rows and 5 columns	2.OA.C.4
FORWARD 1	Evaluate expressions with grouping symbols	5.OA.A.1
FORWARD 2	Write and evaluate expressions with whole-number exponents	6.EE.A.1

SET 11 • 4.OA.B.4

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

Which number is neither prime nor composite?

A. 29

B. 9

C. 1

D. 2

True or false?

51 is a prime number.

A. True

B. False

Is 39 prime or composite? Explain.

Answer: _____

Explanation: _____

List all the prime numbers between 20 and 40.

Primes: _____

True or false?

A composite number has more than two factors.

A. True

B. False

SET 11 • 4.OA.B.4

BACKWARD 1

Find the unknown number in a multiplication or division equation • Grade 3 •
Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$$? \times 3 = 21$$

Unknown: _____

True or false?

If $7 \times ? = 49$, the unknown number is 7.

A. True

B. False

$$9 \times ? = 54$$

A. 7

B. 45

C. 6

D. 5

Find the unknown number.

$$35 \div ? = 5$$

Unknown: _____

True or false?

If $? \times 4 = 32$, the unknown number is 9.

A. True

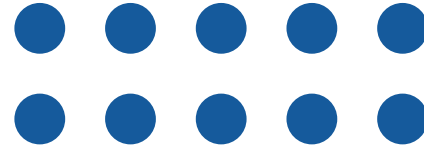
B. False

SET 11 • 4.OA.B.4

BACKWARD 2

Arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation for the array.



Equation: _____

True or false?

5 dots can make a rectangle only as 1 row of 5 or 5 rows of 1.

A. True

B. False

Which array has 12 dots?

- A.** 3 rows of 4
- B.** 4 rows of 4
- C.** 3 rows of 5
- D.** 2 rows of 5

How many dots are in an array with 4 rows of 4?

Dots: _____

True or false?

An array with 2 rows of 3 dots has 5 dots.

- A. True
- B. False

SET 11 • 4.OA.B.4

FORWARD 1

Evaluate expressions with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times 3 \times 7$$

Value: _____

Evaluate.

$$2 \times (5 \times 5)$$

Value: _____

Evaluate.

$$(2 \times 2) \times (3 \times 3)$$

A. 10

B. 36

C. 24

D. 12

True or false?

$$3 \times (2 \times 11) = 66$$

A. True

B. False

True or false?

$$[2 \times (3 + 5)] \times 2 = 22$$

A. True

B. False

SET 11 • 4.OA.B.4

FORWARD 2

Write and evaluate expressions with whole-number exponents • Grade 6 •
Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^3 \times 3$$

Value: _____

Write $2 \times 2 \times 2 \times 2 \times 5$ using an exponent.

Expression: _____

Which expression equals 72?

A. 6^2

B. $2^2 \times 3^3$

C. $2^3 \times 3^2$

D. $2 \times 3 \times 6$

True or false?

$$3^2 \times 5 = 45$$

A. True

B. False

True or false?

$$2^5 = 10$$

A. True

B. False

Generate a number or shape pattern that follows a given rule

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Arithmetic patterns	3.OA.D.9
BACKWARD 2	Add or subtract 10 or 100 mentally	2.NBT.B.8
FORWARD 1	Form ordered pairs from two patterns	5.OA.B.3
FORWARD 2	Write an equation for a pattern with dependent and independent v...	6.EE.C.9

SET 12 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 7." The pattern starts at 4.
Write the first five terms.

Terms: _____

The rule is "multiply by 3." The pattern starts at 2.
Write the first four terms.

Terms: _____

Which pattern follows the rule "subtract 6," starting at 50?

A. 50, 45, 40, 35

B. 50, 44, 38, 32

C. 50, 56, 62, 68

D. 44, 38, 32, 26

True or false?

The rule "add 9," starting at 15, gives 15, 24, 33, 43.

A. True

B. False

In a shape pattern, each figure has 2 more squares than the one before. The 1st figure has 3 squares.

How many squares are in the 6th figure?

Squares: _____

SET 12 • 4.OA.C.5

BACKWARD 1

Arithmetic patterns • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

What comes next?

12, 16, 20, 24, ____

Next: _____

True or false?

The pattern 5, 10, 15, 20 adds 5 each time.

A. True

B. False

What is the rule for 30, 27, 24, 21?

- A.** Add 3
- B.** Multiply by 3
- C.** Subtract 3
- D.** Subtract 2

What comes next?

7, 14, 21, 28, ____

Next: _____

True or false?

The pattern 2, 4, 8, 16 adds 2 each time.

A. True

B. False

SET 12 • 4.OA.C.5

BACKWARD 2

Add or subtract 10 or 100 mentally • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8

What is 10 more than 347?

Answer: _____

True or false?

100 less than 612 is 512.

A. True

B. False

What is 100 more than 285?

- A.** 295
- B.** 385
- C.** 185
- D.** 286

What is 10 less than 503?

Answer: _____

True or false?

10 more than 790 is 890.

A. True

B. False

SET 12 • 4.OA.C.5

FORWARD 1

Form ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 1 and adds 1. Pattern B starts at 5 and adds 5.

Write the first four ordered pairs (A, B).

Ordered pairs: _____

True or false?

Pattern A is 0, 3, 6, 9 and Pattern B is 0, 9, 18, 27.
Each B term is 3 times the matching A term.

A. True

B. False

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8.

What is the B term when the A term is 6?

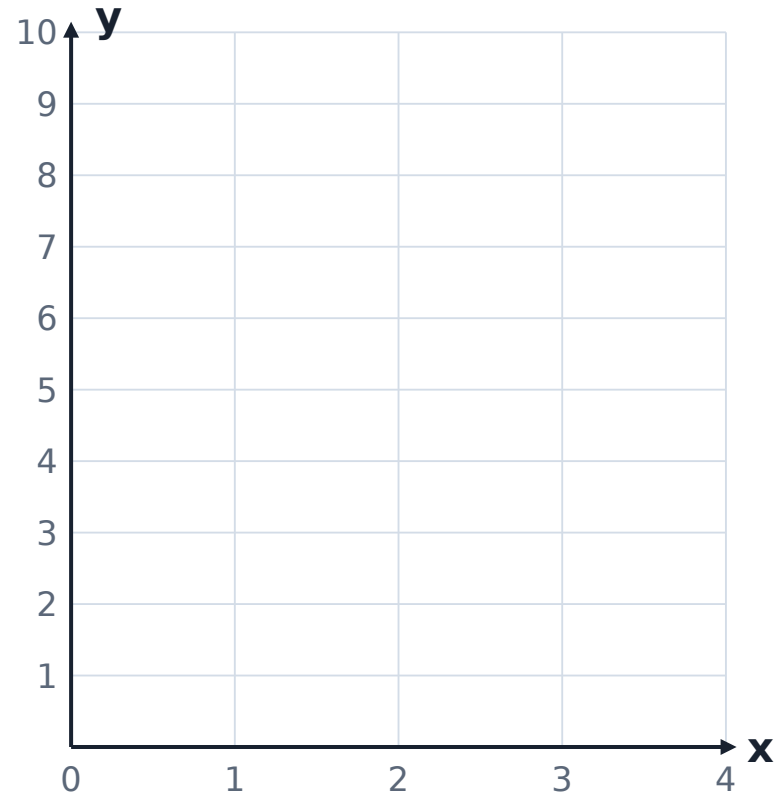
A. 14

B. 12

C. 48

D. 24

Graph the ordered pairs (0, 0), (1, 3), (2, 6), and (3, 9) from two patterns.



True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. The 4th pair is (20, 40).

A. True

B. False

SET 12 • 4.OA.C.5

FORWARD 2

Write an equation for a pattern with dependent and independent variables •
Grade 6 • Expressions & Equations • 6.EE.C.9

Figure 1 has 4 tiles, and each figure has 4 more tiles than the one before.

Write an equation for the tiles t in figure n .

Equation: _____

In $t = 4n$, which variable depends on the other?

- A.** t
- B.** n
- C.** 4
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	6	12	18	24

Equation: _____

True or false?

In $c = 8h$, when h increases by 1, c increases by 8.

- A. True
- B. False

True or false?

The table x : 1, 2, 3 and y : 3, 5, 7 fits $y = 3x$.

A. True

B. False

Identify features of a pattern that the rule does not state, and explain them

MAIN	Grade 4 main question	4.OA.C.5
BACKWARD 1	Explain patterns in the multiplication and addition tables	3.OA.D.9
BACKWARD 2	Odd and even numbers	2.OA.C.3
FORWARD 1	Explain the relationship between two patterns	5.OA.B.3
FORWARD 2	Evaluate expressions for a given value of the variable	6.EE.A.2.c

SET 13 • 4.OA.C.5

MAIN

Grade 4 main question • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 4," starting at 2: 2, 6, 10, 14, 18.

What do you notice about every term? Explain why it happens.

Feature: _____

Why: _____

The rule is "add 5," starting at 3: 3, 8, 13, 18, 23.

Which statement is true?

- A.** Every term is a multiple of 5.
- B.** Every term is even.
- C.** The ones digits alternate between 3 and 8.
- D.** Every term is odd.

True or false?

The rule "add 2," starting at 7, makes only odd numbers.

A. True

B. False

True or false?

The rule "add 6," starting at 1, makes terms that alternate between odd and even.

- A. True
- B. False

The rule is "multiply by 2," starting at 3: 3, 6, 12, 24, 48.
After the first term, are the terms odd or even? Explain why.

Answer: _____

Why: _____

SET 13 • 4.OA.C.5

BACKWARD 1

Explain patterns in the multiplication and addition tables • Grade 3 •
Operations & Algebraic Thinking • 3.OA.D.9

True or false?

4 times any whole number is even.

A. True

B. False

Why is every product in the 6s row of the multiplication table even?

- A.** The 6s row has 10 products.
- B.** 6 is greater than 5.
- C.** Every number is even.
- D.** 6 is 2×3 , so every product is 2 times a number.

The 10s row is 10, 20, 30, 40, ...

What is the same about every product?

Answer: _____

True or false?

Every product in the 9s row of the multiplication table is odd.

- A. True
- B. False

Is the sum of two odd numbers odd or even?

Answer: _____

SET 13 • 4.OA.C.5

BACKWARD 2

Odd and even numbers • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

True or false?

11 is odd.

A. True

B. False

Is 16 odd or even?

Answer: _____

Which number is odd?

A. 10

B. 8

C. 14

D. 17

Write 18 as the sum of two equal addends.

Sum: _____

True or false?

Adding two even numbers can give an odd number.

A. True

B. False

SET 13 • 4.OA.C.5

FORWARD 1

Explain the relationship between two patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6. Pattern B starts at 0 and adds 2.

Explain why each term of A is 3 times the matching term of B.

Explanation: _____

True or false?

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8. Each B term is 2 times the A term because B adds twice as much at every step.

A. True

B. False

Pattern A starts at 2 and adds 2. Pattern B starts at 2 and adds 6.

Why is B NOT always 3 times A?

- A.** A has more terms.
- B.** Both rules add even numbers.
- C.** B adds less than A.
- D.** They start at the same number, not at 0.

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

What is the B term when the A term is 35?

Term: _____

True or false?

Two patterns start at 0 and add 5 and 10. Each B term is 5 more than the matching A term.

A. True

B. False

SET 13 • 4.OA.C.5

FORWARD 2

Evaluate expressions for a given value of the variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $4n + 2$ when $n = 6$.

Value: _____

True or false?

The expression $2n$ gives an even number for every whole number n .

A. True

B. False

The n th term of 5, 9, 13, 17, ... is $4n + 1$.

What is the 10th term?

- A.** 40
- B.** 45
- C.** 41
- D.** 14

Evaluate $3n - 1$ when $n = 12$.

Value: _____

True or false?

When $n = 5$, $2n + 1 = 12$.

A. True

B. False

A digit in one place is worth ten times what it is worth in the place to its right

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	Hundreds, tens, and ones	2.NBT.A.1
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Place value in decimals: 10 times and 1/10 of a digit	5.NBT.A.1
FORWARD 2	Write and evaluate powers of 10 with exponents	6.EE.A.1

SET 14 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

True or false?

In 4,440, the 4 in the hundreds place is worth 10 times the 4 in the tens place.

- A. True
- B. False

The 8 in 8,325 is worth how many times the 8 in 825?

- A.** 10 times
- B.** 8 times
- C.** The same
- D.** 100 times

What is the value of the 6 in 62,904?

Value: _____

Write the number that is 10 times as much as 3,070.

Number: _____

True or false?

In 5,555, the 5 in the tens place is worth 10 times the 5 in the hundreds place.

- A.** True
- B.** False

SET 14 • 4.NBT.A.1

BACKWARD 1

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

How many tens make 1 hundred?

Tens: _____

True or false?

In 472, the 7 means 7 tens.

A. True

B. False

What is the value of the 3 in 360?

A. 3,000

B. 30

C. 3

D. 300

Write the number with 5 hundreds, 0 tens, and 8 ones.

Number: _____

True or false?

In 914, the 1 means 1 hundred.

A. True

B. False

SET 14 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$6 \times 40$$

Product: _____

True or false?

$$3 \times 70 = 210$$

A. True

B. False

$$9 \times 50 = ?$$

- A.** 45
- B.** 59
- C.** 4,500
- D.** 450

Multiply.

$$8 \times 30$$

Product: _____

True or false?

$$5 \times 80 = 4,000$$

A. True

B. False

SET 14 • 4.NBT.A.1

FORWARD 1

Place value in decimals: 10 times and $\frac{1}{10}$ of a digit • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1

True or false?

In 3.33, the 3 in the ones place is worth 10 times the 3 in the tenths place.

- A. True
- B. False

What is $\frac{1}{10}$ of 70?

Answer: _____

In 6.66, the 6 in the hundredths place is worth what part of the 6 in the tenths place?

A. $\frac{1}{10}$

B. The same amount

C. 10 times as much

D. $\frac{1}{100}$

What is 10 times 0.4?

Answer: _____

True or false?

In 2.22, the 2 in the tenths place is worth 10 times the 2 in the ones place.

A. True

B. False

SET 14 • 4.NBT.A.1

FORWARD 2

Write and evaluate powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $10 \times 10 \times 10$ using an exponent.

Exponent form: _____

True or false?

$$10^4 = 10,000$$

A. True

B. False

Evaluate.

$$10^2$$

- A.** 12
- B.** 20
- C.** 1,000
- D.** 100

Evaluate.

$$4 \times 10^3$$

Value: _____

True or false?

10^5 has 4 zeros.

A. True

B. False

Use place value and division to explain why one number is ten times another

MAIN	Grade 4 main question	4.NBT.A.1
BACKWARD 1	A hundred is a bundle of ten tens	2.NBT.A.1.a
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Divide whole numbers and decimals by powers of 10	5.NBT.A.2
FORWARD 2	Divide multi-digit decimals	6.NS.B.3

SET 15 • 4.NBT.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Use place value to find $5,000 \div 500$.

Quotient: _____

True or false?

$900 \div 90 = 10$ because 9 hundreds is 10 times 9 tens.

A. True

B. False

Which statement explains why $3,000 \div 300 = 10$?

- A.** $3,000 - 300 = 10$
- B.** 3 thousands is 10 times 3 hundreds.
- C.** 3,000 has one more zero.
- D.** $300 \div 3,000 = 10$

4,000 is how many times as much as 400? Explain using place value.

Answer: _____

Explanation: _____

True or false?

$$60 \div 6 = 100$$

A. True

B. False

SET 15 • 4.NBT.A.1

BACKWARD 1

A hundred is a bundle of ten tens • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1.a

True or false?

100 can be thought of as a bundle of 10 tens.

A. True

B. False

How many tens make 1 hundred?

Tens: _____

Which shows the same amount as 1 hundred?

- A.** 10 tens
- B.** 1 ten
- C.** 100 tens
- D.** 10 ones

Fill in the blank.

10 tens = ____ hundred

Blank: _____

True or false?

A bundle of 10 tens is the same amount as 1,000.

A. True

B. False

SET 15 • 4.NBT.A.1

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations
in Base Ten • 3.NBT.A.3

Multiply.

$$4 \times 90$$

Product: _____

True or false?

$$7 \times 60 = 420$$

A. True

B. False

$$2 \times 80 = ?$$

- A.** 160
- B.** 82
- C.** 16
- D.** 1,600

Multiply.

$$9 \times 20$$

Product: _____

True or false?

$$6 \times 50 = 3,000$$

A. True

B. False

SET 15 • 4.NBT.A.1

FORWARD 1

Divide whole numbers and decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$$4,500 \div 100$$

Quotient: _____

True or false?

$$7,000 \div 10^3 = 7$$

A. True

B. False

Divide.

$$36 \div 10$$

A. 360

B. 3.6

C. 36

D. 0.36

Divide.

$$820 \div 10^2$$

Quotient: _____

True or false?

$$50 \div 10 = 500$$

A. True

B. False

SET 15 • 4.NBT.A.1

FORWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.5 \div 0.45$$

Quotient: _____

True or false?

$$0.8 \div 0.08 = 10$$

A. True

B. False

Divide.

$$6 \div 0.6$$

A. 10

B. 0.1

C. 1

D. 100

Divide.

$$2.4 \div 0.3$$

Quotient: _____

True or false?

$$1.2 \div 0.12 = 100$$

A. True

B. False

Read and write multi-digit whole numbers using numerals and number names

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Read and write numbers to 1000	2.NBT.A.3
BACKWARD 2	Hundreds, tens, and ones	2.NBT.A.1
FORWARD 1	Read and write decimals to thousandths	5.NBT.A.3.a
FORWARD 2	Use positive and negative numbers to describe quantities — near...	6.NS.C.5

SET 16 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write the number in standard form.

forty-six thousand, two hundred eight

Number: _____

Write 309,051 in words.

Words: _____

Which is 72,016 in words?

- A.** seventy-two thousand, sixteen
- B.** seven thousand, two hundred sixteen
- C.** seventy-two thousand, one hundred six
- D.** seventy-two thousand, one hundred sixty

True or false?

Six hundred thousand, forty is written 600,400.

- A.** True
- B.** False

Write the number in standard form.

ninety thousand, nine hundred

Number: _____

SET 16 • 4.NBT.A.2

BACKWARD 1

Read and write numbers to 1000 • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.3

Write the number in standard form.

four hundred seventeen

Number: _____

True or false?

Six hundred five is written 650.

A. True

B. False

Which is 238 in words?

- A.** two hundred thirty-eight
- B.** three hundred twenty-eight
- C.** two hundred eighty-three
- D.** twenty-three eight

Write 702 in words.

Words: _____

True or false?

Nine hundred ninety is written 990.

- A. True
- B. False

SET 16 • 4.NBT.A.2

BACKWARD 2

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

What digit is in the tens place of 586?

Digit: _____

True or false?

In 249, the 2 is worth 200.

A. True

B. False

Which number has 7 hundreds, 3 tens, and 0 ones?

A. 730

B. 703

C. 73

D. 370

How many hundreds are in 600?

Hundreds: _____

True or false?

In 381, the 8 is worth 8.

A. True

B. False

SET 16 • 4.NBT.A.2

FORWARD 1

Read and write decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a

Write the number in standard form.

four and twenty-five thousandths

Number: _____

True or false?

0.6 is read "six tenths."

A. True

B. False

Which is 3.48 in words?

- A.** thirty-four and eight tenths
- B.** three hundred forty-eight
- C.** three and forty-eight tenths
- D.** three and forty-eight hundredths

Write 12.307 in words.

Words: _____

True or false?

Five and nine hundredths is written 5.9.

A. True

B. False

FORWARD 2

Use positive and negative numbers to describe quantities • Grade 6 • The Number System • 6.NS.C.5

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A diver is 30 feet below sea level.

Write the diver's position as an integer.

Integer: _____

True or false?

A gain of 8 yards can be written as +8.

A. True

B. False

Which integer shows a temperature of 12 degrees below zero?

A. -12

B. 12

C. 0

D. -21

A deposit of \$45 is written as +45.

How is a withdrawal of \$45 written?

Integer: _____

True or false?

–5 means 5 more than zero.

A. True

B. False

Write multi-digit whole numbers in expanded form

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Expanded form for numbers to 1000	2.NBT.A.3
BACKWARD 2	Two-digit numbers as tens and ones	1.NBT.B.2
FORWARD 1	Write decimals in expanded form	5.NBT.A.3.a
FORWARD 2	Write numbers using powers of 10 with exponents	6.EE.A.1

SET 17 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write 7,345 in expanded form.

Expanded form: _____

Write the number.

$$50,000 + 2,000 + 80 + 6$$

Number: _____

Which is the expanded form of 40,209?

A. $40,000 + 2,000 + 9$

B. $40,000 + 20 + 9$

C. $40,000 + 200 + 9$

D. $4,000 + 200 + 9$

True or false?

$$600,000 + 3,000 + 70 = 630,070$$

A. True

B. False

Write 18,504 in expanded form.

Expanded form: _____

SET 17 • 4.NBT.A.2

BACKWARD 1

Expanded form for numbers to 1000 • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3

Write 846 in expanded form.

Expanded form: _____

True or false?

$$300 + 9 = 390$$

A. True

B. False

Which number is $500 + 20 + 7$?

- A.** 527
- B.** 572
- C.** 257
- D.** 5,207

Write the number.

$$900 + 60$$

Number: _____

True or false?

$$200 + 40 + 1 = 241$$

A. True

B. False

SET 17 • 4.NBT.A.2

BACKWARD 2

Two-digit numbers as tens and ones • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.2

How many tens and ones are in 47?

Answer: _____

True or false?

63 is 6 tens and 3 ones.

A. True

B. False

Which number is 8 tens and 2 ones?

- A.** 28
- B.** 10
- C.** 82
- D.** 802

Write the number that is 5 tens and 0 ones.

Number: _____

True or false?

91 is 1 ten and 9 ones.

A. True

B. False

FORWARD 1

Write decimals in expanded form • Grade 5 • Number & Operations in Base Ten
• 5.NBT.A.3.a

Write 3.46 in expanded form.

Expanded form: _____

True or false?

$$2 + 0.5 + 0.03 = 2.53$$

A. True

B. False

Which number is $7 \times 1 + 2 \times \frac{1}{10} + 9 \times \frac{1}{1000}$?

A. 7.029

B. 7.209

C. 72.9

D. 7.29

Write the number.

$$5 \times 10 + 1 \times 1 + 8 \times \frac{1}{100}$$

Number: _____

True or false?

$$4 \times \frac{1}{10} + 1 \times \frac{1}{100} = 0.041$$

A. True

B. False

SET 17 • 4.NBT.A.2

FORWARD 2

Write numbers using powers of 10 with exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $6,000 + 400 + 70 + 2$ using powers of 10.

Expression: _____

True or false?

$$3 \times 10^4 = 30,000$$

A. True

B. False

Which number equals $5 \times 10^3 + 8 \times 10$?

- A.** 580
- B.** 5,008
- C.** 5,800
- D.** 5,080

Evaluate.

$$2 \times 10^3 + 9 \times 10^2$$

Value: _____

True or false?

$$7 \times 10^2 = 70$$

A. True

B. False

Compare multi-digit whole numbers using $>$, $=$, and $<$

MAIN	Grade 4 main question	4.NBT.A.2
BACKWARD 1	Compare three-digit numbers	2.NBT.A.4
BACKWARD 2	Compare two-digit numbers	1.NBT.B.3
FORWARD 1	Compare decimals to thousandths	5.NBT.A.3.b
FORWARD 2	Order rational numbers using their positions on a number line	6.NS.C.7.a

SET 18 • 4.NBT.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.2

Write $>$, $=$, or $<$.

45,621 ____ 45,612

Symbol: _____

Which comparison is true?

A. $80,099 > 80,100$

B. $80,099 = 80,100$

C. $80,100 < 80,099$

D. $80,099 < 80,100$

True or false?

305,000 < 350,000

A. True

B. False

Order from least to greatest.

27,450 24,750 27,045

Order: _____

True or false?

9,998 > 10,001

A. True

B. False

SET 18 • 4.NBT.A.2

BACKWARD 1

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

482 428

Symbol: _____

True or false?

$$309 < 390$$

A. True

B. False

Which number is greatest?

A. 170

B. 699

C. 701

D. 710

Write $>$, $=$, or $<$.

555 555

Symbol: _____

True or false?

$$640 > 664$$

A. True

B. False

SET 18 • 4.NBT.A.2

BACKWARD 2

Compare two-digit numbers • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Write $>$, $=$, or $<$.

67 76

Symbol: _____

True or false?

$$58 > 49$$

A. True

B. False

Which number is less than 35?

A. 35

B. 53

C. 29

D. 40

Write $>$, $=$, or $<$.

90 19

Symbol: _____

True or false?

$$42 < 24$$

A. True

B. False

SET 18 • 4.NBT.A.2

FORWARD 1

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Write $>$, $=$, or $<$.

0.45 0.405

Symbol: _____

True or false?

$$3.09 < 3.1$$

A. True

B. False

Which decimal is greatest?

- A.** 2.5
- B.** 2.09
- C.** 2.405
- D.** 2.48

Write $>$, $=$, or $<$.

6.70 6.7

Symbol: _____

True or false?

$$0.125 > 0.13$$

A. True

B. False

SET 18 • 4.NBT.A.2

FORWARD 2

Order rational numbers using their positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

Write $>$, $=$, or $<$.

$$-8 \text{ ____ } -3$$

Symbol: _____

True or false?

$$-2 > -10$$

A. True

B. False

Which number is least?

A. 0

B. -5

C. 5

D. -15

Order from least to greatest.

4, -6, 0, -1

Order: _____

True or false?

$$-7 > 2$$

A. True

B. False

Round multi-digit whole numbers to any place

MAIN	Grade 4 main question	4.NBT.A.3
BACKWARD 1	Round to the nearest 10 or 100	3.NBT.A.1
BACKWARD 2	Skip-count by 5s, 10s, and 100s	2.NBT.A.2
FORWARD 1	Round decimals to any place	5.NBT.A.4
FORWARD 2	Compute with decimals and round the result — nearest related (n...	6.NS.B.3

SET 19 • 4.NBT.A.3

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 46,829 to the nearest thousand.

Rounded: _____

Round 352,618 to the nearest ten thousand.

Rounded: _____

Round 7,950 to the nearest hundred.

A. 7,900

B. 7,960

C. 7,000

D. 8,000

True or false?

128,476 rounded to the nearest hundred thousand is 200,000.

- A. True
- B. False

A stadium has 63,548 seats.

Round the number of seats to the nearest thousand.

Rounded: _____

SET 19 • 4.NBT.A.3

BACKWARD 1

Round to the nearest 10 or 100 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.1

Round 84 to the nearest ten.

Rounded: _____

True or false?

652 rounded to the nearest hundred is 700.

A. True

B. False

Round 345 to the nearest ten.

A. 400

B. 350

C. 300

D. 340

Round 961 to the nearest hundred.

Rounded: _____

True or false?

26 rounded to the nearest ten is 20.

A. True

B. False

SET 19 • 4.NBT.A.3

BACKWARD 2

Skip-count by 5s, 10s, and 100s • Grade 2 • Number & Operations in Base Ten
• 2.NBT.A.2

Skip-count by 100s.

300, 400, 500, ____

Next: _____

True or false?

Counting by 10s from 470 gives 480, 490, 500.

A. True

B. False

What comes next when counting by 5s?

85, 90, 95, ____

A. 96

B. 105

C. 100

D. 99

Skip-count by 10s.

620, 630, 640, ____

Next: _____

True or false?

Counting by 100s from 700 gives 800, 900, 990.

A. True

B. False

SET 19 • 4.NBT.A.3

FORWARD 1

Round decimals to any place • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.468 to the nearest tenth.

Rounded: _____

Round 14.729 to the nearest whole number.

Rounded: _____

Round 0.638 to the nearest hundredth.

- A.** 0.6
- B.** 0.64
- C.** 0.7
- D.** 0.63

True or false?

5.95 rounded to the nearest tenth is 6.0.

A. True

B. False

True or false?

8.249 rounded to the nearest tenth is 8.3.

A. True

B. False

FORWARD 2

Compute with decimals and round the result • Grade 6 • The Number System •
6.NS.B.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Divide. Round the quotient to the nearest tenth.

$$10 \div 3$$

Quotient: _____

True or false?

7.25×4.2 , rounded to the nearest whole number, is 30.

- A. True
- B. False

Divide. Round the quotient to the nearest hundredth.

$$2 \div 7$$

A. 0.27

B. 0.29

C. 0.28

D. 0.3

Multiply. Round the product to the nearest whole number.

$$3.6 \times 2.4$$

Product: _____

True or false?

$15 \div 4$, rounded to the nearest whole number, is 3.

- A. True
- B. False

Add multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Add within 1000	3.NBT.A.2
BACKWARD 2	Add by place value, making a ten or a hundred	2.NBT.B.7
FORWARD 1	Add decimals to hundredths	5.NBT.B.7
FORWARD 2	Add multi-digit decimals with the standard algorithm	6.NS.B.3

SET 20 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to add. Show your work.

$$4,587 + 2,946$$

Sum: _____

Use the standard algorithm to add. Show your work.

$$38,209 + 15,894$$

Sum: _____

Add.

$$6,078 + 3,945$$

- A.** 9,023
- B.** 9,923
- C.** 10,013
- D.** 10,023

A library had 12,486 books. It got 3,758 more.

Use the standard algorithm to find how many books it has now. Show your work.

Books: _____

True or false?

In the standard algorithm for $3,685 + 1,476$, you regroup 11 ones as 1 ten and 1 one.

- A. True
- B. False

SET 20 • 4.NBT.B.4

BACKWARD 1

Add within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Add.

$$356 + 478$$

Sum: _____

True or false?

$$609 + 295 = 904$$

A. True

B. False

$$147 + 268 = ?$$

- A.** 415
- B.** 315
- C.** 405
- D.** 425

Add.

$$523 + 389$$

Sum: _____

True or false?

$$480 + 360 = 740$$

A. True

B. False

SET 20 • 4.NBT.B.4

BACKWARD 2

Add by place value, making a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

7 ones and 5 ones make 1 ten and 2 ones.

A. True

B. False

Rename 16 tens as hundreds and tens.

Answer: _____

Add the hundreds, the tens, and the ones.

$$248 + 135$$

A. 483

B. 373

C. 383

D. 313

Add.

$$200 + 70 + 4 + 300 + 20 + 9$$

Sum: _____

True or false?

12 ones is the same as 2 tens.

A. True

B. False

SET 20 • 4.NBT.B.4

FORWARD 1

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$35.68 + 7.45$$

Sum: _____

Add.

$$4.9 + 12.36$$

Sum: _____

Add.

$$0.75 + 0.48$$

- A.** 1.13
- B.** 12.3
- C.** 1.23
- D.** 0.123

True or false?

$$6.07 + 3.9 = 9.97$$

A. True

B. False

True or false?

$$2.5 + 1.75 = 3.80$$

A. True

B. False

SET 20 • 4.NBT.B.4

FORWARD 2

Add multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.406 + 75.89$$

Sum: _____

Add.

$$9.075 + 0.98$$

Sum: _____

Add.

$$46.3 + 8.759$$

A. 55.059

B. 131.89

C. 55.159

D. 54.059

True or false?

$$0.625 + 0.375 = 1$$

A. True

B. False

True or false?

$$12.48 + 3.6 = 12.84$$

A. True

B. False

Subtract multi-digit whole numbers using the standard algorithm

MAIN	Grade 4 main question	4.NBT.B.4
BACKWARD 1	Subtract within 1000	3.NBT.A.2
BACKWARD 2	Subtract by breaking apart a ten or a hundred	2.NBT.B.7
FORWARD 1	Subtract decimals to hundredths	5.NBT.B.7
FORWARD 2	Subtract multi-digit decimals with the standard algorithm	6.NS.B.3

SET 21 • 4.NBT.B.4

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Use the standard algorithm to subtract. Show your work.

$$8,253 - 4,678$$

Difference: _____

Use the standard algorithm to subtract. Show your work.

$$60,000 - 27,438$$

Difference: _____

Subtract.

$$7,012 - 3,456$$

A. 4,656

B. 3,556

C. 4,444

D. 3,566

A school raised \$15,250. It spent \$8,795.

Use the standard algorithm to find how much money is left. Show your work.

Money left: _____

True or false?

To find $5,042 - 1,317$ with the standard algorithm, you rename 1 ten as 10 ones.

- A. True
- B. False

SET 21 • 4.NBT.B.4

BACKWARD 1

Subtract within 1000 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Subtract.

$$732 - 485$$

Difference: _____

True or false?

$$600 - 278 = 322$$

A. True

B. False

$$514 - 296 = ?$$

A. 222

B. 228

C. 218

D. 318

Subtract.

$$905 - 467$$

Difference: _____

True or false?

$$840 - 390 = 550$$

A. True

B. False

SET 21 • 4.NBT.B.4

BACKWARD 2

Subtract by breaking apart a ten or a hundred • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

True or false?

To find $52 - 8$, you can rename 1 ten as 10 ones.

- A. True
- B. False

Rename 4 hundreds 2 tens as 3 hundreds and how many tens?

Tens: _____

Which shows 300 renamed so you can subtract 126?

- A.** 2 hundreds 10 ones
- B.** 3 hundreds 10 tens
- C.** 30 tens 1 one
- D.** 2 hundreds 9 tens 10 ones

Subtract.

$$400 - 158$$

Difference: _____

True or false?

6 tens 3 ones = 5 tens 3 ones

A. True

B. False

SET 21 • 4.NBT.B.4

FORWARD 1

Subtract decimals to hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

Subtract.

$$42.5 - 18.76$$

Difference: _____

Subtract.

$$9 - 3.45$$

Difference: _____

Subtract.

$$6.03 - 2.8$$

A. 3.22

B. 3.75

C. 4.23

D. 3.23

True or false?

$$10 - 0.25 = 9.75$$

A. True

B. False

True or false?

$$5.4 - 2.15 = 3.35$$

A. True

B. False

SET 21 • 4.NBT.B.4

FORWARD 2

Subtract multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$$100 - 36.875$$

Difference: _____

Subtract.

$$15.06 - 8.3$$

Difference: _____

Subtract.

$$4.2 - 1.375$$

A. 2.825

B. 2.925

C. 3.825

D. 3.175

True or false?

$$50.5 - 0.505 = 49.995$$

A. True

B. False

True or false?

$$8 - 0.08 = 0.72$$

A. True

B. False

Multiply a whole number of up to four digits by a one-digit number

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Multiplication facts within 100	3.OA.C.7
BACKWARD 2	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
FORWARD 1	Multiply multi-digit whole numbers using the standard algorithm	5.NBT.B.5
FORWARD 2	Multiply multi-digit decimals	6.NS.B.3

SET 22 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$4,236 \times 7$$

Product: _____

Multiply.

$$608 \times 5$$

Product: _____

Multiply.

$$3,049 \times 6$$

- A.** 3,055
- B.** 18,254
- C.** 18,924
- D.** 18,294

A theater sells 1,275 tickets each night for 8 nights.

How many tickets does it sell?

Tickets: _____

True or false?

$$2,513 \times 4 = 10,042$$

A. True

B. False

SET 22 • 4.NBT.B.5

BACKWARD 1

Multiplication facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

True or false?

$$9 \times 8 = 72$$

A. True

B. False

$$4 \times 7 = ?$$

A. 24

B. 32

C. 11

D. 28

Multiply.

$$6 \times 9$$

Product: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

SET 22 • 4.NBT.B.5

BACKWARD 2

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$7 \times 80$$

Product: _____

True or false?

$$5 \times 60 = 300$$

A. True

B. False

$$3 \times 90 = ?$$

A. 27

B. 2,700

C. 93

D. 270

Multiply.

$$4 \times 70$$

Product: _____

True or false?

$$8 \times 40 = 3,200$$

A. True

B. False

SET 22 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$3,214 \times 26$$

Product: _____

Multiply.

$$507 \times 48$$

Product: _____

Multiply.

$$1,250 \times 34$$

A. 42,500

B. 8,750

C. 42,050

D. 41,500

Multiply.

$$289 \times 75$$

Product: _____

True or false?

$$406 \times 52 = 21,112$$

A. True

B. False

SET 22 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$4.236 \times 7$$

Product: _____

Multiply.

$$6.08 \times 0.5$$

Product: _____

Multiply.

$$30.49 \times 0.6$$

- A.** 18.94
- B.** 18.294
- C.** 182.94
- D.** 1.8294

True or false?

$$2.5 \times 1.6 = 4$$

A. True

B. False

True or false?

$$0.7 \times 0.08 = 0.56$$

A. True

B. False

Multiply two two-digit numbers

MAIN

Grade 4 main question

4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10

3.NBT.A.3

BACKWARD 2

Break apart a factor to multiply

3.OA.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm

5.NBT.B.5

FORWARD 2

Multiply multi-digit decimals

6.NS.B.3

SET 23 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 27$$

Product: _____

Multiply.

$$58 \times 63$$

Product: _____

Multiply.

$$46 \times 19$$

A. 414

B. 884

C. 864

D. 874

A box holds 24 cans.

How many cans are in 36 boxes?

Cans: _____

True or false?

$$72 \times 15 = 1,008$$

A. True

B. False

SET 23 • 4.NBT.B.5

BACKWARD 1

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$3 \times 40$$

Product: _____

True or false?

$$6 \times 90 = 540$$

A. True

B. False

$$8 \times 20 = ?$$

- A.** 28
- B.** 1,600
- C.** 160
- D.** 16

Multiply.

$$9 \times 70$$

Product: _____

True or false?

$$4 \times 50 = 2,000$$

A. True

B. False

SET 23 • 4.NBT.B.5

BACKWARD 2

Break apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

True or false?

$$8 \times 7 = 8 \times 5 + 8 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 9 = 6 \times 5 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 4×8 ?

A. $4 \times 5 \times 3$

B. $4 \times 5 + 4 \times 3$

C. $4 \times 5 + 3$

D. $4 + 5 \times 3$

Find 7×6 by adding $7 \times 3 + 7 \times 3$.

Product: _____

True or false?

$$9 \times 6 = 9 \times 3 + 3$$

A. True

B. False

SET 23 • 4.NBT.B.5

FORWARD 1

Multiply multi-digit whole numbers using the standard algorithm • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$427 \times 53$$

Product: _____

Multiply.

$$86 \times 145$$

Product: _____

Multiply.

$$312 \times 64$$

A. 18,968

B. 19,968

C. 19,868

D. 1,248

Multiply.

$$2,005 \times 18$$

Product: _____

True or false?

$$95 \times 95 = 9,025$$

A. True

B. False

SET 23 • 4.NBT.B.5

FORWARD 2

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.4 \times 2.7$$

Product: _____

Multiply.

$$0.58 \times 6.3$$

Product: _____

Multiply.

$$4.6 \times 0.19$$

- A.** 4.79
- B.** 0.874
- C.** 0.0874
- D.** 8.74

True or false?

$$2.4 \times 3.6 = 8.64$$

A. True

B. False

True or false?

$$7.2 \times 1.5 = 1.08$$

A. True

B. False

Illustrate and explain multiplication with equations, arrays, and area models

MAIN	Grade 4 main question	4.NBT.B.5
BACKWARD 1	Use area to show the distributive property	3.MD.C.7.c
BACKWARD 2	Break apart an array to multiply	3.OA.B.5
FORWARD 1	Use place value and area models to multiply decimals	5.NBT.B.7
FORWARD 2	Use the distributive property to write a sum as a product	6.NS.B.4

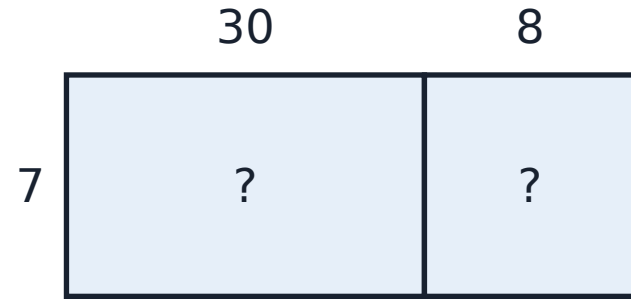
SET 24 • 4.NBT.B.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Complete the area model for 7×38 by writing the area of each part.

Then write the product.



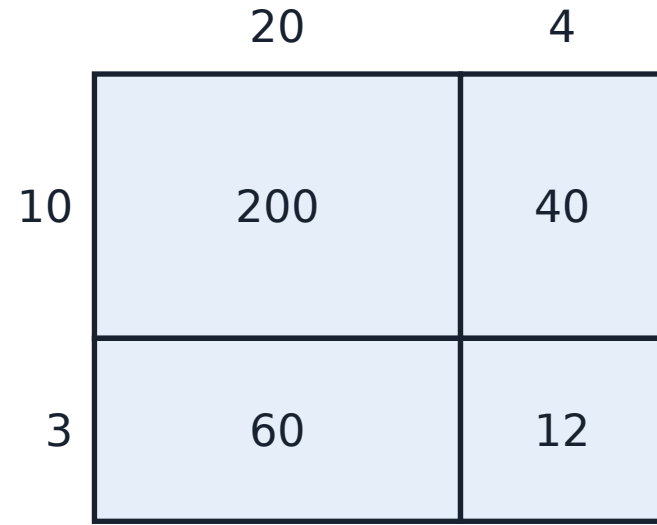
Product: _____

The area model shows
 24×13 .

Write the four partial
products and the total.

Partial products: _____

Total: _____



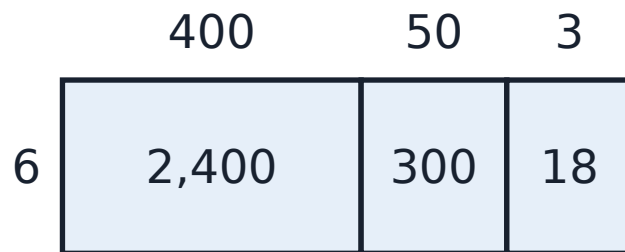
Which equation matches the area model?

A. $6 \times 453 = 2,400 + 50 + 3$

B. $6 \times 453 = 2,400 + 300 + 18$

C. $6 \times 453 = 6 \times 400 + 53$

D. $6 + 453 = 2,718$



Use an area model or equations to multiply. Show your model or equations.

$$9 \times 2,146$$

Product: _____

Explain why $15 \times 12 = 15 \times 10 + 15 \times 2$.

Explanation: _____

SET 24 • 4.NBT.B.5

BACKWARD 1

Use area to show the distributive property • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

A rectangle that is 4 by $(5 + 3)$ has an area of $4 \times 5 + 4 \times 3$.

A. True

B. False

A 6-by-9 rectangle is split into a 6-by-5 part and a 6-by-4 part.

What is the total area?

Area: _____

A 3-by-8 rectangle is split into a 3-by-5 part and a 3-by-3 part.

Which expression gives its area?

A. $3 \times 8 + 3$

B. $3 \times 5 \times 3$

C. $3 + 5 + 3$

D. $3 \times 5 + 3 \times 3$

Fill in the blank.

$$7 \times 9 = 7 \times 5 + 7 \times \underline{\quad}$$

Blank: _____

True or false?

$$5 \times 7 = 5 \times 4 + 3$$

A. True

B. False

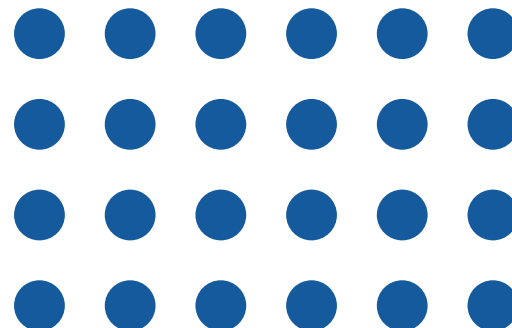
SET 24 • 4.NBT.B.5

BACKWARD 2

Break apart an array to multiply • Grade 3 • Operations & Algebraic Thinking •
3.OA.B.5

Write a multiplication equation for the array.

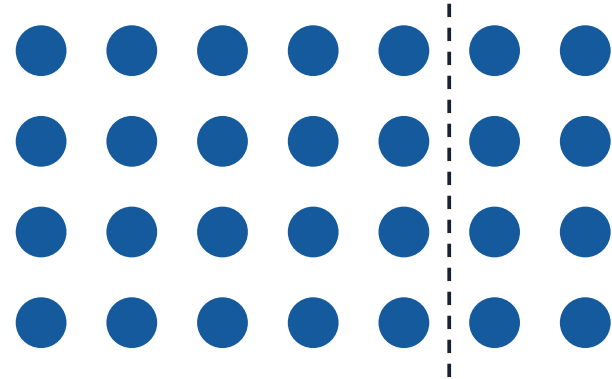
Equation: _____



The dashed line splits the array into two parts.

Which equation matches?

- A.** $4 \times 7 = 4 + 5 + 2$
- B.** $4 \times 7 = 4 \times 5 \times 2$
- C.** $4 \times 7 = 4 \times 5 + 4 \times 2$
- D.** $4 + 7 = 4 \times 5 + 2$



True or false?

An array of 3 rows of 9 can be split into 3 rows of 5 and 3 rows of 4.

A. True

B. False

An array of 6 rows of 8 is split into 6 rows of 5 and 6 rows of 3.

How many dots are in the whole array?

Dots: _____

True or false?

An array of 5 rows of 6, split into two arrays of 5 rows of 3, has 33 dots.

- A. True
- B. False

SET 24 • 4.NBT.B.5

FORWARD 1

Use place value and area models to multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

An area model splits 3×2.4 into 3×2 and 3×0.4 .

What is the product?

Product: _____

True or false?

$$4 \times 1.25 = 4 \times 1 + 4 \times 0.25 = 5$$

A. True

B. False

Which shows 6×3.5 broken apart by place value?

A. $6 \times 3 \times 0.5$

B. $6 \times 3 + 0.5$

C. $6 + 3 + 0.5$

D. $6 \times 3 + 6 \times 0.5$

Multiply.

$$5 \times 0.36$$

Product: _____

True or false?

$$2 \times 4.3 = 8.3$$

A. True

B. False

SET 24 • 4.NBT.B.5

FORWARD 2

Use the distributive property to write a sum as a product • Grade 6 • The Number System • 6.NS.B.4

Use the factor 4 to write $36 + 8$ as a product.

Expression: _____

True or false?

$$7(10 + 3) = 70 + 21$$

A. True

B. False

Which expression equals 5×47 ?

A. $5(40 + 7)$

B. $5 \times 40 + 7$

C. $5 + 40 \times 7$

D. $(5 + 40) \times 7$

Use the greatest common factor to write $24 + 60$ as a product.

Expression: _____

True or false?

$$3(12 + 5) = 36 + 5$$

A. True

B. False

Divide up to four-digit dividends by one-digit divisors, with remainders

MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division facts within 100	3.OA.C.7
BACKWARD 2	Share equally	3.OA.A.2
FORWARD 1	Divide with two-digit divisors	5.NBT.B.6
FORWARD 2	Divide multi-digit numbers with the standard algorithm	6.NS.B.2

SET 25 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$952 \div 7$$

Quotient: _____

Divide.

$$4,815 \div 5$$

Quotient: _____

Divide.

$$2,184 \div 6$$

- A.** 364
- B.** 354
- C.** 2,178
- D.** 374

Divide. Write the quotient and the remainder.

$$1,357 \div 4$$

Quotient: _____

Remainder: _____

True or false?

$$3,600 \div 9 = 40$$

A. True

B. False

SET 25 • 4.NBT.B.6

BACKWARD 1

Division facts within 100 • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

True or false?

$$72 \div 8 = 9$$

A. True

B. False

$$54 \div 6 = ?$$

A. 48

B. 7

C. 8

D. 9

Divide.

$$27 \div 3$$

Quotient: _____

True or false?

$$63 \div 9 = 8$$

A. True

B. False

SET 25 • 4.NBT.B.6

BACKWARD 2

Share equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

28 marbles are shared equally by 4 children.

How many marbles does each child get?

Marbles: _____

True or false?

When 45 is split into 5 equal groups, each group has 9.

A. True

B. False

Which expression shows 32 shared equally among 8 people?

A. $32 - 8$

B. 32×8

C. $32 \div 8$

D. $8 \div 32$

48 stickers are put into groups of 6.

How many groups are there?

Groups: _____

True or false?

20 shared equally by 4 people gives each person 16.

A. True

B. False

SET 25 • 4.NBT.B.6

FORWARD 1

Divide with two-digit divisors • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,736 \div 28$$

Quotient: _____

Divide.

$$5,904 \div 72$$

Quotient: _____

Divide.

$$3,360 \div 48$$

A. 70

B. 7

C. 68

D. 700

Divide. Write the quotient and the remainder.

$$2,000 \div 37$$

Quotient: _____

Remainder: _____

True or false?

$$1,125 \div 25 = 54$$

A. True

B. False

SET 25 • 4.NBT.B.6

FORWARD 2

Divide multi-digit numbers with the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$9,516 \div 39$$

Quotient: _____

Divide.

$$27,648 \div 72$$

Quotient: _____

Divide.

$$15,525 \div 45$$

- A.** 355
- B.** 345
- C.** 3,450
- D.** 335

Divide.

$$8,184 \div 62$$

Quotient: _____

True or false?

$$20,808 \div 51 = 408$$

A. True

B. False

Illustrate and explain division with equations, area models, and place value

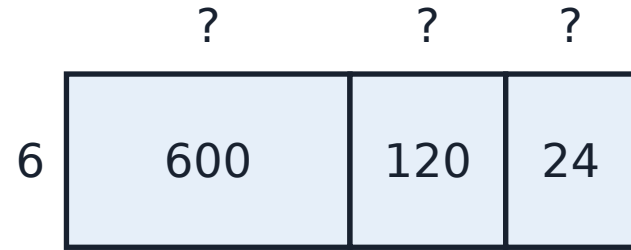
MAIN	Grade 4 main question	4.NBT.B.6
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Find a missing side from the area of a rectangle	3.MD.C.7.b
FORWARD 1	Explain division with two-digit divisors using area models and equ...	5.NBT.B.6
FORWARD 2	Connect the standard division algorithm to partial quotients	6.NS.B.2

SET 26 • 4.NBT.B.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Complete the area model for $744 \div 6$ by writing the missing length of each part. Then write the quotient.



Quotient: _____

Explain how these equations show $852 \div 4$.

$$4 \times 200 = 800$$

$$4 \times 13 = 52$$

$$800 + 52 = 852$$

Quotient: _____

Explanation: _____

Ana finds $3,515 \div 5$ by thinking $3,500 \div 5 = 700$ and $15 \div 5 = 3$.

What is the quotient?

A. 7,003

B. 73

C. 703

D. 730

Use an area model or equations to divide. Show your model or equations.

$$1,792 \div 8$$

Quotient: _____

True or false?

To find $636 \div 3$, you can divide each place:
 $600 \div 3 = 200$, $30 \div 3 = 10$, and $6 \div 3 = 2$. The quotient is 212.

- A. True
- B. False

SET 26 • 4.NBT.B.6

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $56 \div 8$ by thinking $8 \times ? = 56$.

Quotient: _____

True or false?

$36 \div 9 = 4$ because $9 \times 4 = 36$.

A. True

B. False

Which fact helps you find $45 \div 5$?

A. $5 \times 9 = 45$

B. $45 - 5 = 40$

C. $9 + 5 = 14$

D. $5 \times 5 = 25$

Find the unknown factor.

$$7 \times ? = 63$$

Factor: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 26 • 4.NBT.B.6

BACKWARD 2

Find a missing side from the area of a rectangle • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle has an area of 48 square units. One side is 6 units long.

How long is the other side?

Length: _____

True or false?

A rectangle that is 7 units by 5 units has an area of 35 square units.

- A.** True
- B.** False

A rectangle has an area of 36 square feet and a width of 4 feet.

What is its length?

A. 32 feet

B. 40 feet

C. 9 feet

D. 8 feet

What is the area of a rectangle that is 8 cm by 6 cm?

Area: _____

True or false?

A rectangle with an area of 30 square units and one side of 5 units has another side of 25 units.

A. True

B. False

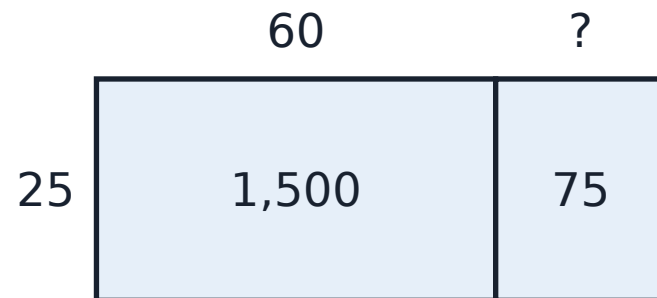
SET 26 • 4.NBT.B.6

FORWARD 1

Explain division with two-digit divisors using area models and equations •
Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

The area model shows
 $1,575 \div 25$.

What is the quotient?



Quotient: _____

True or false?

An area model for $2,016 \div 48$ can use parts with areas 1,920 and 96, so the quotient is $40 + 2 = 42$.

- A.** True
- B.** False

Which equations show $1,410 \div 30$?

A. $1,410 - 30 = 1,380$

B. $30 \times 40 = 1,200$; $30 \times 7 = 210$; $40 + 7 = 47$

C. $30 \times 47 = 1,140$

D. $30 \times 4 = 120$; $30 \times 7 = 210$; $4 + 7 = 11$

Divide.

$$2,262 \div 39$$

Quotient: _____

True or false?

An area model for $1,344 \div 32$ with parts 32×40 and 32×3 shows a quotient of 43.

A. True

B. False

SET 26 • 4.NBT.B.6

FORWARD 2

Connect the standard division algorithm to partial quotients • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$6,912 \div 24$$

Quotient: _____

In the standard algorithm for $4,284 \div 12$, the first step uses $12 \times 300 = 3,600$.

What does the 300 stand for?

- A.** Part of the quotient
- B.** The remainder
- C.** Part of the divisor
- D.** The dividend

True or false?

The steps of the standard algorithm for $3,705 \div 15$ match the partial quotients $200 + 40 + 7$.

- A.** True
- B.** False

Divide.

$$10,488 \div 76$$

Quotient: _____

True or false?

$$5,616 \div 52 = 118$$

A. True

B. False

Explain why fractions are equivalent using visual fraction models

MAIN	Grade 4 main question	4.NF.A.1
BACKWARD 1	Recognize equivalent fractions with a visual model	3.NF.A.3.b
BACKWARD 2	Equivalent fractions are at the same point on a number line	3.NF.A.3.a
FORWARD 1	Use equivalent fractions to add fractions with unlike denominators	5.NF.A.1
FORWARD 2	Find equivalent ratios	6.RP.A.3.a

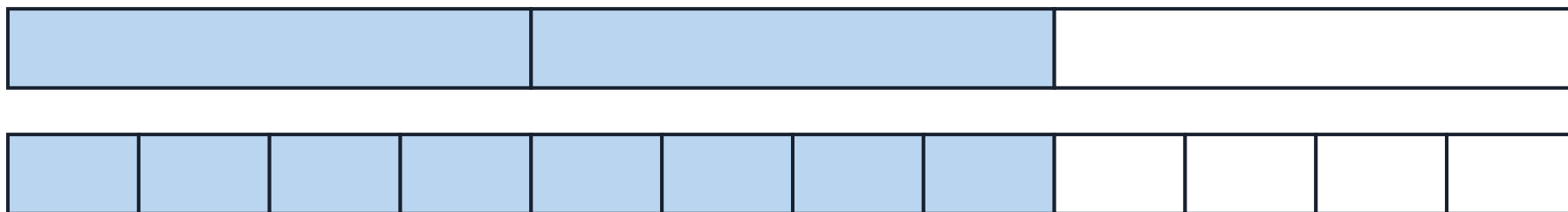
SET 27 • 4.NF.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

The bars show $\frac{2}{3}$ and $\frac{8}{12}$ of the same whole.

Explain how the model shows that $\frac{2}{3} = \frac{8}{12}$.



Explanation: _____

A bar shows $\frac{3}{5}$ shaded. Each fifth is cut into 2 equal parts.

Which fraction now names the shaded part?

A. $\frac{6}{10}$

B. $\frac{3}{10}$

C. $\frac{6}{5}$

D. $\frac{5}{10}$

True or false?

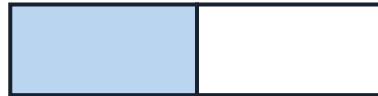
$\frac{3}{4} = \frac{9}{12}$ because the numerator and the denominator were both multiplied by 3.

A. True

B. False

The bar shows $\frac{1}{2}$ shaded.

Draw lines to split each part into 3 equal parts. Then write the equivalent fraction.



Equivalent fraction: _____

True or false?

$\frac{2}{5} = \frac{4}{5}$ because the numerator was multiplied by 2.

A. True

B. False

SET 27 • 4.NF.A.1

BACKWARD 1

Recognize equivalent fractions with a visual model • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

True or false?

$$\frac{1}{2} = \frac{2}{4}$$

A. True

B. False

The bars show the same whole.

What fraction of the second bar names the same amount as $\frac{1}{3}$?



Fraction: _____

Which fraction is equivalent to $\frac{3}{4}$?

A. $\frac{3}{8}$

B. $\frac{4}{6}$

C. $\frac{6}{8}$

D. $\frac{4}{3}$

Find the missing number.

$$\frac{1}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{2}{3} = \frac{2}{6}$$

A. True

B. False

SET 27 • 4.NF.A.1

BACKWARD 2

Equivalent fractions are at the same point on a number line • Grade 3 •
Number & Operations–Fractions • 3.NF.A.3.a

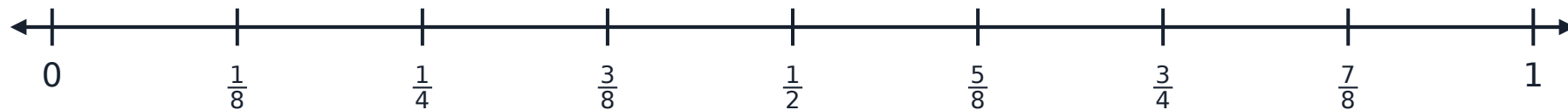
True or false?

$\frac{1}{2}$ and $\frac{2}{4}$ are at the same point on a number line.

A. True

B. False

Which fraction is at the same point on the number line as $\frac{3}{4}$?



A. $\frac{7}{8}$

B. $\frac{3}{8}$

C. $\frac{4}{8}$

D. $\frac{6}{8}$

Name a fraction in eighths that is at the same point as $\frac{1}{2}$.

Fraction: _____

True or false?

$\frac{1}{3}$ and $\frac{2}{6}$ are at different points on a number line.

A. True

B. False

What fraction in sixths is at the same point as $\frac{2}{3}$?

Fraction: _____

SET 27 • 4.NF.A.1

FORWARD 1

Use equivalent fractions to add fractions with unlike denominators • Grade 5 •
Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{3} + \frac{1}{4}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{10} = \frac{7}{10}$$

A. True

B. False

Add.

$$\frac{1}{2} + \frac{3}{8}$$

A. $\frac{7}{8}$

B. $\frac{4}{8}$

C. $\frac{3}{16}$

D. $\frac{4}{10}$

Add.

$$\frac{5}{6} + \frac{1}{12}$$

Sum: _____

True or false?

$$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$$

A. True

B. False

SET 27 • 4.NF.A.1

FORWARD 2

Find equivalent ratios • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Find the missing value.

$$2 : 3 = 8 : ?$$

Missing value: _____

True or false?

3 : 4 and 9 : 12 are equivalent ratios.

A. True

B. False

Which ratio is equivalent to 5 : 2?

- A.** 5 : 4
- B.** 7 : 4
- C.** 15 : 6
- D.** 10 : 2

The table shows equivalent ratios.

Find the missing value.

Cups of flour	2	4	6
Cups of milk	3	6	?

Missing value: _____

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

Generate equivalent fractions

MAIN	Grade 4 main question	4.NF.A.1
BACKWARD 1	Generate simple equivalent fractions	3.NF.A.3.b
BACKWARD 2	Whole numbers written as fractions	3.NF.A.3.c
FORWARD 1	Explain equivalent fractions as multiplying by a fraction equal to 1	5.NF.B.5.b
FORWARD 2	Write fractions as percents	6.RP.A.3.c

SET 28 • 4.NF.A.1

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{5} = \frac{?}{10}$$

Missing number: _____

Find the missing number.

$$\frac{8}{12} = \frac{2}{?}$$

Missing number: _____

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{6}$

B. $\frac{6}{8}$

C. $\frac{8}{12}$

D. $\frac{4}{12}$

True or false?

$$\frac{30}{100} = \frac{3}{100}$$

A. True

B. False

Write two fractions that are equivalent to $\frac{1}{4}$.

Fractions: _____

SET 28 • 4.NF.A.1

BACKWARD 1

Generate simple equivalent fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

Find the missing number.

$$\frac{1}{3} = \frac{?}{6}$$

Missing number: _____

True or false?

$$\frac{3}{6} = \frac{1}{2}$$

A. True

B. False

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. $\frac{2}{8}$

Find the missing number.

$$\frac{2}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

A. True

B. False

SET 28 • 4.NF.A.1

BACKWARD 2

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 3 as a fraction with a denominator of 1.

Fraction: _____

True or false?

$$\frac{4}{4} = 1$$

A. True

B. False

Which fraction is equal to 2?

A. $\frac{2}{2}$

B. $\frac{3}{6}$

C. $\frac{6}{3}$

D. $\frac{2}{3}$

Write 1 as a fraction with a denominator of 8.

Fraction: _____

True or false?

$$\frac{6}{6} = 6$$

A. True

B. False

SET 28 • 4.NF.A.1

FORWARD 1

Explain equivalent fractions as multiplying by a fraction equal to 1 • Grade 5 •
Number & Operations–Fractions • 5.NF.B.5.b

True or false?

$$\frac{3}{4} \times \frac{2}{2} = \frac{6}{8}, \text{ and } \frac{6}{8} = \frac{3}{4} \text{ because } \frac{2}{2} = 1.$$

A. True

B. False

To rewrite $\frac{2}{5}$ as $\frac{6}{15}$, by which whole number must you multiply BOTH its numerator and its denominator?

- A.** 3
- B.** 15
- C.** 6
- D.** 5

By which whole number must you multiply BOTH the numerator and the denominator of $\frac{5}{8}$ to write $\frac{20}{32}$?

Whole number: _____

True or false?

$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$ shows that $\frac{1}{2} = \frac{3}{8}$.

A. True

B. False

Explain why multiplying a fraction by $\frac{5}{5}$ does not change its value.

Explanation: _____

SET 28 • 4.NF.A.1

FORWARD 2

Write fractions as percents • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

Write $\frac{3}{4}$ as a percent.

Percent: _____

True or false?

$$\frac{2}{5} = 40\%$$

A. True

B. False

Which percent is equal to $\frac{7}{10}$?

- A.** 0.7%
- B.** 17%
- C.** 70%
- D.** 7%

Write 35% as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{1}{8} = 18\%$$

A. True

B. False

Compare two fractions by finding a common denominator or a common numerator

MAIN	Grade 4 main question	4.NF.A.2
BACKWARD 1	Compare fractions with the same numerator or the same denomin...	3.NF.A.3.d
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Compare the size of a product to the size of one factor	5.NF.B.5.a
FORWARD 2	Compare and order rational numbers	6.NS.C.7.a

SET 29 • 4.NF.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Write $>$, $=$, or $<$.

$$\frac{3}{4} \text{ — } \frac{5}{8}$$

Symbol: _____

Write $>$, $=$, or $<$.

$$\frac{2}{5} \text{ — } \frac{3}{10}$$

Symbol: _____

Which comparison is true?

A. $\frac{3}{8} > \frac{3}{5}$

B. $\frac{3}{5} > \frac{3}{8}$

C. $\frac{3}{5} < \frac{3}{8}$

D. $\frac{3}{5} = \frac{3}{8}$

Compare $\frac{5}{6}$ and $\frac{7}{12}$. Write $>$, $=$, or $<$, and explain how you know.

Comparison: _____

Explanation: _____

True or false?

$$\frac{4}{10} < \frac{35}{100}$$

A. True

B. False

SET 29 • 4.NF.A.2

BACKWARD 1

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{3}{8} \text{ — } \frac{5}{8}$$

Symbol: _____

True or false?

$$\frac{1}{3} > \frac{1}{6}$$

A. True

B. False

Which fraction is greatest?

A. $\frac{2}{6}$

B. $\frac{2}{3}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

Write $>$, $=$, or $<$.

$$\frac{4}{6} \text{ — } \frac{4}{8}$$

Symbol: _____

True or false?

$$\frac{3}{4} < \frac{2}{4}$$

A. True

B. False

SET 29 • 4.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{5}{6}$ is closer to 1 than $\frac{1}{6}$ is.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



A. 4

B. 7

C. 8

D. 9

What fraction is at point B?



B = _____

True or false?

$\frac{2}{2}$ is at the same point as 2 on a number line.

A. True

B. False

SET 29 • 4.NF.A.2

FORWARD 1

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

True or false?

$\frac{3}{4} \times 9$ is less than 9.

A. True

B. False

Which product is greater than 12?

A. $\frac{5}{4} \times 12$

B. $\frac{3}{4} \times 12$

C. $\frac{1}{2} \times 12$

D. 1×12

Without multiplying, is $\frac{7}{8} \times 20$ greater than, less than, or equal to 20?

Answer: _____

True or false?

$\frac{6}{6} \times 15$ is greater than 15.

A. True

B. False

Without multiplying, which is greater: $\frac{2}{3} \times 30$ or $\frac{4}{3} \times 30$?

Answer: _____

SET 29 • 4.NF.A.2

FORWARD 2

Compare and order rational numbers • Grade 6 • The Number System •
6.NS.C.7.a

Write $>$, $=$, or $<$.

$$-\frac{1}{2} \text{ — } -\frac{3}{4}$$

Symbol: _____

True or false?

$$-\frac{2}{3} < \frac{1}{3}$$

A. True

B. False

Which number is least?

A. $-\frac{5}{6}$

B. 0

C. $\frac{1}{4}$

D. $-\frac{1}{2}$

Order from least to greatest.

$$\frac{1}{2}, -\frac{3}{4}, -\frac{1}{4}$$

Order: _____

True or false?

$$-\frac{1}{4} < -\frac{1}{2}$$

A. True

B. False

Compare fractions using the benchmark $\frac{1}{2}$, and only when they refer to the same whole

MAIN	Grade 4 main question	4.NF.A.2
BACKWARD 1	Compare fractions of the same whole	3.NF.A.3.d
BACKWARD 2	Equal parts of a shape	3.G.A.2
FORWARD 1	Use benchmark fractions to estimate and check sums and differe...	5.NF.A.2
FORWARD 2	Write and interpret statements of order for rational numbers in c...	6.NS.C.7.b

SET 30 • 4.NF.A.2

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{2}{5}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{4}{10}$

Compare $\frac{3}{8}$ and $\frac{4}{6}$ by comparing each one to $\frac{1}{2}$. Write $>$, $=$, or $<$.

Comparison: _____

Reason: _____

True or false?

$$\frac{5}{12} < \frac{1}{2}$$

A. True

B. False

True or false?

Ben ate $\frac{1}{2}$ of a small pizza and Kim ate $\frac{1}{2}$ of a large pizza. They ate the same amount of pizza.

- A. True
- B. False

Explain why $\frac{2}{3}$ of a small cake and $\frac{2}{3}$ of a large cake are not the same amount of cake.

Explanation: _____

SET 30 • 4.NF.A.2

BACKWARD 1

Compare fractions of the same whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

True or false?

$\frac{1}{4}$ of a pizza is less than $\frac{1}{2}$ of the same pizza.

A. True

B. False

Write $>$, $=$, or $<$.

$$\frac{2}{6} \text{ — } \frac{5}{6}$$

Symbol: _____

Which fraction of the same bar is greatest?

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

Write $>$, $=$, or $<$.

$$\frac{3}{4} \text{ — } \frac{3}{6}$$

Symbol: _____

True or false?

$\frac{1}{8}$ of a ribbon is more than $\frac{1}{3}$ of the same ribbon.

A. True

B. False

SET 30 • 4.NF.A.2

BACKWARD 2

Equal parts of a shape • Grade 3 • Geometry • 3.G.A.2

A square is split into 4 equal parts.
What fraction of its area is each part?

Fraction: _____

True or false?

A rectangle cut into 6 equal parts has parts that are each $\frac{1}{6}$ of its area.

- A.** True
- B.** False

A circle is cut into 8 equal parts.

What fraction of the area is 1 part?

A. $\frac{8}{1}$

B. $\frac{7}{8}$

C. $\frac{1}{4}$

D. $\frac{1}{8}$

What fraction of the shape is shaded?



Fraction: _____

True or false?

A shape cut into 2 parts of different sizes is cut into halves.

A. True

B. False

SET 30 • 4.NF.A.2

FORWARD 1

Use benchmark fractions to estimate and check sums and differences • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

$\frac{3}{5} + \frac{4}{7}$ is greater than 1, because each fraction is greater than $\frac{1}{2}$.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{1}{12}$?

- A.** About 1
- B.** About 0
- C.** About 2
- D.** About $\frac{1}{2}$

Use benchmark fractions to decide whether $\frac{2}{5} + \frac{3}{8}$ is more or less than 1.

Answer: _____

Reason: _____

True or false?

$\frac{5}{6} - \frac{1}{4}$ is about 1, because both fractions are close to 1.

A. True

B. False

Use the benchmarks 0, $\frac{1}{2}$, and 1 to estimate $\frac{9}{10} + \frac{5}{12}$.

Estimate: _____

SET 30 • 4.NF.A.2

FORWARD 2

Write and interpret statements of order for rational numbers in context • Grade 6 • The Number System • 6.NS.C.7.b

Write an inequality that compares $-\frac{3}{4}^{\circ}\text{C}$ and $-\frac{1}{2}^{\circ}\text{C}$.

Inequality: _____

True or false?

A temperature of $-\frac{1}{4}^{\circ}\text{F}$ is warmer than $-\frac{3}{4}^{\circ}\text{F}$.

A. True

B. False

Which statement is true?

A. $-\frac{2}{3} < -\frac{1}{3}$

B. $-\frac{2}{3} > -\frac{1}{3}$

C. $-\frac{1}{3} < -\frac{2}{3}$

D. $\frac{1}{3} < -\frac{2}{3}$

A diver at $-2\frac{1}{2}$ m is deeper than a diver at $-1\frac{1}{2}$ m.

Write an inequality for the two positions.

Inequality: _____

True or false?

$$-\frac{1}{8} < -\frac{7}{8}$$

A. True

B. False

Understand adding and subtracting fractions as joining and separating parts of the same whole

MAIN	Grade 4 main question	4.NF.B.3.a
BACKWARD 1	A fraction is a number of unit-fraction parts	3.NF.A.1
BACKWARD 2	Add to, take from, put together, and take apart within 20	1.OA.A.1
FORWARD 1	Add fractions with unlike denominators	5.NF.A.1
FORWARD 2	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7

SET 31 • 4.NF.B.3.a

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.a

A pizza is cut into 8 equal slices. Jo eats 3 slices and Al eats 2 slices.

What fraction of the pizza did they eat in all?

Fraction: _____

True or false?

$\frac{2}{5} + \frac{1}{5}$ means joining 2 fifths and 1 fifth of the same whole, which makes $\frac{3}{5}$.

A. True

B. False

A ribbon is cut into 10 equal pieces. 3 pieces are used and 7 are left.

Which equation shows this?

A. $\frac{10}{10} - \frac{3}{10} = \frac{7}{20}$

B. $\frac{3}{10} - \frac{10}{10} = \frac{7}{10}$

C. $\frac{10}{10} - \frac{3}{10} = \frac{7}{0}$

D. $\frac{10}{10} - \frac{3}{10} = \frac{7}{10}$

True or false?

$\frac{3}{4} + \frac{1}{4} = \frac{4}{8}$, because you add the numerators and add the denominators.

A. True

B. False

Shade the bar to show $\frac{2}{6} + \frac{3}{6}$.

Then write the sum.

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Sum: _____

SET 31 • 4.NF.B.3.a

BACKWARD 1

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

What fraction is 2 parts of size $\frac{1}{3}$?

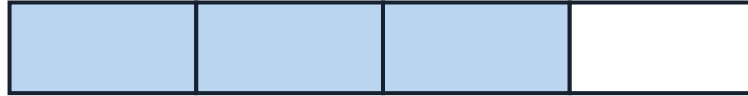
A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{3}{2}$

D. $\frac{2}{6}$

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{4}{6}$ means 6 parts of size $\frac{1}{4}$.

A. True

B. False

SET 31 • 4.NF.B.3.a

BACKWARD 2

Add to, take from, put together, and take apart within 20 • Grade 1 •
Operations & Algebraic Thinking • 1.OA.A.1

Sam has 7 marbles. He gets 6 more.

How many marbles does he have?

Marbles: _____

True or false?

There are 15 birds. 8 fly away. 7 birds are left.

A. True

B. False

9 red apples and 5 green apples are put together.

How many apples are there?

A. 14

B. 13

C. 4

D. 45

A box has 12 crayons. 4 are taken out.

How many crayons are left?

Crayons: _____

True or false?

6 cats and 7 dogs make 12 pets.

A. True

B. False

SET 31 • 4.NF.B.3.a

FORWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{1}{4} + \frac{2}{3}$$

Sum: _____

Add.

$$\frac{3}{10} + \frac{1}{2}$$

Sum: _____

Add.

$$\frac{3}{8} + \frac{1}{2}$$

A. $\frac{4}{10}$

B. $\frac{4}{8}$

C. $\frac{7}{8}$

D. $1\frac{1}{8}$

True or false?

$$\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

A. True

B. False

True or false?

$$\frac{1}{5} + \frac{1}{2} = \frac{2}{7}$$

A. True

B. False

SET 31 • 4.NF.B.3.a

FORWARD 2

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{4} = \frac{3}{4}$$

x = _____

True or false?

The solution of $y - \frac{2}{5} = \frac{1}{5}$ is $y = \frac{3}{5}$.

A. True

B. False

Solve for n.

$$n + \frac{3}{8} = 1$$

A. $1\frac{3}{8}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{8}{5}$

Solve for m.

$$\frac{5}{6} = m + \frac{1}{6}$$

m = _____

True or false?

The solution of $x + \frac{1}{3} = \frac{2}{3}$ is $x = 1$.

A. True

B. False

Decompose a fraction into a sum of fractions with the same denominator, and justify it with a model

MAIN	Grade 4 main question	4.NF.B.3.b
BACKWARD 1	Unit fractions build a fraction	3.NF.A.1
BACKWARD 2	Break apart numbers to 10 into pairs in more than one way	K.OA.A.3
FORWARD 1	Add mixed numbers with unlike denominators	5.NF.A.1
FORWARD 2	Divide fractions by thinking "how many parts of this size?"	6.NS.A.1

SET 32 • 4.NF.B.3.b

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $\frac{4}{5}$ as a sum of unit fractions.

Sum: _____

Write $\frac{7}{10}$ as a sum of two fractions with a denominator of 10, in two different ways.

Way 1: _____

Way 2: _____

Which sum is equal to $\frac{5}{6}$?

A. $\frac{1}{6} + \frac{5}{6}$

B. $\frac{2}{6} + \frac{3}{6}$

C. $\frac{2}{3} + \frac{3}{3}$

D. $\frac{4}{6} + \frac{2}{6}$

True or false?

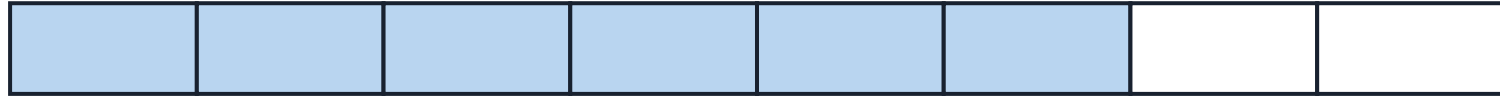
$$1\frac{2}{3} = \frac{3}{3} + \frac{2}{3}$$

A. True

B. False

The bar shows $\frac{6}{8}$ shaded.

Circle a group of 2 eighths and a group of 4 eighths to show $\frac{6}{8} = \frac{2}{8} + \frac{4}{8}$. Then write another way to decompose $\frac{6}{8}$.



Another way: _____

SET 32 • 4.NF.B.3.b

BACKWARD 1

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{8}$ s make $\frac{7}{8}$?

Answer: _____

True or false?

$$\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

A. True

B. False

What fraction is 5 parts of size $\frac{1}{6}$?

A. $\frac{5}{1}$

B. $\frac{5}{6}$

C. $\frac{1}{6}$

D. $\frac{6}{6}$

Write $\frac{2}{3}$ as a sum of unit fractions.

Sum: _____

True or false?

$$\frac{1}{2} + \frac{1}{2} = \frac{2}{4}$$

A. True

B. False

SET 32 • 4.NF.B.3.b

BACKWARD 2

Break apart numbers to 10 into pairs in more than one way • Grade K •
Operations & Algebraic Thinking • K.OA.A.3

Write 7 as the sum of two numbers in two different ways.

Way 1: _____

Way 2: _____

True or false?

$$6 = 2 + 4$$

A. True

B. False

Which pair of numbers makes 9?

A. 5 and 3

B. 5 and 4

C. 7 and 3

D. 6 and 4

Fill in the blank.

$$8 = 5 + \underline{\quad}$$

Blank: _____

True or false?

$$10 = 6 + 3$$

A. True

B. False

SET 32 • 4.NF.B.3.b

FORWARD 1

Add mixed numbers with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$1\frac{1}{2} + 2\frac{1}{3}$$

Sum: _____

Add.

$$3\frac{3}{4} + 1\frac{1}{6}$$

Sum: _____

Add.

$$2\frac{2}{5} + 1\frac{1}{2}$$

A. $3\frac{3}{7}$

B. $4\frac{1}{10}$

C. $3\frac{9}{10}$

D. $3\frac{3}{10}$

True or false?

$$2\frac{1}{2} + 1\frac{3}{4} = 4\frac{1}{4}$$

A. True

B. False

True or false?

$$1\frac{2}{3} + 1\frac{1}{6} = 2\frac{3}{9}$$

A. True

B. False

SET 32 • 4.NF.B.3.b

FORWARD 2

Divide fractions by thinking "how many parts of this size?" • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{8}$ s are in $\frac{3}{4}$?

Answer: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = 4$$

A. True

B. False

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

A. 10

B. $\frac{25}{72}$

C. $\frac{1}{2}$

D. 2

Divide.

$$\frac{9}{10} \div \frac{3}{10}$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

Add mixed numbers with like denominators

MAIN

Grade 4 main question

4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions

3.NF.A.3.c

BACKWARD 2

Fractions greater than 1 on a number line

3.NF.A.2

FORWARD 1

Add mixed numbers with unlike denominators

5.NF.A.1

FORWARD 2

Solve equations of the form $x + p = q$ with mixed numbers

6.EE.B.7

SET 33 • 4.NF.B.3.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.c

Add.

$$2\frac{3}{8} + 1\frac{4}{8}$$

Sum: _____

Add.

$$3\frac{4}{5} + 2\frac{3}{5}$$

Sum: _____

Add.

$$1\frac{5}{6} + 4\frac{2}{6}$$

A. $6\frac{1}{6}$

B. $5\frac{7}{12}$

C. $6\frac{7}{12}$

D. $5\frac{1}{6}$

True or false?

$$2\frac{1}{4} + 1\frac{3}{4} = 3\frac{4}{8}$$

A. True

B. False

A plant grew $1\frac{7}{10}$ inches in May and $2\frac{6}{10}$ inches in June.

How much did it grow in all?

Growth: _____

SET 33 • 4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 2 as a fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{3}{3} = 1$$

A. True

B. False

Which fraction is equal to 1?

A. $\frac{0}{6}$

B. $\frac{6}{6}$

C. $\frac{6}{1}$

D. $\frac{1}{6}$

How many halves are in 3 wholes?

Halves: _____

True or false?

$$\frac{8}{4} = 4$$

A. True

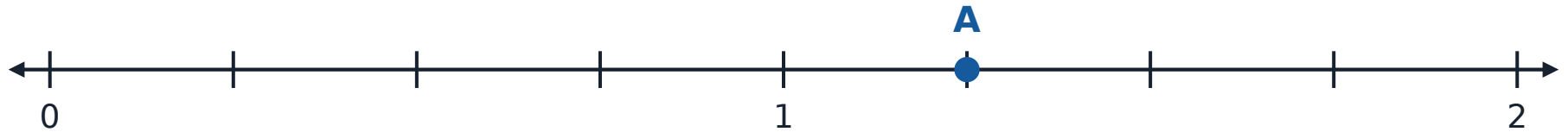
B. False

SET 33 • 4.NF.B.3.c

BACKWARD 2

Fractions greater than 1 on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A? Write it as a fraction or a mixed number.



A = _____

True or false?

$\frac{3}{2}$ is between 1 and 2 on a number line.

A. True

B. False

Which fraction is at the same point as 1 on a number line?

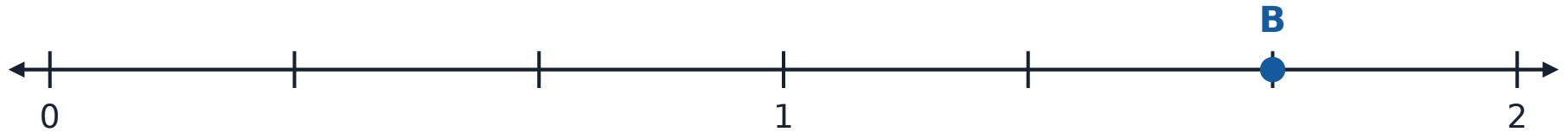
A. $\frac{1}{4}$

B. $\frac{4}{1}$

C. $\frac{3}{4}$

D. $\frac{4}{4}$

What number is at point B? Write it as a fraction or a mixed number.



B = _____

True or false?

$\frac{7}{8}$ is greater than 1.

A. True

B. False

SET 33 • 4.NF.B.3.c

FORWARD 1

Add mixed numbers with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$3\frac{1}{4} + 1\frac{2}{3}$$

Sum: _____

Add.

$$2\frac{5}{6} + 1\frac{1}{2}$$

Sum: _____

Add.

$$1\frac{3}{10} + 2\frac{1}{2}$$

A. $3\frac{4}{5}$

B. $4\frac{4}{5}$

C. $3\frac{2}{5}$

D. $3\frac{4}{12}$

True or false?

$$4\frac{3}{8} + 2\frac{1}{4} = 6\frac{5}{8}$$

A. True

B. False

True or false?

$$1\frac{1}{2} + 1\frac{1}{3} = 2\frac{2}{5}$$

A. True

B. False

SET 33 • 4.NF.B.3.c

FORWARD 2

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 1\frac{1}{2} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $y - 2\frac{1}{3} = 1\frac{2}{3}$ is $y = 4$.

A. True

B. False

Solve for n.

$$n + 3\frac{3}{8} = 6$$

A. $2\frac{3}{8}$

B. $2\frac{5}{8}$

C. $3\frac{5}{8}$

D. $9\frac{3}{8}$

A board was $6\frac{1}{2}$ feet long. After a piece was cut off, $4\frac{3}{4}$ feet were left.

Write and solve an equation for the length p of the piece.

Equation: _____

$p =$ _____

True or false?

The solution of $x + 2\frac{1}{2} = 5$ is $x = 3\frac{1}{2}$.

A. True

B. False

Subtract mixed numbers with like denominators

MAIN

Grade 4 main question

4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions

3.NF.A.3.c

BACKWARD 2

Subtract within 100

2.NBT.B.5

FORWARD 1

Subtract mixed numbers with unlike denominators

5.NF.A.1

FORWARD 2

Solve equations with subtraction and mixed numbers

6.EE.B.7

SET 34 • 4.NF.B.3.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.c

Subtract.

$$5\frac{7}{8} - 2\frac{3}{8}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{3}{5}$$

Difference: _____

Subtract.

$$6 - 2\frac{5}{6}$$

A. $3\frac{5}{6}$

B. $4\frac{5}{6}$

C. $4\frac{1}{6}$

D. $3\frac{1}{6}$

True or false?

$$3\frac{2}{10} - 1\frac{7}{10} = 1\frac{5}{10}$$

A. True

B. False

A jug had $3\frac{1}{4}$ liters of water. Lee poured out $1\frac{3}{4}$ liters.

How much water is left?

Water left: _____

SET 34 • 4.NF.B.3.c

BACKWARD 1

Whole numbers written as fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

Write 1 as a fraction with a denominator of 6.

Fraction: _____

True or false?

$$\frac{4}{4} = 1$$

A. True

B. False

Which fraction is equal to 3?

A. $\frac{4}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{4}$

D. $\frac{12}{4}$

How many thirds are in 2 wholes?

Thirds: _____

True or false?

$$\frac{2}{2} = 2$$

A. True

B. False

SET 34 • 4.NF.B.3.c

BACKWARD 2

Subtract within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$62 - 37$$

Difference: _____

True or false?

$$80 - 26 = 54$$

A. True

B. False

$$71 - 48 = ?$$

A. 119

B. 23

C. 33

D. 27

Subtract.

$$95 - 59$$

Difference: _____

True or false?

$$43 - 18 = 35$$

A. True

B. False

SET 34 • 4.NF.B.3.c

FORWARD 1

Subtract mixed numbers with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$5\frac{1}{2} - 2\frac{1}{3}$$

Difference: _____

Subtract.

$$4 - 1\frac{3}{8}$$

Difference: _____

Subtract.

$$3\frac{1}{4} - 1\frac{2}{3}$$

A. $2\frac{1}{12}$

B. $1\frac{7}{12}$

C. $1\frac{5}{12}$

D. $2\frac{7}{12}$

True or false?

$$6\frac{1}{2} - 3\frac{3}{4} = 2\frac{3}{4}$$

A. True

B. False

True or false?

$$2\frac{5}{6} - 1\frac{1}{2} = 1\frac{4}{4}$$

A. True

B. False

SET 34 • 4.NF.B.3.c

FORWARD 2

Solve equations with subtraction and mixed numbers • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x - 2\frac{1}{4} = 3\frac{1}{2}$$

x = _____

True or false?

The solution of $y + 1\frac{5}{6} = 4$ is $y = 2\frac{1}{6}$.

A. True

B. False

Solve for n.

$$7\frac{1}{3} - n = 2\frac{2}{3}$$

A. $4\frac{2}{3}$

B. 10

C. $4\frac{1}{3}$

D. $5\frac{1}{3}$

A tank had some water. After $3\frac{1}{2}$ gallons were used, $5\frac{1}{4}$ gallons were left.

Write and solve an equation for the starting amount w .

Equation: _____

$w =$ _____

True or false?

The solution of $x - 1\frac{1}{2} = 1\frac{1}{2}$ is $x = 0$.

A. True

B. False

Solve word problems by adding fractions and mixed numbers with like denominators

MAIN	Grade 4 main question	4.NF.B.3.d
BACKWARD 1	Addition word problems within 100	2.OA.A.1
BACKWARD 2	Fractions of a whole	3.NF.A.1
FORWARD 1	Solve word problems by adding fractions with unlike denominators	5.NF.A.2
FORWARD 2	Solve word problems by dividing fractions	6.NS.A.1

SET 35 • 4.NF.B.3.d

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.d

Mia ran $\frac{3}{10}$ mile in the morning and $\frac{5}{10}$ mile in the afternoon.

How far did she run in all?

Distance: _____

A jug had $\frac{4}{12}$ liter of juice. Tom poured in $\frac{5}{12}$ liter more.

How much juice is in the jug now?

Juice: _____

Ty used $1\frac{2}{4}$ cups of rice on Monday and $2\frac{3}{4}$ cups on Tuesday.

How many cups did he use in all?

A. $3\frac{1}{4}$ cups

B. $3\frac{5}{8}$ cups

C. $4\frac{1}{4}$ cups

D. $4\frac{5}{8}$ cups

True or false?

Ann sews a $\frac{3}{8}$ -yard ribbon to a $\frac{4}{8}$ -yard ribbon, end to end. The new ribbon is $\frac{7}{8}$ yard long.

A. True

B. False

A painter used $\frac{2}{6}$ of a can of paint on a door and $\frac{3}{6}$ of the can on a fence.

What fraction of the can did the painter use in all?

Fraction: _____

SET 35 • 4.NF.B.3.d

BACKWARD 1

Addition word problems within 100 • Grade 2 • Operations & Algebraic Thinking
• 2.OA.A.1

A shelf has 46 books. 18 more books are put on it.
How many books are on the shelf now?

Books: _____

True or false?

Kim had 25 stickers and got 17 more. She has 42 stickers.

- A.** True
- B.** False

A class has 14 boys and 15 girls.

How many students are in the class?

A. 39

B. 19

C. 1

D. 29

Jo had 50 cents. She found 23 more cents.
How much money does she have now?

Money: _____

True or false?

A bus had 36 people. 9 more got on. Now 44 people are on the bus.

A. True

B. False

SET 35 • 4.NF.B.3.d

BACKWARD 2

Fractions of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A pie is cut into 6 equal pieces. 5 pieces are eaten.
What fraction of the pie is eaten?

Fraction: _____

True or false?

A pizza is cut into 8 equal slices and 3 are left. $\frac{3}{8}$ of the pizza is left.

- A. True
- B. False

A bar is cut into 4 equal parts.

What fraction of the bar is 1 part?

A. $\frac{3}{4}$

B. $\frac{1}{3}$

C. $\frac{1}{4}$

D. $\frac{4}{1}$

A garden has 3 equal sections. Beans fill 2 of them.
What fraction of the garden is beans?

Fraction: _____

True or false?

A board cut into 2 equal pieces has pieces that are each $\frac{1}{3}$ of the board.

A. True

B. False

SET 35 • 4.NF.B.3.d

FORWARD 1

Solve word problems by adding fractions with unlike denominators • Grade 5 •
Number & Operations–Fractions • 5.NF.A.2

Ana walked $\frac{1}{2}$ mile and then $\frac{1}{3}$ mile.

How far did she walk in all?

Distance: _____

Ben drank $\frac{2}{3}$ cup of milk and then $\frac{1}{4}$ cup more.

How much milk did he drink in all?

Milk: _____

Lu read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of it on Tuesday.

What fraction of the book did she read?

A. $\frac{4}{12}$

B. $\frac{5}{8}$

C. $\frac{4}{8}$

D. $\frac{3}{32}$

True or false?

A $\frac{1}{2}$ -meter board and a $\frac{1}{3}$ -meter board laid end to end are $\frac{5}{6}$ meter long.

A. True

B. False

True or false?

Sam ate $\frac{1}{2}$ of a pizza and Lee ate $\frac{1}{3}$ of it. They ate $\frac{2}{5}$ of the pizza.

A. True

B. False

SET 35 • 4.NF.B.3.d

FORWARD 2

Solve word problems by dividing fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{1}{4}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

A $\frac{9}{10}$ -mile path is split into $\frac{3}{10}$ -mile sections. There are 3 sections.

- A.** True
- B.** False

How many $\frac{2}{3}$ -foot pieces can be cut from 4 feet of ribbon?

A. 6

B. 8

C. $\frac{1}{6}$

D. $\frac{8}{3}$

$\frac{3}{4}$ pound of nuts is shared equally among 3 bags.

How much is in each bag?

Pounds: _____

True or false?

$\frac{1}{2}$ cup of oats fills a $\frac{1}{6}$ -cup scoop exactly 2 times.

A. True

B. False

Solve word problems by subtracting fractions and mixed numbers with like denominators

MAIN	Grade 4 main question	4.NF.B.3.d
BACKWARD 1	Subtraction word problems within 100	2.OA.A.1
BACKWARD 2	Whole numbers written as fractions	3.NF.A.3.c
FORWARD 1	Solve word problems by subtracting fractions with unlike denomi...	5.NF.A.2
FORWARD 2	Write and solve equations of the form $x + p = q$ for fraction word...	6.EE.B.7

SET 66 • 4.NF.B.3.d

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.d

A jug had $\frac{11}{12}$ liter of juice. Tom drank $\frac{4}{12}$ liter.

How much juice is left?

Juice left: _____

True or false?

A ribbon is $\frac{7}{8}$ yard long. Ann cuts off $\frac{3}{8}$ yard. $\frac{4}{8}$ yard is left.

- A.** True
- B.** False

A painter used $\frac{2}{6}$ of a can of paint on a door and $\frac{3}{6}$ of the can on a fence.

What fraction of the can is left?

Fraction: _____

A trail is $3\frac{1}{4}$ miles long. Rosa has hiked $1\frac{3}{4}$ miles.

How much farther does she have to hike?

A. $1\frac{1}{4}$ miles

B. $2\frac{2}{4}$ miles

C. $1\frac{2}{4}$ miles

D. 5 miles

A bag held $2\frac{1}{8}$ pounds of flour. Dee used $\frac{5}{8}$ pound.

How much flour is left?

Flour left: _____

SET 66 • 4.NF.B.3.d

BACKWARD 1

Subtraction word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A shelf has 46 books. 18 are taken.

How many books are left?

Books: _____

Jo had 50 cents. She spent 23 cents.

How much money is left?

Money left: _____

True or false?

A bus had 36 people. 9 got off. 25 people are still on the bus.

A. True

B. False

A tree had 72 apples. 35 apples fell.

How many apples are still on the tree?

A. 37

B. 107

C. 43

D. 47

True or false?

Lee had 64 cards and gave away 27. He has 37 cards left.

A. True

B. False

SET 66 • 4.NF.B.3.d

BACKWARD 2

Whole numbers written as fractions • Grade 3 • Number &
Operations–Fractions • 3.NF.A.3.c

Write 1 as a fraction with a denominator of 8.

Fraction: _____

True or false?

$$\frac{6}{6} = 1$$

A. True

B. False

Which fraction is equal to 2?

A. $\frac{2}{4}$

B. $\frac{8}{4}$

C. $\frac{4}{4}$

D. $\frac{4}{8}$

How many fourths are in 1 whole?

Fourths: _____

True or false?

$$\frac{3}{3} = 3$$

A. True

B. False

SET 66 • 4.NF.B.3.d

FORWARD 1

Solve word problems by subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A jug had $\frac{3}{4}$ gallon of milk. Ben used $\frac{2}{3}$ gallon.

How much milk is left?

Milk left: _____

True or false?

A board is $\frac{5}{6}$ meter long. $\frac{1}{2}$ meter is cut off. $\frac{1}{3}$ meter is left.

- A. True
- B. False

A pitcher holds $\frac{7}{8}$ quart of lemonade. Kai pours out $\frac{1}{2}$ quart.

How much lemonade is left?

A. $\frac{6}{8}$ quart

B. $\frac{5}{8}$ quart

C. $\frac{1}{8}$ quart

D. $\frac{3}{8}$ quart

Ari has $\frac{5}{6}$ hour to practice. He has practiced for $\frac{1}{4}$ hour.

How much longer does he have to practice?

Time: _____

True or false?

A rope is $\frac{9}{10}$ meter long. $\frac{2}{5}$ meter is cut off. $\frac{7}{10}$ meter is left.

A. True

B. False

SET 66 • 4.NF.B.3.d

FORWARD 2

Write and solve equations of the form $x + p = q$ for fraction word problems •
Grade 6 • Expressions & Equations • 6.EE.B.7

A jug had $\frac{11}{12}$ liter of juice. After Tom drank some, $\frac{4}{12}$ liter was left.

Write and solve an equation for the amount j that Tom drank.

Equation: _____

$j =$ _____

True or false?

The solution of $x + \frac{2}{5} = \frac{9}{10}$ is $x = \frac{1}{2}$.

A. True

B. False

A trail is $4\frac{1}{2}$ miles long. After hiking h miles, Rosa has $1\frac{3}{4}$ miles left.

Which value of h solves $h + 1\frac{3}{4} = 4\frac{1}{2}$?

A. $3\frac{1}{4}$

B. $2\frac{1}{4}$

C. $2\frac{3}{4}$

D. $6\frac{1}{4}$

Solve for y.

$$y + \frac{3}{8} = 2$$

$$y = \underline{\hspace{10cm}}$$

True or false?

The solution of $n + \frac{1}{6} = \frac{2}{3}$ is $n = \frac{1}{3}$.

A. True

B. False

Understand a fraction a/b as a multiple of $1/b$

MAIN	Grade 4 main question	4.NF.B.4.a
BACKWARD 1	A fraction is a number of unit-fraction parts	3.NF.A.1
BACKWARD 2	Products as equal groups	3.OA.A.1
FORWARD 1	Find a fraction of a fractional amount	5.NF.B.4.a
FORWARD 2	Divide a fraction by a unit fraction	6.NS.A.1

SET 36 • 4.NF.B.4.a

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.a

Write $\frac{7}{8}$ as a whole number times $\frac{1}{8}$.

Expression: _____

Which equation shows $\frac{3}{5}$ as a multiple of $\frac{1}{5}$?

A. $\frac{3}{5} = \frac{1}{5} \div 3$

B. $\frac{3}{5} = 3 + \frac{1}{5}$

C. $\frac{3}{5} = 5 \times \frac{1}{3}$

D. $\frac{3}{5} = 3 \times \frac{1}{5}$

True or false?

$$\frac{9}{4} = 9 \times \frac{1}{4}$$

A. True

B. False

Shade the tape diagram to show $4 \times \frac{1}{6}$.

Then write the product as a fraction.

--	--	--	--	--	--

Fraction: _____

True or false?

$$\frac{6}{10} = 10 \times \frac{1}{6}$$

A. True

B. False

SET 36 • 4.NF.B.4.a

BACKWARD 1

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

True or false?

$\frac{5}{8}$ is 5 parts of size $\frac{1}{8}$.

A. True

B. False

What fraction is 4 parts of size $\frac{1}{6}$?

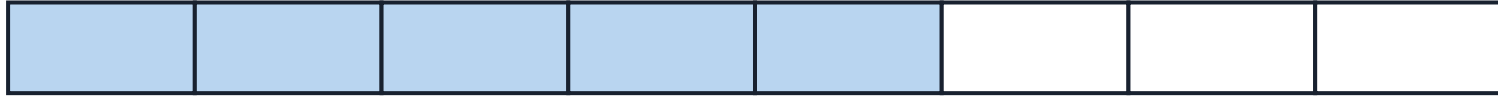
A. $\frac{4}{8}$

B. $\frac{1}{6}$

C. $\frac{4}{6}$

D. $\frac{6}{4}$

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{2}{3}$ is 3 parts of size $\frac{1}{2}$.

A. True

B. False

SET 36 • 4.NF.B.4.a

BACKWARD 2

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

What is the total in 5 groups of 3?

Total: _____

True or false?

4×6 can mean 4 groups of 6.

A. True

B. False

Which expression shows 7 groups of 2?

A. 7×2

B. $2 \div 7$

C. $7 - 2$

D. $7 + 2$

Write a multiplication expression for 3 groups of 9.

Expression: _____

True or false?

6×3 means $6 + 3$.

A. True

B. False

SET 36 • 4.NF.B.4.a

FORWARD 1

Find a fraction of a fractional amount • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{3}{4}$$

Product: _____

True or false?

$\frac{1}{4}$ of $\frac{4}{5}$ is $\frac{1}{5}$.

A. True

B. False

What is $\frac{2}{5}$ of $\frac{5}{6}$?

A. $\frac{12}{25}$

B. $\frac{25}{12}$

C. $\frac{7}{11}$

D. $\frac{1}{3}$

What is $\frac{1}{3}$ of $\frac{3}{4}$ yard? Think of $\frac{3}{4}$ as 3 parts of size $\frac{1}{4}$.

Answer: _____

True or false?

$$\frac{1}{2} \times \frac{2}{3} = \frac{3}{5}$$

A. True

B. False

SET 36 • 4.NF.B.4.a

FORWARD 2

Divide a fraction by a unit fraction • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{5}$ s are in $\frac{4}{5}$?

Answer: _____

True or false?

$$\frac{7}{8} \div \frac{1}{8} = 7$$

A. True

B. False

Divide.

$$\frac{3}{4} \div \frac{1}{8}$$

A. $\frac{3}{32}$

B. 6

C. 3

D. $\frac{1}{6}$

Divide.

$$2 \div \frac{1}{3}$$

Quotient: _____

True or false?

$$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$$

A. True

B. False

Multiply a fraction by a whole number:

$$n \times (a/b) = (n \times a)/b$$

MAIN

Grade 4 main question

4.NF.B.4.b

BACKWARD 1

Multiply three numbers in any grouping

3.OA.B.5

BACKWARD 2

A fraction is a number of unit-fraction parts

3.NF.A.1

FORWARD 1

Multiply a fraction by a fraction

5.NF.B.4.a

FORWARD 2

Divide a fraction by a fraction

6.NS.A.1

SET 37 • 4.NF.B.4.b

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$4 \times \frac{2}{3}$$

Product: _____

Multiply.

$$5 \times \frac{3}{8}$$

Product: _____

Which is equal to $3 \times \frac{4}{5}$?

A. $12 \times \frac{1}{5}$

B. $7 \times \frac{1}{5}$

C. $3 \times \frac{1}{5}$

D. $\frac{12}{15}$

True or false?

$$6 \times \frac{2}{12} = \frac{12}{72}$$

A. True

B. False

Explain why $2 \times \frac{3}{10} = 6 \times \frac{1}{10}$.

Explanation: _____

SET 37 • 4.NF.B.4.b

BACKWARD 1

Multiply three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 5 = 2 \times (3 \times 5)$$

A. True

B. False

Fill in the blank.

$$4 \times (2 \times 3) = (4 \times 2) \times \underline{\quad}$$

Blank: _____

Which is equal to $3 \times 2 \times 4$?

A. 6×4

B. 5×4

C. 9×4

D. $3 + 8$

Multiply.

$$2 \times 4 \times 5$$

Product: _____

True or false?

$$(3 \times 2) \times 4 = 3 + 2 + 4$$

A. True

B. False

SET 37 • 4.NF.B.4.b

BACKWARD 2

A fraction is a number of unit-fraction parts • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

True or false?

$\frac{6}{8}$ is 6 parts of size $\frac{1}{8}$.

A. True

B. False

What fraction is 3 parts of size $\frac{1}{4}$?

A. $\frac{1}{4}$

B. $\frac{3}{4}$

C. $\frac{3}{8}$

D. $\frac{4}{3}$

What fraction is 5 parts of size $\frac{1}{6}$?

Fraction: _____

True or false?

$\frac{3}{6}$ is 6 parts of size $\frac{1}{3}$.

A. True

B. False

SET 37 • 4.NF.B.4.b

FORWARD 1

Multiply a fraction by a fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{3}{5}$$

Product: _____

True or false?

$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

A. True

B. False

Multiply.

$$\frac{3}{4} \times \frac{2}{5}$$

A. $\frac{3}{10}$

B. $\frac{5}{9}$

C. $\frac{5}{20}$

D. $\frac{6}{9}$

What is $\frac{1}{3}$ of $\frac{6}{7}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{1}{2} = \frac{3}{7}$$

A. True

B. False

SET 37 • 4.NF.B.4.b

FORWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{1}{6}$$

Quotient: _____

True or false?

$$\frac{3}{4} \div \frac{3}{8} = 2$$

A. True

B. False

Divide.

$$\frac{4}{5} \div \frac{2}{5}$$

A. 2

B. $\frac{1}{2}$

C. 4

D. $\frac{8}{25}$

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{8} = \frac{1}{16}$$

A. True

B. False

Solve word problems by multiplying a fraction by a whole number

MAIN	Grade 4 main question	4.NF.B.4.c
BACKWARD 1	Equal-groups word problems within 100	3.OA.A.3
BACKWARD 2	Jump by a unit fraction on a number line	3.NF.A.2
FORWARD 1	Solve real-world problems by multiplying fractions and mixed num...	5.NF.B.6
FORWARD 2	Find a percent of a quantity	6.RP.A.3.c

SET 38 • 4.NF.B.4.c

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.c

Each glass holds $\frac{3}{4}$ cup of milk.

How much milk is in 6 glasses?

Milk: _____

A snail crawls $\frac{2}{5}$ meter each hour.

How far does it crawl in 4 hours?

Distance: _____

Each of 3 friends eats $\frac{5}{8}$ of a sandwich.

How many sandwiches do they eat in all?

A. $1\frac{7}{8}$

B. $\frac{8}{8}$

C. $\frac{15}{24}$

D. $3\frac{5}{8}$

True or false?

A runner runs $\frac{7}{10}$ mile each day for 5 days. She runs $3\frac{5}{10}$ miles in all.

A. True

B. False

A recipe uses $\frac{2}{3}$ cup of flour for each batch.

How much flour do 5 batches use, and between which two whole numbers is that amount?

Flour: _____

Between: _____

SET 38 • 4.NF.B.4.c

BACKWARD 1

Equal-groups word problems within 100 • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Each bag holds 6 oranges.

How many oranges are in 7 bags?

Oranges: _____

True or false?

9 boxes with 4 pens in each box hold 36 pens.

A. True

B. False

A spider has 8 legs.

How many legs do 5 spiders have?

A. 35

B. 45

C. 13

D. 40

Each row has 9 seats.

How many seats are in 3 rows?

Seats: _____

True or false?

6 packs of 5 cards is 11 cards.

A. True

B. False

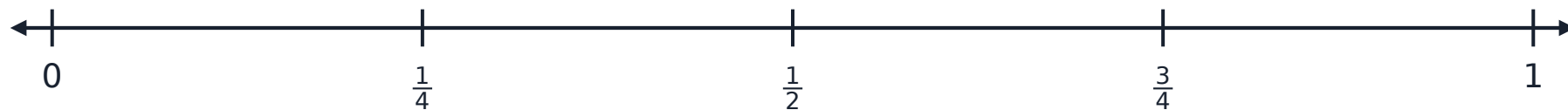
SET 38 • 4.NF.B.4.c

BACKWARD 2

Jump by a unit fraction on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Start at 0 and make 3 jumps of $\frac{1}{4}$ on the number line.

Where do you land?



Answer: _____

True or false?

5 jumps of $\frac{1}{8}$ from 0 land on $\frac{5}{8}$.

A. True

B. False

How many jumps of $\frac{1}{3}$ from 0 reach 1?

A. 3

B. 2

C. 6

D. 1

Where do 4 jumps of $\frac{1}{6}$ from 0 land?

Answer: _____

True or false?

2 jumps of $\frac{1}{2}$ from 0 land on $\frac{2}{4}$.

A. True

B. False

SET 38 • 4.NF.B.4.c

FORWARD 1

Solve real-world problems by multiplying fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is $7\frac{1}{2}$ m long. $\frac{3}{5}$ of its length is planted.

How many meters are planted?

Meters: _____

A bottle holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the bottle?

A. $\frac{1}{2}$ liter

B. $2\frac{1}{6}$ liters

C. 1 liter

D. $\frac{3}{4}$ liter

True or false?

A board is $4\frac{1}{2}$ feet long. Half of the board is $2\frac{1}{4}$ feet long.

- A.** True
- B.** False

True or false?

$\frac{2}{3}$ of a $4\frac{1}{2}$ -mile trail is 2 miles.

A. True

B. False

SET 38 • 4.NF.B.4.c

FORWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 36?

Answer: _____

True or false?

50% of 18 is 9.

A. True

B. False

What is 75% of 40?

- A.** 10
- B.** 35
- C.** 25
- D.** 30

A class has 20 students. 40% of them walk to school.
How many students walk?

Students: _____

True or false?

10% of 70 is 10.

A. True

B. False

Write a fraction with denominator 10 as an equivalent fraction with denominator 100

MAIN

Grade 4 main question

4.NF.C.5

BACKWARD 1

Simple equivalent fractions

3.NF.A.3.b

BACKWARD 2

A hundred is ten tens

2.NBT.A.1.a

FORWARD 1

Add decimals to hundredths

5.NBT.B.7

FORWARD 2

Write tenths and hundredths as percents

6.RP.A.3.c

SET 39 • 4.NF.C.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Write $\frac{6}{10}$ as an equivalent fraction with a denominator of 100.

Fraction: _____

Find the missing number.

$$\frac{4}{10} = \frac{?}{100}$$

Missing number: _____

Which fraction is equivalent to $\frac{9}{10}$?

A. $\frac{90}{100}$

B. $\frac{19}{100}$

C. $\frac{9}{100}$

D. $\frac{99}{100}$

True or false?

$$\frac{50}{100} = \frac{5}{100}$$

A. True

B. False

Write $\frac{70}{100}$ as an equivalent fraction with a denominator of 10.

Fraction: _____

SET 39 • 4.NF.C.5

BACKWARD 1

Simple equivalent fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{2}{4} = \frac{4}{8}$$

A. True

B. False

Which fraction is equal to $\frac{2}{3}$?

A. $\frac{3}{6}$

B. $\frac{2}{6}$

C. $\frac{3}{2}$

D. $\frac{4}{6}$

Find the missing number.

$$\frac{3}{4} = \frac{6}{?}$$

Missing number: _____

True or false?

$$\frac{1}{3} = \frac{3}{6}$$

A. True

B. False

SET 39 • 4.NF.C.5

BACKWARD 2

A hundred is ten tens • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1.a

How many tens are in 100?

Tens: _____

True or false?

10 tens make 1 hundred.

A. True

B. False

A hundred flat is cut into rods of ten. How many rods are there?

- A.** 1
- B.** 10
- C.** 20
- D.** 100

Mia has 9 bundles of ten straws.

How many more bundles of ten does she need to make 1 hundred?

Bundles: _____

True or false?

10 tens make 1,000.

A. True

B. False

SET 39 • 4.NF.C.5

FORWARD 1

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$0.6 + 0.25$$

Sum: _____

True or false?

$$0.3 + 0.07 = 0.37$$

A. True

B. False

Add.

$$0.4 + 0.58$$

- A.** 0.62
- B.** 9.8
- C.** 0.98
- D.** 0.098

Add.

$$1.7 + 0.15$$

Sum: _____

True or false?

$$0.5 + 0.05 = 0.10$$

A. True

B. False

SET 39 • 4.NF.C.5

FORWARD 2

Write tenths and hundredths as percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{6}{10}$ as a percent.

Percent: _____

True or false?

$$\frac{45}{100} = 45\%$$

A. True

B. False

Which percent is equal to $\frac{3}{10}$?

- A.** 3%
- B.** 13%
- C.** 30%
- D.** 0.3%

Write 8% as a fraction with a denominator of 100.

Fraction: _____

True or false?

$$\frac{9}{10} = 9\%$$

A. True

B. False

Add two fractions with denominators 10 and 100

MAIN	Grade 4 main question	4.NF.C.5
BACKWARD 1	Add tens and ones within 100	2.NBT.B.5
BACKWARD 2	Simple equivalent fractions	3.NF.A.3.b
FORWARD 1	Add decimals with tenths and hundredths in context	5.NBT.B.7
FORWARD 2	Add multi-digit decimals with the standard algorithm	6.NS.B.3

SET 67 • 4.NF.C.5

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

$$\frac{7}{10} + \frac{9}{100} = ?$$

A. $\frac{79}{100}$

B. $\frac{16}{110}$

C. $\frac{79}{10}$

D. $\frac{16}{100}$

Add.

$$\frac{8}{10} + \frac{15}{100}$$

Sum: _____

True or false?

$$\frac{3}{10} + \frac{4}{100} = \frac{34}{100}$$

A. True

B. False

A jar is $\frac{4}{10}$ full of sand. Ali pours in another $\frac{35}{100}$ of a jar.

What fraction of the jar is full now?

Fraction: _____

SET 67 • 4.NF.C.5

BACKWARD 1

Add tens and ones within 100 • Grade 2 • Number & Operations in Base Ten •
2.NBT.B.5

Add.

$$20 + 45$$

Sum: _____

True or false?

$$70 + 9 = 79$$

A. True

B. False

$$80 + 15 = ?$$

A. 23

B. 95

C. 815

D. 85

Add.

$$30 + 34$$

Sum: _____

True or false?

$$40 + 35 = 85$$

A. True

B. False

SET 67 • 4.NF.C.5

BACKWARD 2

Simple equivalent fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

Find the missing number.

$$\frac{1}{2} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{3}{6} = \frac{1}{2}$$

A. True

B. False

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{2}{4}$

B. $\frac{1}{8}$

C. $\frac{4}{1}$

D. $\frac{2}{8}$

Find the missing number.

$$\frac{2}{3} = \frac{?}{6}$$

Missing number: _____

True or false?

$$\frac{2}{8} = \frac{1}{2}$$

A. True

B. False

SET 67 • 4.NF.C.5

FORWARD 1

Add decimals with tenths and hundredths in context • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

A beetle crawls 0.4 meter and then 0.35 meter.

How far does it crawl in all?

Distance: _____

True or false?

$$0.8 + 0.12 = 0.92$$

A. True

B. False

A cup holds 0.25 liter of water. Ty pours in 0.5 liter more.

How much water is in the cup now?

A. 0.30 liter

B. 0.075 liter

C. 0.75 liter

D. 7.5 liters

Add.

$$2.6 + 0.08$$

Sum: _____

True or false?

$$0.9 + 0.09 = 0.18$$

A. True

B. False

SET 67 • 4.NF.C.5

FORWARD 2

Add multi-digit decimals with the standard algorithm • Grade 6 • The Number System • 6.NS.B.3

Add.

$$12.6 + 3.45$$

Sum: _____

True or false?

$$7.08 + 2.9 = 9.98$$

A. True

B. False

Add.

$$0.75 + 18.3$$

- A.** 0.933
- B.** 19.05
- C.** 19.8
- D.** 18.105

Add.

$$45.2 + 6.875$$

Sum: _____

True or false?

$$3.4 + 1.25 = 4.59$$

A. True

B. False

Write fractions with denominators 10 or 100 as decimals

MAIN	Grade 4 main question	4.NF.C.6
BACKWARD 1	Dimes, pennies, and dollars	2.MD.C.8
BACKWARD 2	Hundreds, tens, and ones	2.NBT.A.1
FORWARD 1	Write thousandths as decimals	5.NBT.A.3.a
FORWARD 2	Write decimals as percents	6.RP.A.3.c

SET 40 • 4.NF.C.6

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{47}{100}$ as a decimal.

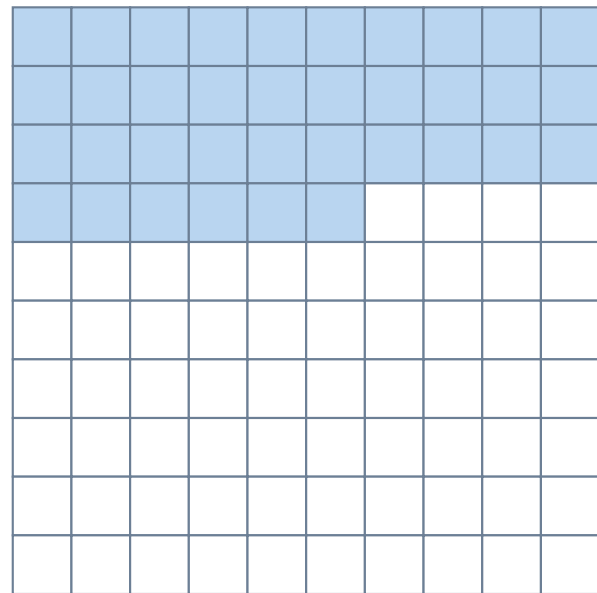
Decimal: _____

Write 0.8 as a fraction with a denominator of 10.

Fraction: _____

What decimal does the shaded part of the grid show?

- A.** 0.36
- B.** 3.6
- C.** 36
- D.** 0.036



True or false?

$$\frac{5}{100} = 0.5$$

A. True

B. False

A ribbon is $\frac{93}{100}$ meter long.

Write its length as a decimal.

Length: _____

SET 40 • 4.NF.C.6

BACKWARD 1

Dimes, pennies, and dollars • Grade 2 • Measurement & Data • 2.MD.C.8

How many cents are 3 dimes and 4 pennies?

Cents: _____

True or false?

10 dimes are worth 1 dollar.

A. True

B. False

Which coins make 25 cents?

- A.** 2 dimes and 5 pennies
- B.** 2 dimes and 2 pennies
- C.** 25 dimes
- D.** 1 dime and 5 pennies

How many pennies make 1 dollar?

Pennies: _____

True or false?

7 dimes are worth 7 cents.

A. True

B. False

SET 40 • 4.NF.C.6

BACKWARD 2

Hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.1

How many hundreds, tens, and ones are in 245?

Answer: _____

True or false?

In 307, the 0 means 0 tens.

A. True

B. False

What is the value of the 5 in 452?

- A.** 45
- B.** 500
- C.** 5
- D.** 50

Write the number that is 6 tens and 2 ones.

Number: _____

True or false?

In 180, the 8 means 8 ones.

A. True

B. False

SET 40 • 4.NF.C.6

FORWARD 1

Write thousandths as decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write $\frac{125}{1000}$ as a decimal.

Decimal: _____

True or false?

$$0.009 = \frac{9}{1000}$$

A. True

B. False

Which decimal is equal to $\frac{4}{1000}$?

- A.** 0.04
- B.** 0.004
- C.** 4.000
- D.** 0.4

Write 0.375 as a fraction with a denominator of 1,000.

Fraction: _____

True or false?

$$\frac{72}{1000} = 0.72$$

A. True

B. False

SET 40 • 4.NF.C.6

FORWARD 2

Write decimals as percents • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

Write 0.35 as a percent.

Percent: _____

True or false?

$$0.07 = 7\%$$

A. True

B. False

Which decimal is equal to 60%?

A. 0.6

B. 60

C. 0.06

D. 6.0

Write 120% as a decimal.

Decimal: _____

True or false?

$$0.5 = 5\%$$

A. True

B. False

Locate decimals to hundredths on a number line

MAIN	Grade 4 main question	4.NF.C.6
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Round decimals using place value	5.NBT.A.4
FORWARD 2	Locate positive and negative decimals on a number line	6.NS.C.6.c

SET 41 • 4.NF.C.6

MAIN

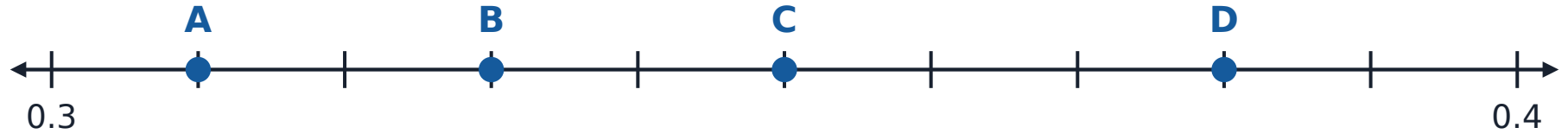
Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal is at point A?



A = _____

Which point is at 0.35?



A. Point A

B. Point B

C. Point C

D. Point D

Mark and label 0.4 and 0.85 on the number line.



True or false?

0.5 is halfway between 0 and 1 on a number line.

A. True

B. False

What decimal is at point B?



B = _____

SET 41 • 4.NF.C.6

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

$\frac{1}{2}$ is halfway between 0 and 1 on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



A. 2

B. 5

C. 3

D. 4

What fraction is at point B?



B = _____

True or false?

$\frac{3}{4}$ is to the left of $\frac{1}{4}$ on a number line.

A. True

B. False

SET 41 • 4.NF.C.6

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, 12 is to the right of 9.

A. True

B. False

Which number is halfway between 20 and 30 on a number line?

- A.** 24
- B.** 25
- C.** 50
- D.** 15

What number is at point Q?



Q = _____

True or false?

The distance from 3 to 8 on a number line is 11.

A. True

B. False

SET 41 • 4.NF.C.6

FORWARD 1

Round decimals using place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 2.68 to the nearest tenth.

Rounded: _____

True or false?

On a number line, 4.17 is closer to 4.2 than to 4.1.

- A. True
- B. False

Which number is closest to 5?

A. 5.1

B. 4.8

C. 5.09

D. 4.96

Round 0.734 to the nearest hundredth.

Rounded: _____

True or false?

6.25 rounded to the nearest whole number is 7.

A. True

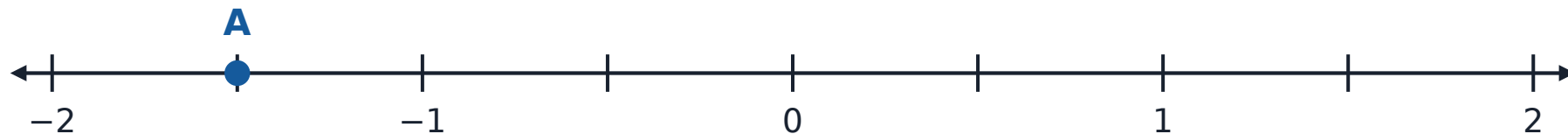
B. False

SET 41 • 4.NF.C.6

FORWARD 2

Locate positive and negative decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



A = _____

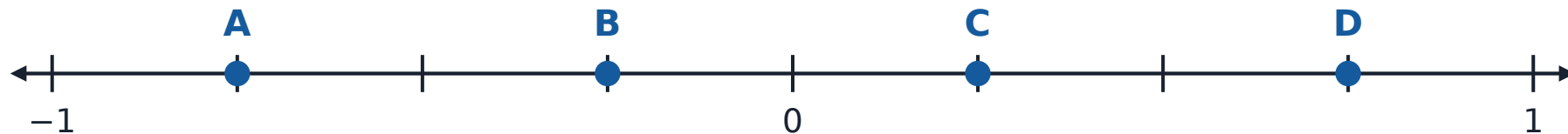
True or false?

-0.5 is halfway between -1 and 0 on a number line.

A. True

B. False

Which point is at -0.75 ?



A. Point A

B. Point B

C. Point C

D. Point D

Mark and label -1.2 and 0.8 on the number line.



True or false?

-1.25 is to the right of -1 on a number line.

A. True

B. False

Compare two decimals to hundredths and justify the comparison

MAIN	Grade 4 main question	4.NF.C.7
BACKWARD 1	Compare fractions with the same numerator or the same denomin...	3.NF.A.3.d
BACKWARD 2	Compare three-digit numbers	2.NBT.A.4
FORWARD 1	Compare decimals to thousandths	5.NBT.A.3.b
FORWARD 2	Write and interpret statements of order for decimals in context	6.NS.C.7.b

SET 42 • 4.NF.C.7

MAIN

Grade 4 main question • Grade 4 • Number & Operations–Fractions • 4.NF.C.7

Write $>$, $=$, or $<$.

0.6 0.58

Symbol: _____

Which comparison is true?

A. $0.1 < 0.09$

B. $0.09 < 0.1$

C. $0.09 > 0.1$

D. $0.09 = 0.1$

True or false?

$$0.40 = 0.4$$

A. True

B. False

Compare 0.3 and 0.25. Write $>$, $=$, or $<$, and explain using place value or a model.

Comparison: _____

Explanation: _____

True or false?

Jo has 0.5 of a large pizza and Sam has 0.5 of a small pizza. They have the same amount of pizza.

A. True

B. False

SET 42 • 4.NF.C.7

BACKWARD 1

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Write $>$, $=$, or $<$.

$$\frac{5}{8} \text{ — } \frac{3}{8}$$

Symbol: _____

True or false?

$$\frac{1}{6} < \frac{1}{2}$$

A. True

B. False

Which fraction is least?

A. $\frac{2}{6}$

B. $\frac{2}{3}$

C. $\frac{2}{8}$

D. $\frac{2}{4}$

Write $>$, $=$, or $<$.

$$\frac{2}{4} \text{ — } \frac{2}{4}$$

Symbol: _____

True or false?

$$\frac{1}{8} > \frac{1}{4}$$

A. True

B. False

SET 42 • 4.NF.C.7

BACKWARD 2

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Write $>$, $=$, or $<$.

630 603

Symbol: _____

True or false?

$$458 < 485$$

A. True

B. False

Which number is least?

A. 901

B. 910

C. 291

D. 199

Write $>$, $=$, or $<$.

777 787

Symbol: _____

True or false?

$$350 > 530$$

A. True

B. False

SET 42 • 4.NF.C.7

FORWARD 1

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

Write $>$, $=$, or $<$.

0.307 0.37

Symbol: _____

True or false?

$$4.5 > 4.485$$

A. True

B. False

Which decimal is least?

- A.** 0.1
- B.** 0.19
- C.** 0.109
- D.** 0.099

Write $>$, $=$, or $<$.

2.500 2.5

Symbol: _____

True or false?

$$0.62 < 0.602$$

A. True

B. False

SET 42 • 4.NF.C.7

FORWARD 2

Write and interpret statements of order for decimals in context • Grade 6 • The Number System • 6.NS.C.7.b

Write an inequality that compares -1.5°C and -0.5°C .

Inequality: _____

True or false?

A depth of -3.2 m is deeper than a depth of -2.3 m.

A. True

B. False

Which statement is true?

A. $0.3 < -0.8$

B. $-0.3 < -0.8$

C. $-0.8 < -0.3$

D. $-0.8 > -0.3$

Order from least to greatest.

-0.4 , 0.2 , -1.1

Order: _____

True or false?

$$-2.75 > -2.5$$

A. True

B. False

Know the relative sizes of measurement units within one system

MAIN	Grade 4 main question	4.MD.A.1
BACKWARD 1	Grams, kilograms, and liters	3.MD.A.2
BACKWARD 2	Measure with two different units	2.MD.A.2
FORWARD 1	Convert measurement units within one system	5.MD.A.1
FORWARD 2	Use ratio reasoning to convert measurement units	6.RP.A.3.d

SET 43 • 4.MD.A.1

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a river?

- A.** Centimeters
- B.** Kilometers
- C.** Grams
- D.** Liters

True or false?

1 kilogram is 1,000 times as heavy as 1 gram.

A. True

B. False

Convert.

5 kilometers = ____ meters

Meters: _____

Which is heaviest?

- A.** 1 gram
- B.** 1 kilogram
- C.** 100 grams
- D.** 500 grams

True or false?

1 liter is less than 1 milliliter.

A. True

B. False

SET 43 • 4.MD.A.1

BACKWARD 1

Grams, kilograms, and liters • Grade 3 • Measurement & Data • 3.MD.A.2

Which object has a mass of about 1 kilogram?

- A.** A grape
- B.** A car
- C.** A large book
- D.** A paper clip

True or false?

A spoonful of water is about 1 liter.

A. True

B. False

Would you measure the mass of a pencil in grams or in kilograms?

Unit: _____

True or false?

A bucket can hold about 10 liters of water.

A. True

B. False

A bag of flour has a mass of 2 kg. A bag of sugar has a mass of 3 kg.

What is their total mass?

Mass: _____

SET 43 • 4.MD.A.1

BACKWARD 2

Measure with two different units • Grade 2 • Measurement & Data • 2.MD.A.2

True or false?

A desk measured in inches gives a larger number than the same desk measured in feet.

- A.** True
- B.** False

A table is 2 meters long.

About how many centimeters long is it?

- A.** 200
- B.** 20
- C.** 2,000
- D.** 2

A rope is 3 feet long.

Is it longer or shorter than 3 inches?

Answer: _____

True or false?

A book measured in centimeters gives a smaller number than the same book measured in meters.

A. True

B. False

Which is longer: 1 foot or 1 inch?

Answer: _____

SET 43 • 4.MD.A.1

FORWARD 1

Convert measurement units within one system • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

3.5 kilometers = ____ meters

Meters: _____

True or false?

2,500 grams = 2.5 kilograms

A. True

B. False

How many ounces are in $2\frac{1}{2}$ pounds?

- A.** 32
- B.** 20
- C.** 36
- D.** 40

Convert.

90 minutes = ____ hours

Hours: _____

True or false?

400 centimeters = 40 meters

A. True

B. False

SET 43 • 4.MD.A.1

FORWARD 2

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 54 inches?

Feet: _____

True or false?

If 1 mile is about 1.6 km, then 5 miles is about 8 km.

- A. True
- B. False

There are 4 quarts in 1 gallon.

How many gallons are in 26 quarts?

A. 104

B. $6\frac{1}{2}$

C. 30

D. 22

There are 2 cups in 1 pint.

How many pints are in 15 cups?

Pints: _____

True or false?

If 1 inch = 2.54 cm, then 10 inches = 2.54 cm.

- A. True
- B. False

Express a larger unit in terms of a smaller unit and record the pairs in a table

MAIN	Grade 4 main question	4.MD.A.1
BACKWARD 1	Multiply one-digit numbers by multiples of 10	3.NBT.A.3
BACKWARD 2	Patterns in a table	3.OA.D.9
FORWARD 1	Convert from a smaller unit to a larger unit	5.MD.A.1
FORWARD 2	Use tables of equivalent ratios to convert units	6.RP.A.3.d

SET 44 • 4.MD.A.1

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.1

Complete the table.

Meters	1	2	3	4
Centimeters	100	200	?	?

Missing values: _____

How many ounces are in 5 pounds?

Ounces: _____

A song is 3 minutes long.

How many seconds is that?

A. 300

B. 30

C. 180

D. 63

True or false?

4 liters = 400 milliliters

A. True

B. False

Complete the table.

Liters	1	2	3	4
Milliliters	1,000	?	3,000	?

Missing values: _____

SET 44 • 4.MD.A.1

BACKWARD 1

Multiply one-digit numbers by multiples of 10 • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

True or false?

$$2 \times 50 = 100$$

A. True

B. False

$$8 \times 60 = ?$$

- A.** 480
- B.** 48
- C.** 4,800
- D.** 68

Multiply.

$$3 \times 70$$

Product: _____

True or false?

$$6 \times 60 = 3,600$$

A. True

B. False

SET 44 • 4.MD.A.1

BACKWARD 2

Patterns in a table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

What comes next?

16, 32, 48, ____

Next: _____

True or false?

In 15, 30, 45, 60, each number is 15 more than the one before.

A. True

B. False

What is the rule for 10, 20, 30, 40?

- A.** Add 10
- B.** Add 1
- C.** Multiply by 2
- D.** Add 20

What comes next?

12, 24, 36, ____

Next: _____

True or false?

The pattern 5, 10, 20, 40 adds 5 each time.

A. True

B. False

SET 44 • 4.MD.A.1

FORWARD 1

Convert from a smaller unit to a larger unit • Grade 5 • Measurement & Data •
5.MD.A.1

Convert.

48 ounces = ____ pounds

Pounds: _____

True or false?

2,000 milliliters = 2 liters

A. True

B. False

Convert.

250 centimeters = ____ meters

A. 2,500

B. 0.25

C. 25

D. 2.5

Convert.

150 seconds = ____ minutes

Minutes: _____

True or false?

36 inches = 4 feet

A. True

B. False

SET 44 • 4.MD.A.1

FORWARD 2

Use tables of equivalent ratios to convert units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

Complete the table.

Yards	1	2	5
Feet	3	6	?

Missing value: _____

True or false?

A table with the pairs (1, 1,000), (2, 2,000), and (3, 3,000) converts kilograms to grams.

- A. True
- B. False

There are 16 ounces in 1 pound.

How many pounds are 40 ounces?

A. 640

B. $2\frac{1}{2}$

C. 24

D. 3

1 liter = 1,000 milliliters.

How many liters are 3,500 milliliters?

Liters: _____

True or false?

Because $1 \text{ meter} = 100 \text{ cm}$, $0.75 \text{ meter} = 7.5 \text{ cm}$.

A. True

B. False

Solve word problems about distances and intervals of time

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Time intervals in minutes	3.MD.A.1
BACKWARD 2	Length word problems within 100	2.MD.B.5
FORWARD 1	Solve multistep problems that involve converting units	5.MD.A.1
FORWARD 2	Solve constant-speed problems	6.RP.A.3.b

SET 45 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A movie started at 3:45 p.m. and ended at 5:20 p.m.

How long was the movie?

Length: _____

Ty biked 2 km 350 m on Monday and 1 km 900 m on Tuesday.

How far did he bike in all? Give the answer in meters.

Meters: _____

A race is 3 km long. Ana has run 1,850 m.

How many more meters must she run?

A. 1,150 m

B. 4,850 m

C. 1,250 m

D. 2,150 m

True or false?

Soccer practice starts at 4:30 p.m. and lasts 1 hour 15 minutes. It ends at 5:45 p.m.

A. True

B. False

A bus trip takes 45 minutes each way.

How long does a round trip take? Write the answer in hours and minutes.

Time: _____

SET 45 • 4.MD.A.2

BACKWARD 1

Time intervals in minutes • Grade 3 • Measurement & Data • 3.MD.A.1

A class starts at 9:10 and ends at 9:55.

How many minutes long is it?

Minutes: _____

True or false?

From 2:30 to 3:15 is 45 minutes.

A. True

B. False

Jo started reading at 7:20 and read for 25 minutes.

What time did she stop?

A. 7:35

B. 7:45

C. 8:45

D. 7:25

A show begins at 4:05 and lasts 50 minutes.
What time does it end?

Time: _____

True or false?

From 10:40 to 11:00 is 40 minutes.

A. True

B. False

SET 45 • 4.MD.A.2

BACKWARD 2

Length word problems within 100 • Grade 2 • Measurement & Data • 2.MD.B.5

A ribbon is 64 cm long. Mia cuts off 28 cm.
How long is the ribbon now?

Length: _____

True or false?

A 45-inch board and a 37-inch board laid end to end are 82 inches long.

A. True

B. False

A path is 90 m long. Sam has walked 55 m.

How far is left?

- A.** 45 m
- B.** 35 m
- C.** 25 m
- D.** 145 m

A tree was 38 feet tall. It grew 6 feet.

How tall is it now?

Height: _____

True or false?

Cutting 17 cm from a 50-cm rope leaves 37 cm.

A. True

B. False

SET 45 • 4.MD.A.2

FORWARD 1

Solve multistep problems that involve converting units • Grade 5 •
Measurement & Data • 5.MD.A.1

A runner ran 1.5 km and then 800 m.
How many meters did she run in all?

Meters: _____

True or false?

A trip of 2 hours 30 minutes takes 150 minutes.

A. True

B. False

A 4-yard rope has 5 feet cut off.

How many feet are left?

- A.** 17 feet
- B.** 1 foot
- C.** 7 feet
- D.** 9 feet

A train ride lasts $2\frac{1}{4}$ hours.

How many minutes is that?

Minutes: _____

True or false?

$$3 \text{ km} - 1,200 \text{ m} = 2,800 \text{ m}$$

A. True

B. False

SET 45 • 4.MD.A.2

FORWARD 2

Solve constant-speed problems • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.b

A car travels 150 miles in 3 hours at a constant speed.
What is its speed in miles per hour?

Speed: _____

True or false?

Walking 6 km in 2 hours is a speed of 3 km per hour.

- A.** True
- B.** False

A cyclist rides 12 miles per hour.

How far does she ride in 2.5 hours?

- A.** 14.5 miles
- B.** 30 miles
- C.** 24 miles
- D.** 36 miles

A train goes 240 km in 4 hours.

At this rate, how long does it take to go 420 km?

Time: _____

True or false?

At 60 miles per hour, a car goes 90 miles in 2 hours.

- A. True
- B. False

Solve word problems about liquid volume, mass, and money

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Mass and liquid volume word problems	3.MD.A.2
BACKWARD 2	Dollars and cents	2.MD.C.8
FORWARD 1	Solve money and measurement problems with decimal operations	5.NBT.B.7
FORWARD 2	Find and compare unit prices	6.RP.A.3.b

SET 46 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A jug holds 2 liters of juice. Ella pours 6 cups of 250 milliliters each.

How many milliliters are left in the jug?

Milliliters: _____

A box of apples has a mass of 3 kg 400 g. Another box has a mass of 2 kg 750 g.

What is their total mass in grams?

Grams: _____

Jo buys 3 notebooks that cost \$2.45 each.

How much does she spend?

A. \$7.35

B. \$6.35

C. \$5.35

D. \$7.45

True or false?

Lee pays for items that cost \$3.60 and \$4.15 with a \$10 bill. His change is \$3.25.

- A. True
- B. False

A cat has a mass of 4 kg. A dog has 6 times as much mass.

What is the dog's mass in grams?

Grams: _____

SET 46 • 4.MD.A.2

BACKWARD 1

Mass and liquid volume word problems • Grade 3 • Measurement & Data •
3.MD.A.2

A pot holds 8 liters. 3 liters are poured out.
How many liters are left?

Liters: _____

True or false?

Four 5-kg bags have a total mass of 20 kg.

A. True

B. False

45 liters of water are shared equally among 9 buckets.

How many liters go in each bucket?

A. 54

B. 9

C. 5

D. 36

A melon has a mass of 900 g. A pear has a mass of 200 g.

How much more mass does the melon have?

Grams: _____

True or false?

Two 35-liter tubs hold 60 liters in all.

A. True

B. False

SET 46 • 4.MD.A.2

BACKWARD 2

Dollars and cents • Grade 2 • Measurement & Data • 2.MD.C.8

How much money is 2 quarters, 1 dime, and 3 pennies?

Cents: _____

True or false?

3 quarters are worth 75 cents.

A. True

B. False

Ben has 2 dollar bills and 4 dimes.

How much money does he have?

A. \$2.04

B. \$2.40

C. \$6.00

D. \$2.10

A toy costs 85 cents. Ann pays with 1 dollar.
How much change does she get?

Change: _____

True or false?

5 nickels and 2 dimes are worth 50 cents.

A. True

B. False

SET 46 • 4.MD.A.2

FORWARD 1

Solve money and measurement problems with decimal operations • Grade 5 •
Number & Operations in Base Ten • 5.NBT.B.7

Pens cost \$1.25 each.

How much do 8 pens cost?

Cost: _____

True or false?

A \$20 bill pays for 3 shirts that cost \$5.99 each.
The change is \$2.03.

- A.** True
- B.** False

A 2.4-liter bottle is shared equally among 6 cups.

How many liters go in each cup?

A. 1.4

B. 0.4

C. 4

D. 0.04

Grapes cost \$2.80 per pound.

How much do 3.5 pounds cost?

Cost: _____

True or false?

A 1.5-kg bag and a 0.75-kg bag have a total mass of 1.125 kg.

A. True

B. False

SET 46 • 4.MD.A.2

FORWARD 2

Find and compare unit prices • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A 12-pack of water costs \$4.80.

What is the price of 1 bottle?

Price: _____

True or false?

4 kg of rice for \$6 costs \$1.50 per kilogram.

A. True

B. False

Which is the best buy?

- A.** 2 liters for \$3.20
- B.** 3 liters for \$4.50
- C.** 4 liters for \$6.40
- D.** 1 liter for \$1.70

5 pounds of apples cost \$8.75.

How much does 1 pound cost?

Cost: _____

True or false?

A pack of 6 juice boxes for \$3.60 costs \$0.70 per box.

- A. True
- B. False

Solve measurement problems with simple fractions or decimals, using number line diagrams

MAIN	Grade 4 main question	4.MD.A.2
BACKWARD 1	Solve time problems with a number line	3.MD.A.1
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Solve measurement problems by adding and subtracting fractions...	5.NF.A.2
FORWARD 2	Find distances between points with the same first or second coor...	6.NS.C.8

SET 47 • 4.MD.A.2

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.2

A board is 6 feet long. Jo cuts off $\frac{3}{4}$ foot.

How long is the board now?

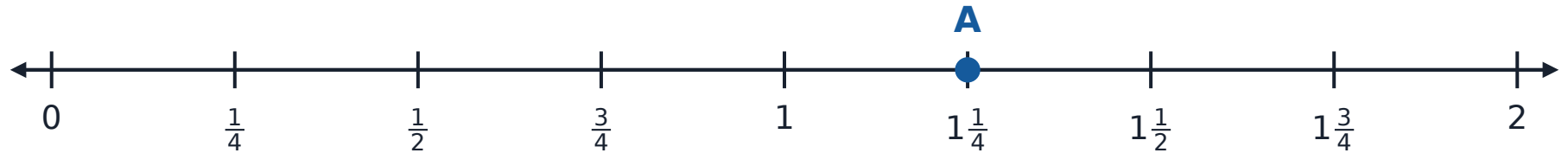
Length: _____

Ana ran 0.5 km before lunch and 0.25 km after lunch.
How far did she run in all?

Distance: _____

The number line shows a 2-mile hike. Lu has hiked to point A.

How much farther must she hike to reach the end?



- A.** $\frac{1}{4}$ mile **B.** 2 miles **C.** $\frac{3}{4}$ mile **D.** $1\frac{1}{4}$ miles

True or false?

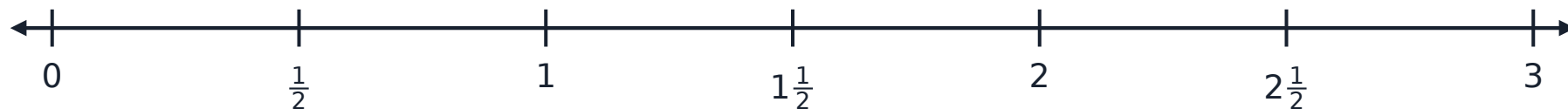
A jug has 2 liters of water. Lee pours out $\frac{1}{4}$ liter.
 $1\frac{3}{4}$ liters are left.

A. True

B. False

A rope is 3 meters long. Use the number line to show $\frac{1}{2}$ meter being cut off.

How much rope is left?



Rope left: _____

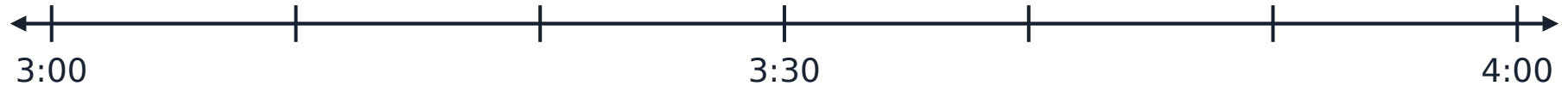
SET 47 • 4.MD.A.2

BACKWARD 1

Solve time problems with a number line • Grade 3 • Measurement & Data •
3.MD.A.1

Ty starts at 3:00 and works for 40 minutes. Use the number line.

What time does he finish?



Time: _____

True or false?

On a number line from 1:00 to 2:00, 1:30 is halfway.

A. True

B. False

A game started at 6:15 and lasted 35 minutes.

What time did it end?

A. 6:40

B. 6:35

C. 7:50

D. 6:50

How many minutes are there from 8:20 to 9:00?

Minutes: _____

True or false?

From 4:45 to 5:05 is 30 minutes.

A. True

B. False

SET 47 • 4.MD.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

$\frac{3}{3}$ and 1 are at the same point on a number line.

A. True

B. False

Which fraction is closest to 0?

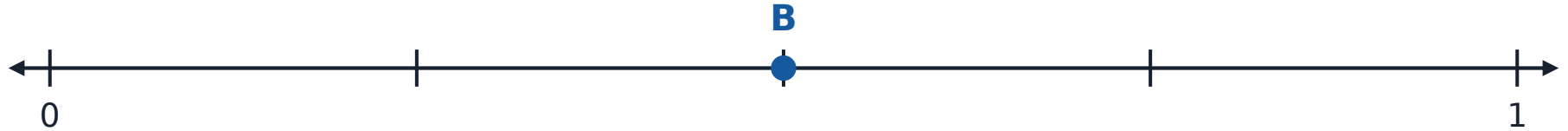
A. $\frac{1}{8}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{7}{8}$

What fraction is at point B?



B = _____

True or false?

$\frac{1}{4}$ is to the right of $\frac{3}{4}$ on a number line.

A. True

B. False

SET 47 • 4.MD.A.2

FORWARD 1

Solve measurement problems by adding and subtracting fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A board is 5 feet long. $\frac{2}{3}$ foot is cut off.

How long is the board now?

Length: _____

True or false?

A jug had $1\frac{1}{2}$ liters. $\frac{3}{4}$ liter was used. $\frac{3}{4}$ liter is left.

A. True

B. False

Ann walked $\frac{1}{2}$ km and then $\frac{3}{5}$ km.

How far did she walk in all?

A. $1\frac{1}{5}$ km

B. $1\frac{1}{10}$ km

C. $\frac{4}{10}$ km

D. $\frac{4}{7}$ km

A rope is $4\frac{1}{2}$ m long. A piece $1\frac{2}{3}$ m long is cut off.

How much rope is left?

Length: _____

True or false?

$$\frac{3}{8} \text{ kg} + \frac{1}{4} \text{ kg} = \frac{4}{12} \text{ kg}$$

A. True

B. False

SET 47 • 4.MD.A.2

FORWARD 2

Find distances between points with the same first or second coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(2, 5)$ and $(2, -3)$.

Distance: _____

True or false?

The distance between $(-4, 1)$ and $(3, 1)$ is 7 units.

A. True

B. False

On a map grid, a school is at $(-2, 4)$ and a park is at $(5, 4)$.

How many units apart are they?

A. 6

B. 7

C. 9

D. 3

Find the distance between $(-6, -2)$ and $(-6, 3)$.

Distance: _____

True or false?

The distance between $(0, -5)$ and $(0, 5)$ is 0 units.

A. True

B. False

Apply the area formula for rectangles

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 1	Area as length times width	3.MD.C.7.b
BACKWARD 3	Find area by tiling a rectangle with unit squares	3.MD.C.7.a
FORWARD 1	Find the area of rectangles with fractional side lengths	5.NF.B.4.b
FORWARD 2	Find the area of triangles and parallelograms	6.G.A.1

SET 48 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

Find the area of the rectangle.

Area: _____



Find the area of the rectangle.

Area: _____



A rectangular rug is 8 feet by 6 feet.

What is its area?

- A.** 96 square feet
- B.** 14 square feet
- C.** 48 square feet
- D.** 28 square feet

True or false?

A square with sides of 7 cm has an area of 28 square cm.

A. True

B. False

A garden is 15 m long and 8 m wide.

What is its area?

Area: _____

SET 48 • 4.MD.A.3

BACKWARD 1

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 units by 7 units.

What is its area?

Area: _____

True or false?

A 5-by-9 rectangle has an area of 45 square units.

A. True

B. False

Which equation gives the area of a 4-by-8 rectangle?

A. $8 - 4 = 4$

B. $4 + 8 = 12$

C. $2 \times (4 + 8) = 24$

D. $4 \times 8 = 32$

Find the area of the rectangle.

Area: _____



True or false?

A 3-by-9 rectangle has an area of 12 square units.

A. True

B. False

SET 48 • 4.MD.A.3

BACKWARD 3

Find area by tiling a rectangle with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

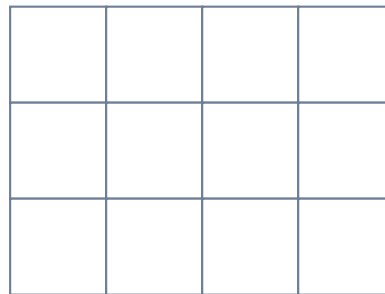
A rectangle covered by 3 rows of 5 square centimeters has an area of 15 square centimeters.

A. True

B. False

Each small square is 1 square unit.

What is the area of the rectangle?



- A.** 12 square units **B.** 14 square units
- C.** 24 square units **D.** 7 square units

A rug is covered by 2 rows of 9 one-foot squares.
What is its area?

Area: _____

True or false?

A rectangle covered by 5 rows of 5 unit squares has an area of 10 square units.

- A. True
- B. False

SET 48 • 4.MD.A.3

FORWARD 1

Find the area of rectangles with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.b

Find the area of a rectangle that is $\frac{3}{4}$ m by $\frac{2}{3}$ m.

Area: _____

True or false?

A rectangle that is $2\frac{1}{2}$ ft by 4 ft has an area of 10 square feet.

- A.** True
- B.** False

A square has sides that are $\frac{1}{2}$ inch long.

What is its area?

A. 2 square inches

B. 1 square inch

C. $\frac{1}{4}$ square inch

D. $\frac{1}{2}$ square inch

Find the area of a rectangle that is $1\frac{1}{2}$ yd by $\frac{2}{3}$ yd.

Area: _____

True or false?

A $\frac{1}{3}$ -by- $\frac{1}{3}$ square has an area of $\frac{2}{3}$ square unit.

A. True

B. False

SET 48 • 4.MD.A.3

FORWARD 2

Find the area of triangles and parallelograms • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

True or false?

A parallelogram with a base of 8 m and a height of 5 m has an area of 40 square meters.

A. True

B. False

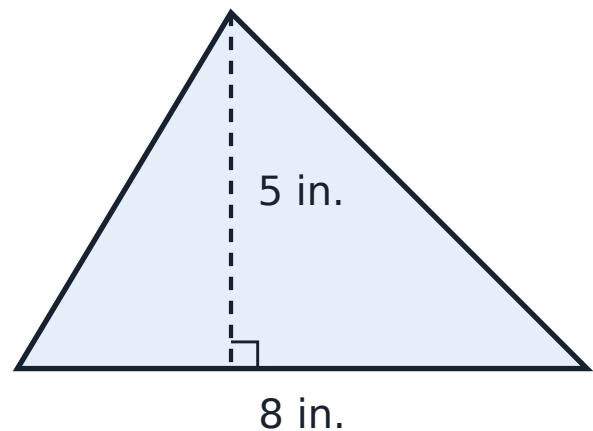
A right triangle has legs that are 4 ft and 9 ft long.

What is its area?

- A.** 26 square feet
- B.** 18 square feet
- C.** 36 square feet
- D.** 13 square feet

Find the area of the triangle.

Area: _____



True or false?

A triangle with a base of 6 units and a height of 4 units has an area of 24 square units.

A. True

B. False

Apply the perimeter formula for rectangles

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 1	Perimeter of polygons	3.MD.D.8
BACKWARD 2	Add up to four two-digit numbers	2.NBT.B.6
FORWARD 1	Find the perimeter of shapes with fractional side lengths	5.NF.A.2
FORWARD 2	Evaluate perimeter formulas for given side lengths	6.EE.A.2.c

SET 68 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

Find the perimeter of the rectangle.

Perimeter: _____



True or false?

A square with sides of 7 cm has a perimeter of 49 cm.

A. True

B. False

A garden is 15 m long and 8 m wide.

What is its perimeter?

Perimeter: _____

A rectangular field is 40 yards long and 25 yards wide.
What is its perimeter?

- A.** 1,000 yards
- B.** 105 yards
- C.** 65 yards
- D.** 130 yards

Write a formula for the perimeter P of a rectangle with length l and width w . Then use it to find the perimeter of a rectangle that is 11 cm by 6 cm.

Formula: _____

Perimeter: _____

SET 68 • 4.MD.A.3

BACKWARD 1

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 5 cm, 7 cm, and 9 cm.
What is its perimeter?

Perimeter: _____

True or false?

A square with sides of 4 m has a perimeter of 16 m.

A. True

B. False

What is the perimeter of a rectangle that is 6 ft by 2 ft?

- A.** 8 feet
- B.** 12 feet
- C.** 16 feet
- D.** 14 feet

A pentagon has five sides that are each 3 inches long.
What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is 5 units by 3 units has a perimeter of 15 units.

A. True

B. False

SET 68 • 4.MD.A.3

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten
• 2.NBT.B.6

Add.

$$12 + 5 + 12 + 5$$

Sum: _____

True or false?

$$25 + 40 + 25 + 40 = 130$$

A. True

B. False

$$9 + 9 + 3 + 3 = ?$$

A. 24

B. 12

C. 21

D. 81

Add.

$$15 + 8 + 15 + 8$$

Sum: _____

True or false?

$$14 + 6 + 14 + 6 = 30$$

A. True

B. False

SET 68 • 4.MD.A.3

FORWARD 1

Find the perimeter of shapes with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A rectangle is $\frac{3}{4}$ m long and $\frac{1}{2}$ m wide.

What is its perimeter?

Perimeter: _____

True or false?

A square with sides of $\frac{2}{3}$ foot has a perimeter of $2\frac{2}{3}$ feet.

- A.** True
- B.** False

A rectangular card is $2\frac{1}{2}$ inches long and $1\frac{1}{4}$ inches wide.

What is its perimeter?

A. $3\frac{3}{4}$ inches

B. $3\frac{1}{8}$ inches

C. $6\frac{1}{2}$ inches

D. $7\frac{1}{2}$ inches

A triangle has sides of $\frac{1}{2}$ yd, $\frac{2}{3}$ yd, and $\frac{5}{6}$ yd.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is $\frac{1}{3}$ m by $\frac{1}{4}$ m has a perimeter of $\frac{7}{12}$ m.

A. True

B. False

SET 68 • 4.MD.A.3

FORWARD 2

Evaluate perimeter formulas for given side lengths • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $P = 2l + 2w$ to find the perimeter when $l = 8.5$ cm and $w = 4$ cm.

Perimeter: _____

True or false?

For a square, $P = 4s$. When $s = 2.25$ m, $P = 9$ m.

A. True

B. False

Use $P = 2(l + w)$.

What is P when $l = 12$ and $w = 7.5$?

A. 19.5

B. 90

C. 39

D. 29

Evaluate $2l + 2w$ when $l = \frac{3}{4}$ and $w = \frac{1}{2}$.

Value: _____

True or false?

When $l = 10$ and $w = 3$, $2l + 2w = 23$.

A. True

B. False

Find an unknown side length from the area of a rectangle

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 2	Find the unknown factor	3.OA.A.4
BACKWARD 3	Find the area of a rectangle by multiplying its side lengths	3.MD.C.7.b
FORWARD 1	Find a missing dimension when the volume is known	5.MD.C.5.b
FORWARD 2	Solve equations of the form $px = q$ for a missing measurement	6.EE.B.7

SET 49 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 72 square feet and a length of 9 feet.

What is its width?

Width: _____

A rectangle has an area of 63 square cm and a length of 9 cm.

What is its width?

Width: _____

A rectangular patio has an area of 96 square meters.
One side is 12 m long.

How long is the other side?

A. 36 m

B. 108 m

C. 84 m

D. 8 m

True or false?

A square has an area of 36 square inches, so each side is 6 inches long.

A. True

B. False

A rectangle has an area of 54 square units and a width of 6 units.

Write an equation with a letter for the length. Then solve it.

Equation: _____

Length: _____

SET 49 • 4.MD.A.3

BACKWARD 2

Find the unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

Find the unknown number.

$$8 \times ? = 64$$

Unknown: _____

True or false?

If $9 \times w = 45$, then $w = 5$.

A. True

B. False

$$? \times 7 = 56$$

A. 7

B. 8

C. 49

D. 63

Find the unknown number.

$$6 \times ? = 42$$

Unknown: _____

True or false?

If $4 \times n = 28$, then $n = 6$.

A. True

B. False

SET 49 • 4.MD.A.3

BACKWARD 3

Find the area of a rectangle by multiplying its side lengths • Grade 3 •
Measurement & Data • 3.MD.C.7.b

A rectangle is 8 cm long and 7 cm wide.

What is its area?

Area: _____

True or false?

A 9-by-6 rectangle has an area of 54 square units.

A. True

B. False

A rug is 5 feet by 8 feet.

What is its area?

- A.** 45 square feet
- B.** 40 square feet
- C.** 13 square feet
- D.** 26 square feet

A garden is 9 m long and 4 m wide.

What is its area?

Area: _____

True or false?

A 7-by-7 square has an area of 14 square units.

- A. True
- B. False

SET 49 • 4.MD.A.3

FORWARD 1

Find a missing dimension when the volume is known • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 60 cubic inches. It is 5 in. long and 3 in. wide.

How tall is it?

Height: _____

True or false?

A prism with a volume of 48 cubic cm and a base area of 12 square cm is 4 cm tall.

- A. True
- B. False

A tank holds 120 cubic feet. Its base is 6 ft by 5 ft.

How deep is the tank?

- A.** 20 feet
- B.** 4 feet
- C.** 10 feet
- D.** 109 feet

A prism has a volume of 90 cubic meters and a height of 5 m.

What is the area of its base?

Base area: _____

True or false?

A box that is 4 units by 3 units at the base and has a volume of 36 cubic units is 4 units tall.

- A. True
- B. False

SET 49 • 4.MD.A.3

FORWARD 2

Solve equations of the form $px = q$ for a missing measurement • Grade 6 •
Expressions & Equations • 6.EE.B.7

A rectangle has an area of 7.5 square meters and a length of 3 m.

Write and solve an equation for its width w .

Equation: _____

$W =$ _____

True or false?

A triangle has an area of 24 square units and a base of 8 units. Solving $4h = 24$ gives a height of 6 units.

- A. True
- B. False

Solve for b.

$$12b = 84$$

A. 1,008

B. 7

C. 72

D. 96

Solve for x.

$$\frac{1}{2}x = 9$$

x = _____

True or false?

The solution of $4w = 30$ is $w = 26$.

A. True

B. False

Find an unknown side length from the perimeter of a rectangle

MAIN

Grade 4 main question

4.MD.A.3

BACKWARD 1

Find an unknown side from the perimeter

3.MD.D.8

BACKWARD 2

Find an unknown addend within 100

2.NBT.B.5

FORWARD 1

Find an unknown side length from a perimeter with fractions

5.NF.A.2

FORWARD 2

Write and solve one-step equations for a missing side length

6.EE.B.7

SET 69 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has a perimeter of 30 cm and a length of 10 cm.

What is its width?

Width: _____

True or false?

A square has a perimeter of 36 inches, so each side is 9 inches long.

A. True

B. False

A rectangular garden has a perimeter of 50 m. Its length is 15 m.

What is its width?

A. 20 m

B. 10 m

C. 25 m

D. 35 m

A rectangle has a perimeter of 64 inches and a width of 12 inches.

Write an equation with a letter for the length. Then solve it.

Equation: _____

Length: _____

True or false?

A rectangle has a perimeter of 26 feet and a length of 8 feet, so its width is 18 feet.

A. True

B. False

SET 69 • 4.MD.A.3

BACKWARD 1

Find an unknown side from the perimeter • Grade 3 • Measurement & Data •
3.MD.D.8

A triangle has a perimeter of 20 cm. Its two known sides add to 14 cm.

How long is the third side?

Length: _____

True or false?

A square with a perimeter of 12 m has sides that are 3 m long.

A. True

B. False

A rectangle has a perimeter of 14 ft. Its two lengths and one width add to 12 ft.

How long is the other width?

A. 2 ft

B. 7 ft

C. 9 ft

D. 4 ft

A rectangle has a perimeter of 18 in. Its two widths and one length add to 13 in.

How long is the other length?

Length: _____

True or false?

A square with a perimeter of 20 units has sides that are 10 units long.

- A. True
- B. False

SET 69 • 4.MD.A.3

BACKWARD 2

Find an unknown addend within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Find the unknown number.

$$20 + ? = 30$$

Unknown: _____

True or false?

If $24 + n = 64$, then $n = 40$.

A. True

B. False

$$? + 30 = 50$$

A. 30

B. 25

C. 80

D. 20

Find the unknown number.

$$? + 16 = 26$$

Unknown: _____

True or false?

If $36 + n = 50$, then $n = 24$.

A. True

B. False

SET 69 • 4.MD.A.3

FORWARD 1

Find an unknown side length from a perimeter with fractions • Grade 5 •
Number & Operations–Fractions • 5.NF.A.2

A rectangle has a perimeter of 5 m. Its length is $1\frac{1}{2}$ m.

What is its width?

Width: _____

True or false?

A square has a perimeter of 3 feet, so each side is $\frac{3}{4}$ foot long.

A. True

B. False

A rectangle has a perimeter of 7 inches. Its width is $1\frac{1}{4}$ inches.

What is its length?

A. $1\frac{1}{4}$ inches

B. $5\frac{3}{4}$ inches

C. $2\frac{1}{4}$ inches

D. $4\frac{1}{2}$ inches

A triangle has a perimeter of 2 yd. Two of its sides are $\frac{1}{2}$ yd and $\frac{2}{3}$ yd long.

How long is the third side?

Length: _____

True or false?

A rectangle has a perimeter of 4 m and a length of $\frac{3}{4}$ m, so its width is $1\frac{1}{2}$ m.

A. True

B. False

SET 69 • 4.MD.A.3

FORWARD 2

Write and solve one-step equations for a missing side length • Grade 6 •
Expressions & Equations • 6.EE.B.7

A square has a perimeter of 18.4 cm.

Write and solve an equation for its side length s .

Equation: _____

$s =$ _____

True or false?

A rectangle has a perimeter of 30 m, and its two widths total 12 m. Solving $2w = 12$ gives $w = 6$ m.

A. True

B. False

A triangle has a perimeter of 15.5 in., and two of its sides total 9.25 in.

Solve $x + 9.25 = 15.5$ for the third side x .

A. 24.75 in.

B. 6.75 in.

C. 6.25 in.

D. 5.25 in.

Solve for w.

$$2w = 13$$

$$w = \underline{\hspace{10cm}}$$

True or false?

A regular hexagon has a perimeter of 27 feet.
Solving $6s = 27$ gives $s = 21$ feet.

A. True

B. False

Make a line plot of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$)

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot with halves and fourths	3.MD.B.4
BACKWARD 2	Line plots of whole-number lengths	2.MD.D.9
FORWARD 1	Make line plots of data given in halves, fourths, and eighths toge...	5.MD.B.2
FORWARD 2	Display numerical data in dot plots	6.SP.B.4

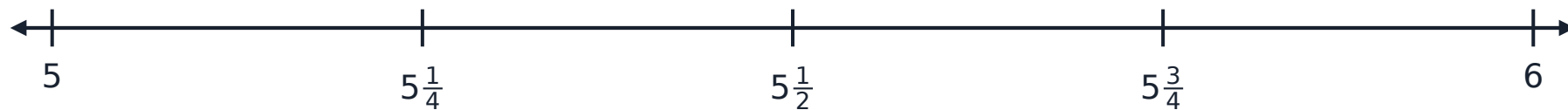
SET 50 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

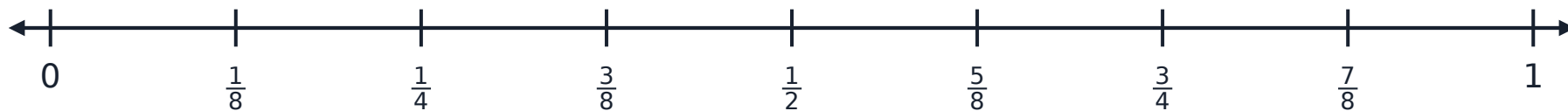
Pencil lengths (inches): $5\frac{1}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$, 6, $5\frac{3}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$

Make a line plot of the data.



Seedling heights (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{8}$, $\frac{1}{8}$

Make a line plot of the data.



Shell lengths (inches): $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1, $\frac{3}{4}$, $\frac{1}{2}$

In a line plot of the data, how many X's go above $\frac{1}{2}$?

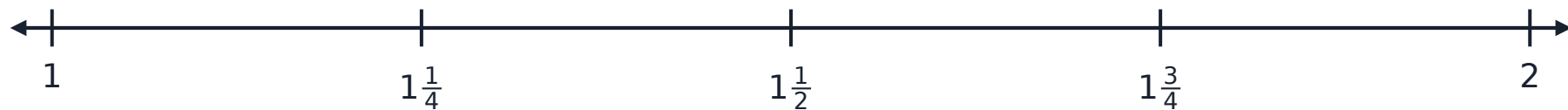
X's: _____

Which number line is best for a line plot of $\frac{1}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, and $\frac{7}{8}$ inch?

- A.** A number line from 1 to 2 marked in fourths
- B.** A number line from 0 to 1 marked in halves
- C.** A number line from 0 to 1 marked in eighths
- D.** A number line from 0 to 8 marked in ones

Ribbon pieces (yards): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, $1\frac{1}{2}$

Make a line plot of the data.



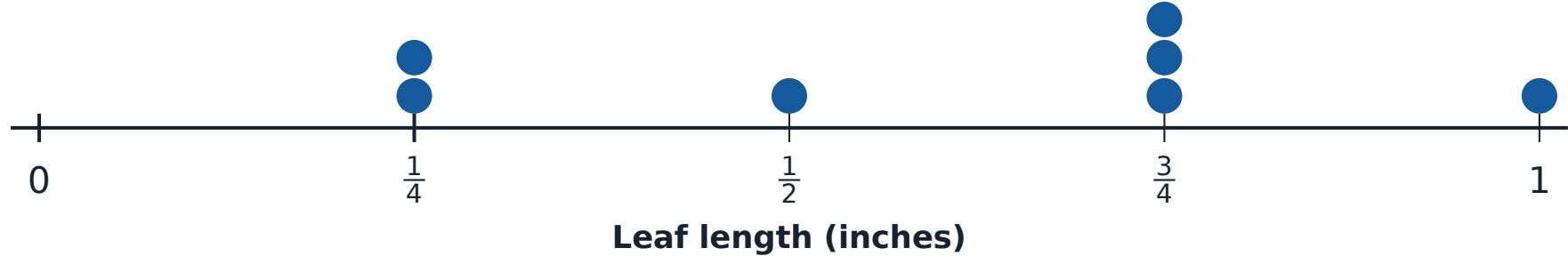
SET 50 • 4.MD.B.4

BACKWARD 1

Read a line plot with halves and fourths • Grade 3 • Measurement & Data •
3.MD.B.4

The line plot shows leaf lengths.

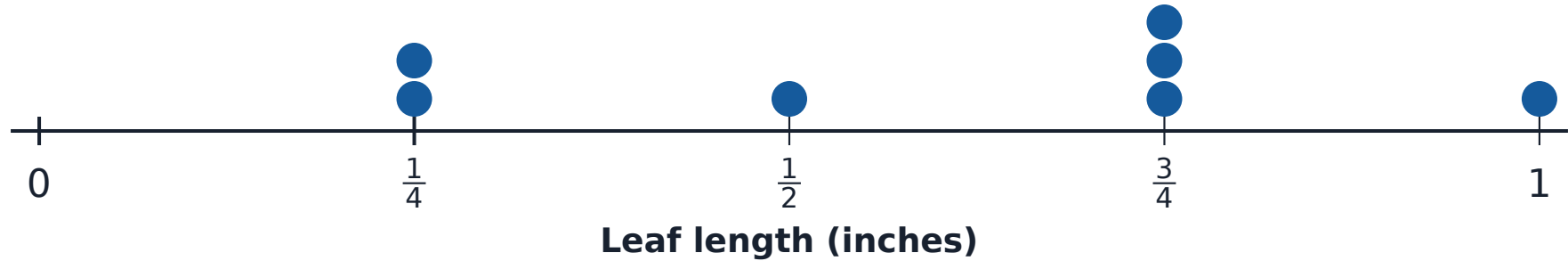
How many leaves are $\frac{3}{4}$ inch long?



Leaves: _____

True or false?

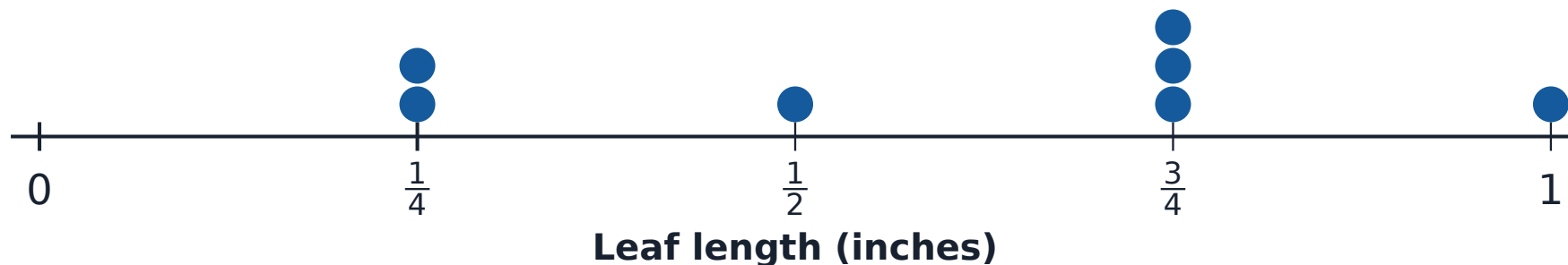
Two leaves are $\frac{1}{4}$ inch long.



A. True

B. False

Which leaf length is most common?



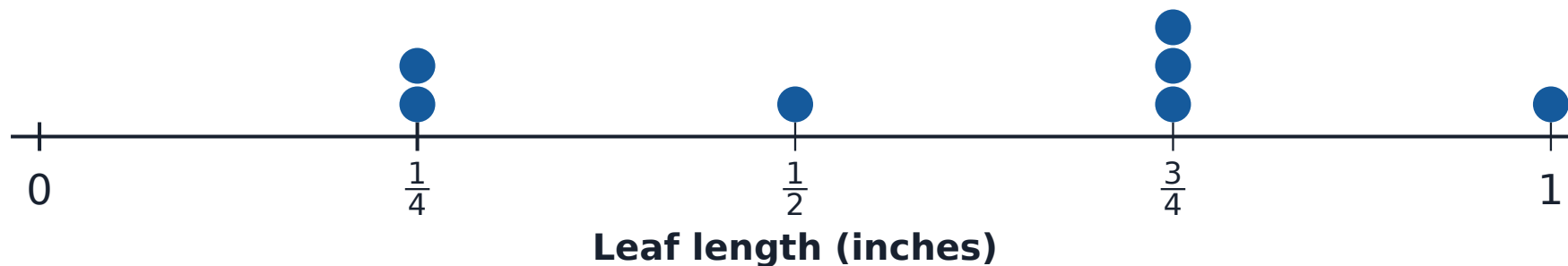
A. $\frac{1}{4}$ inch

B. $\frac{1}{2}$ inch

C. $\frac{3}{4}$ inch

D. 1 inch

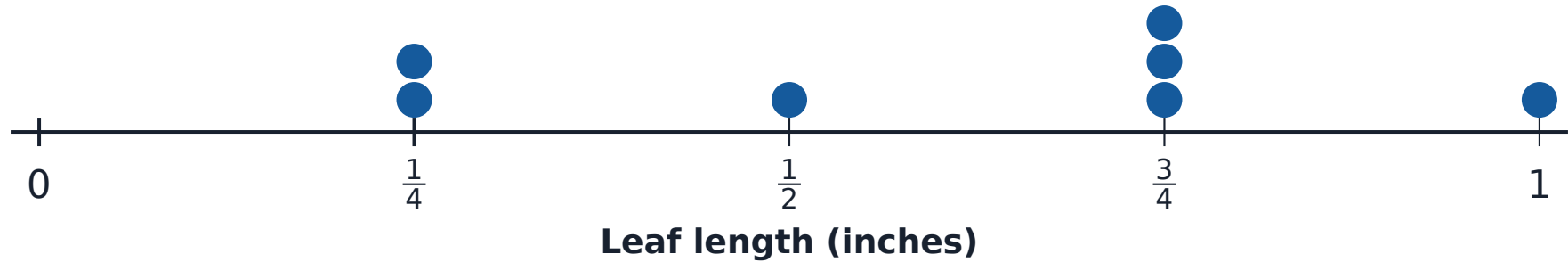
How many leaves were measured?



Leaves: _____

True or false?

No leaf is 1 inch long.



A. True

B. False

SET 50 • 4.MD.B.4

BACKWARD 2

Line plots of whole-number lengths • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (cm): 4, 6, 6, 7, 6

In a line plot of the data, how many X's go above 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line fits a line plot of 2, 3, 3, and 8?

- A.** A number line from 10 to 20
- B.** A number line from 0 to 3
- C.** A number line from 0 to 10
- D.** A number line from 4 to 6

Lengths (inches): 3, 5, 5, 6

Make a line plot of the data.



True or false?

A line plot of 1, 2, 2, 2 has 2 X's above 2.

- A. True
- B. False

SET 50 • 4.MD.B.4

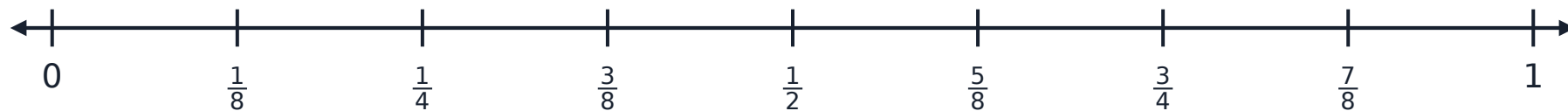
FORWARD 1

Make line plots of data given in halves, fourths, and eighths together • Grade 5

• Measurement & Data • 5.MD.B.2

Bead lengths (inches): $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{7}{8}$

Make a line plot of the data.



True or false?

A number line marked in eighths can show $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{3}{8}$ inch on one line plot.

A. True

B. False

Data (pounds): $\frac{1}{4}$, $\frac{5}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

In a line plot marked in eighths, how many X's go above $\frac{2}{8}$?

A. 2

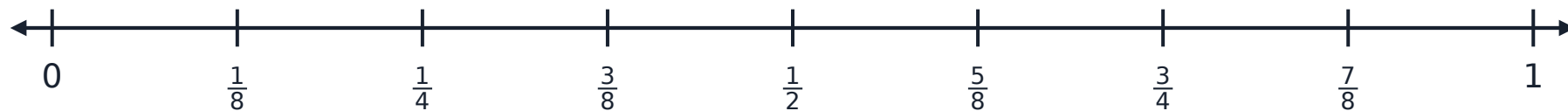
B. 1

C. 4

D. 0

Rain (inches): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$

Make a line plot of the data.



True or false?

In a line plot of $\frac{1}{2}$, $\frac{2}{4}$, and $\frac{4}{8}$ cup, the three X's go above three different marks.

- A. True
- B. False

SET 50 • 4.MD.B.4

FORWARD 2

Display numerical data in dot plots • Grade 6 • Statistics & Probability •
6.SP.B.4

Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data.



True or false?

A histogram groups data values into intervals.

A. True

B. False

Which display shows every individual data value?

- A.** A histogram
- B.** A table of intervals
- C.** A dot plot
- D.** A box plot

Data: 12, 15, 15, 18, 20

In a dot plot of the data, how many dots go above 15?

Dots: _____

True or false?

A dot plot cannot show data with repeated values.

- A.** True
- B.** False

Solve addition and subtraction problems using fraction data in a line plot

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot of lengths in fourths of an inch	3.MD.B.4
BACKWARD 2	Equivalent halves, fourths, and eighths	3.NF.A.3.b
FORWARD 1	Use operations on fractions to solve line plot problems	5.MD.B.2
FORWARD 2	Find the mean of a data set	6.SP.B.5.c

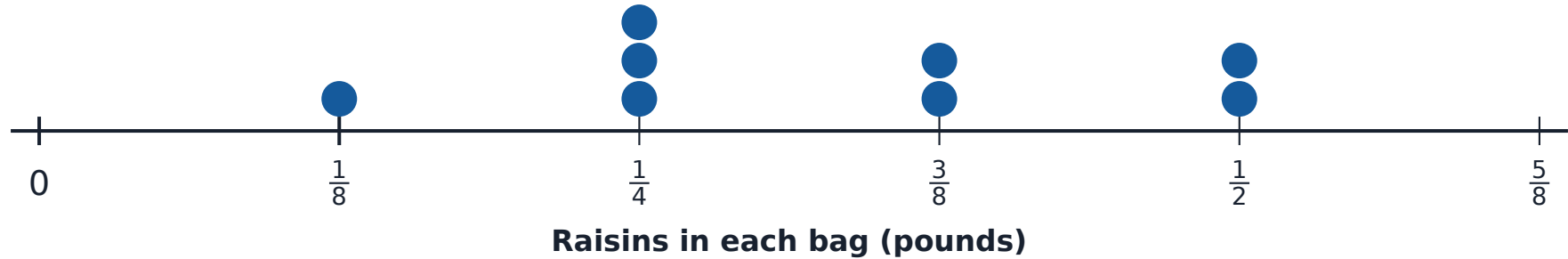
SET 51 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the raisins in some bags.

What is the difference between the heaviest and the lightest bag?



Difference: _____

What is the total weight of the bags that weigh $\frac{1}{4}$ pound each?



Total: _____

How many bags weigh more than $\frac{1}{4}$ pound?



A. 2

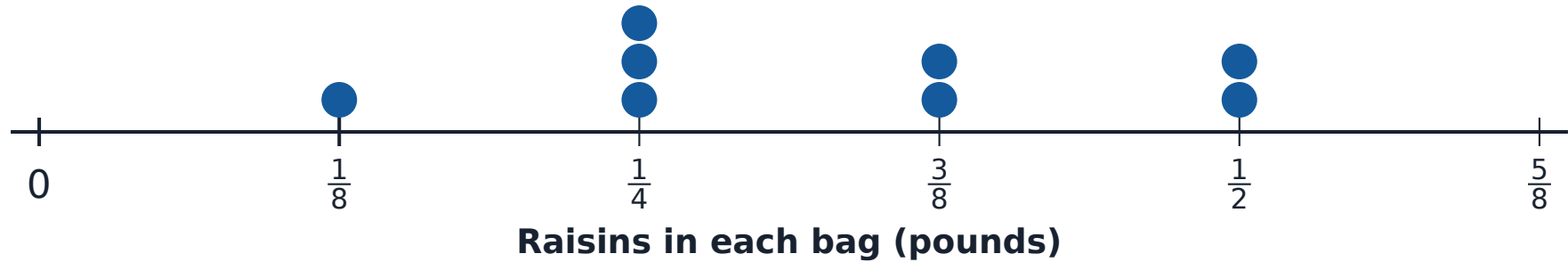
B. 3

C. 5

D. 4

True or false?

The two $\frac{1}{2}$ -pound bags weigh 1 pound in all.



A. True

B. False

What is the total weight of all the bags?



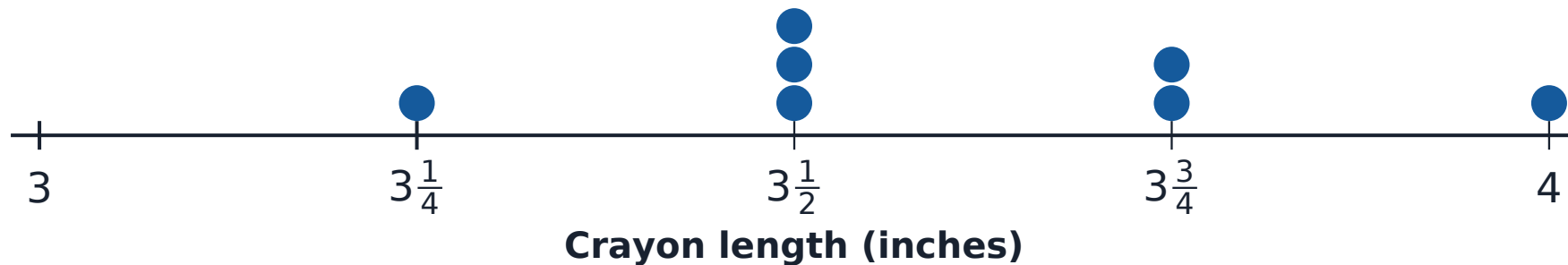
Total: _____

SET 51 • 4.MD.B.4

BACKWARD 1

Read a line plot of lengths in fourths of an inch • Grade 3 • Measurement & Data • 3.MD.B.4

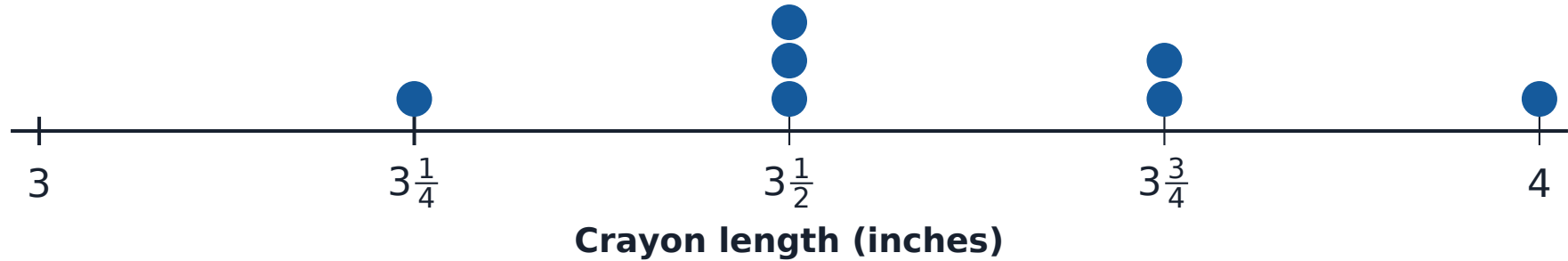
How many crayons are $3\frac{1}{2}$ inches long?



Crayons: _____

True or false?

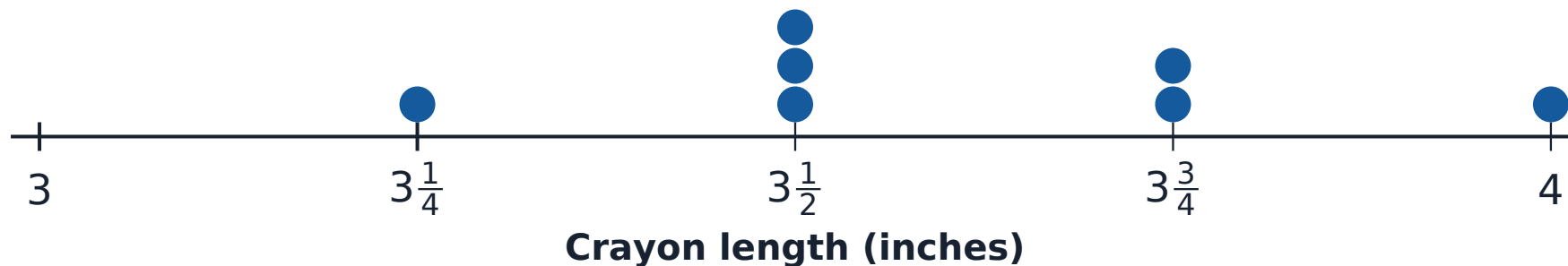
One crayon is 4 inches long.



A. True

B. False

Which length do exactly 2 crayons have?



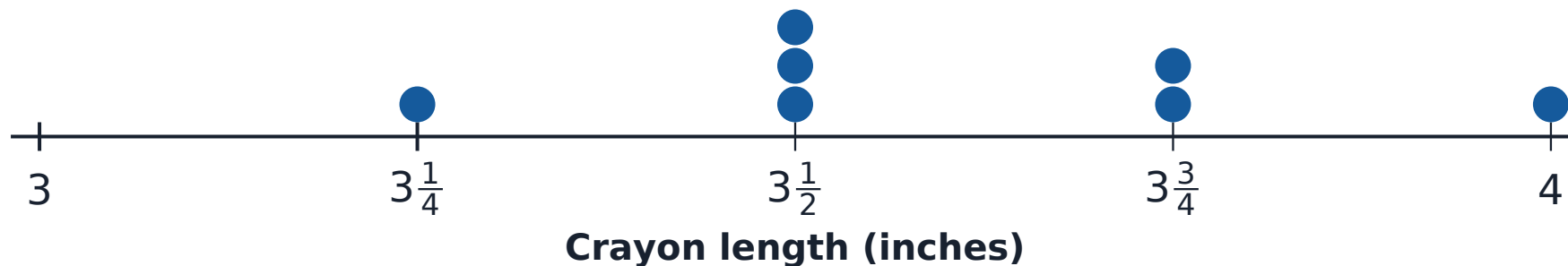
A. $3\frac{1}{2}$
inches

B. 4 inches

C. $3\frac{1}{4}$
inches

D. $3\frac{3}{4}$
inches

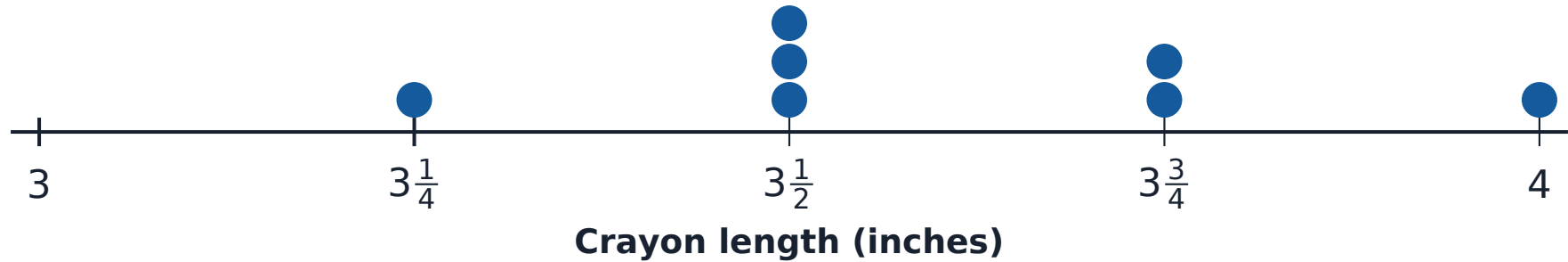
How many crayons were measured?



Crayons: _____

True or false?

The shortest crayon is 3 inches long.



A. True

B. False

SET 51 • 4.MD.B.4

BACKWARD 2

Equivalent halves, fourths, and eighths • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

Write $\frac{1}{2}$ as eighths.

Fraction: _____

True or false?

$$\frac{1}{4} = \frac{2}{8}$$

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{5}{8}$

B. $\frac{6}{8}$

C. $\frac{4}{8}$

D. $\frac{3}{8}$

Find the missing number.

$$\frac{1}{2} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{2}{8} = \frac{1}{2}$$

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 1

Use operations on fractions to solve line plot problems • Grade 5 •
Measurement & Data • 5.MD.B.2

Juice in 3 glasses (cups): $\frac{1}{2}$, $\frac{3}{4}$, $1\frac{1}{4}$

If the juice were shared equally, how much would each glass hold?

Cups: _____

True or false?

Water in beakers (liters): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$. The total is $\frac{3}{4}$ liter.

- A. True
- B. False

Ribbons (yards): $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

What is the total length?

A. $1\frac{3}{4}$ yards

B. $2\frac{1}{4}$ yards

C. $1\frac{1}{2}$ yards

D. 2 yards

Rock masses (kg): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

If the total mass were shared equally by the 4 rocks, what would each rock's mass be?

Mass: _____

True or false?

The total of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ is $\frac{3}{14}$.

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 2

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 6, 8, 10

Mean: _____

True or false?

The mean of 2, 4, and 9 is 5.

A. True

B. False

Find the mean.

$$\frac{1}{2}, \frac{1}{2}, 1, 2$$

A. 4

B. $\frac{1}{2}$

C. 2

D. 1

Five plants grew 3, 5, 5, 6, and 6 cm.

What is the mean growth?

Mean: _____

True or false?

The mean of 10, 10, and 40 is 10.

A. True

B. False

An angle is measured as a fraction of a full circle; a one-degree angle is $\frac{1}{360}$ of a circle

MAIN	Grade 4 main question	4.MD.C.5.a
BACKWARD 1	Equal parts of a shape	3.G.A.2
BACKWARD 2	Turns of the minute hand on a clock	3.MD.A.1
FORWARD 1	Classify triangles by their angles	5.G.B.4
FORWARD 2	Find a percent of a full turn — nearest related (no direct Grade 6 s...	6.RP.A.3.c

SET 52 • 4.MD.C.5.a

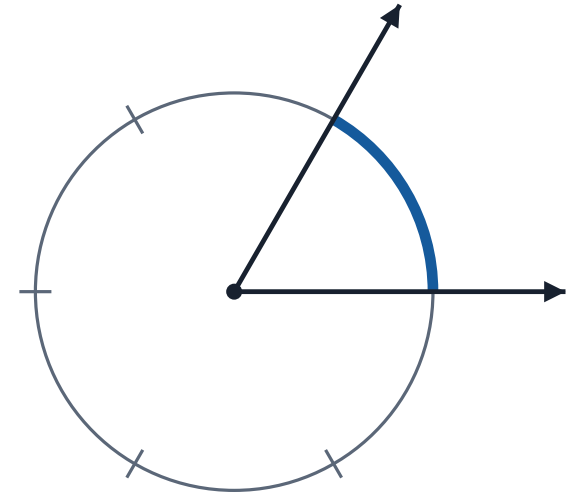
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.5.a

The circle is centered where the rays meet. The thick arc is $\frac{1}{6}$ of the circle.

What is the angle's measure? Explain.

What angle has an arc of $\frac{1}{360}$?



Measure: _____

Explanation: _____

Arc of $\frac{1}{360}$: _____

True or false?

An angle that turns through $\frac{1}{4}$ of a full circle measures 90° .

- A. True
- B. False

What fraction of a full circle does a 180° angle turn through?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{180}$

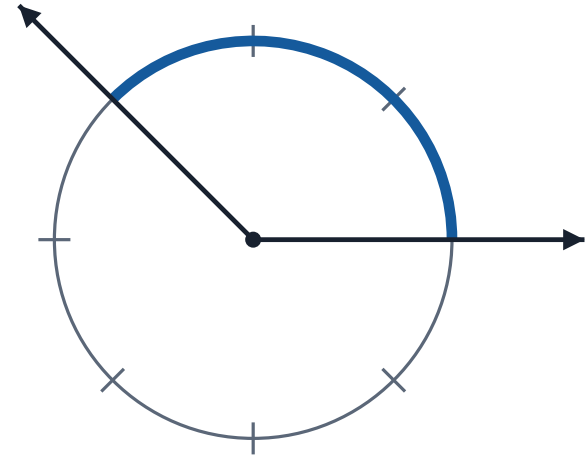
D. $\frac{1}{4}$

The circle is centered at the common endpoint of the two rays and is divided into 8 equal arcs. and is divided into 8 equal arcs.

What fraction of the circle is the thick arc between the rays? What is the measure of the angle?

Fraction: _____

Measure: _____



True or false?

An angle that turns through $\frac{1}{3}$ of a full circle measures 30° .

- A. True
- B. False

SET 52 • 4.MD.C.5.a

BACKWARD 1

Equal parts of a shape • Grade 3 • Geometry • 3.G.A.2

A circle is cut into 4 equal parts.

What fraction of the circle is 1 part?

Fraction: _____

True or false?

A circle cut into 2 equal parts is cut into halves.

A. True

B. False

A circle is cut into 6 equal parts.

What fraction of the circle is each part?

A. $\frac{1}{3}$

B. $\frac{5}{6}$

C. $\frac{6}{1}$

D. $\frac{1}{6}$

Each part of a circle is $\frac{1}{8}$ of the circle.

Into how many equal parts is the circle cut?

Parts: _____

True or false?

A circle cut into 3 parts of different sizes is cut into thirds.

A. True

B. False

SET 52 • 4.MD.C.5.a

BACKWARD 2

Turns of the minute hand on a clock • Grade 3 • Measurement & Data •
3.MD.A.1

The minute hand turns from 12 to 3.
How many minutes pass?

Minutes: _____

True or false?

When the minute hand makes a full turn, 60 minutes pass.

- A. True
- B. False

The minute hand turns halfway around the clock.

How many minutes pass?

A. 60

B. 30

C. 15

D. 12

The minute hand turns from 12 to 9.
How many minutes pass?

Minutes: _____

True or false?

A quarter turn of the minute hand takes 25 minutes.

A. True

B. False

SET 52 • 4.MD.C.5.a

FORWARD 1

Classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

A triangle has angles of 90° , 60° , and 30° .
Classify it by its angles.

Classification: _____

True or false?

A triangle with angles of 70° , 60° , and 50° is an acute triangle.

- A. True
- B. False

A triangle has one angle of 120° .

What kind of triangle is it?

- A.** Right
- B.** Obtuse
- C.** Acute
- D.** Equilateral

Can a triangle have two right angles? Write yes or no.

Answer: _____

True or false?

An obtuse triangle has three angles that are each less than 90° .

- A. True
- B. False

FORWARD 2

Find a percent of a full turn • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

What is 25% of 360° ?

Answer: _____

True or false?

50% of a full turn is 180° .

A. True

B. False

What percent of a full circle is 36° ?

- A.** 10%
- B.** 1%
- C.** 20%
- D.** 36%

What is 75% of 360° ?

Answer: _____

True or false?

20% of 360° is 20° .

A. True

B. False

An angle that turns through n one-degree angles has a measure of n degrees

MAIN	Grade 4 main question	4.MD.C.5.b
BACKWARD 1	A shape covered by n unit squares has an area of n square units	3.MD.C.5.b
BACKWARD 2	Measure length by laying units end to end	1.MD.A.2
FORWARD 1	A solid packed with n unit cubes has a volume of n cubic units —...	5.MD.C.3.b
FORWARD 2	Find volume by packing with unit-fraction cubes — nearest relate...	6.G.A.2

SET 53 • 4.MD.C.5.b

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.5.b

An angle turns through 45 one-degree angles.

What is its measure?

Measure: _____

True or false?

An angle made of 120 one-degree angles measures 120° .

- A. True
- B. False

How many one-degree angles make a 75° angle?

- A.** 15
- B.** 360
- C.** 75
- D.** 285

A fan blade turns through 30 one-degree angles and then 20 more in the same direction.

What angle did it turn through?

Angle: _____

True or false?

A 90° angle is made of 9 one-degree angles.

A. True

B. False

SET 53 • 4.MD.C.5.b

BACKWARD 1

A shape covered by n unit squares has an area of n square units • Grade 3 •
Measurement & Data • 3.MD.C.5.b

A shape is covered by 14 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

A shape covered by 9 unit squares with no gaps has an area of 9 square units.

- A.** True
- B.** False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 20 units
- C.** 40 square units
- D.** 20 square units

How many unit squares cover a shape with an area of 6 square units?

Unit squares: _____

True or false?

A shape covered by 8 unit squares has an area of 16 square units.

- A. True
- B. False

SET 53 • 4.MD.C.5.b

BACKWARD 2

Measure length by laying units end to end • Grade 1 • Measurement & Data •
1.MD.A.2

A pencil is as long as 6 paper clips laid end to end.
How many paper clips long is it?

Paper clips: _____

True or false?

Length units laid end to end must have no gaps or overlaps.

A. True

B. False

A crayon is as long as 4 cubes in a row.

How long is the crayon?

- A.** 5 cubes
- B.** 4 cubes
- C.** 1 cube
- D.** 8 cubes

A book is 9 tiles long. A desk is 3 tiles longer.
How many tiles long is the desk?

Tiles: _____

True or false?

If the cubes overlap when you measure, the count is still correct.

- A.** True
- B.** False

FORWARD 1

A solid packed with n unit cubes has a volume of n cubic units • Grade 5 •
Measurement & Data • 5.MD.C.3.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is filled with 24 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

A solid built from 15 unit cubes has a volume of 15 cubic units.

- A. True
- B. False

A prism has 3 layers with 8 unit cubes in each layer.

What is its volume?

- A.** 24 square units
- B.** 11 cubic units
- C.** 24 cubic units
- D.** 38 cubic units

How many unit cubes fill a box with a volume of 40 cubic units?

Unit cubes: _____

True or false?

A cube with edges 1 unit long has a volume of 6 cubic units.

A. True

B. False

FORWARD 2

Find volume by packing with unit-fraction cubes • Grade 6 • Geometry • 6.G.A.2

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

How many cubes with edges $\frac{1}{2}$ unit long fill a cube with edges 1 unit long?

Cubes: _____

True or false?

A cube with edges $\frac{1}{2}$ unit long has a volume of $\frac{1}{8}$ cubic unit.

- A.** True
- B.** False

A box is packed with 16 cubes that each have edges $\frac{1}{2}$ unit long.

What is its volume?

- A.** 2 cubic units
- B.** 16 cubic units
- C.** 4 cubic units
- D.** 8 cubic units

Find the volume of a prism that is 2 units by $1\frac{1}{2}$ units by $\frac{1}{2}$ unit.

Volume: _____

True or false?

A box that is 1 unit by 1 unit by $\frac{1}{2}$ unit holds exactly 2 cubes with edges $\frac{1}{2}$ unit long.

- A. True
- B. False

Measure angles in whole-number degrees using a protractor

MAIN	Grade 4 main question	4.MD.C.6
BACKWARD 1	Measure length with a ruler	2.MD.A.1
BACKWARD 2	Shapes with a given number of angles	2.G.A.1
FORWARD 1	Attributes of a category belong to its subcategories	5.G.B.3
FORWARD 2	Write and solve $x + p = q$ to find an unknown angle — nearest re...	6.EE.B.7

SET 54 • 4.MD.C.6

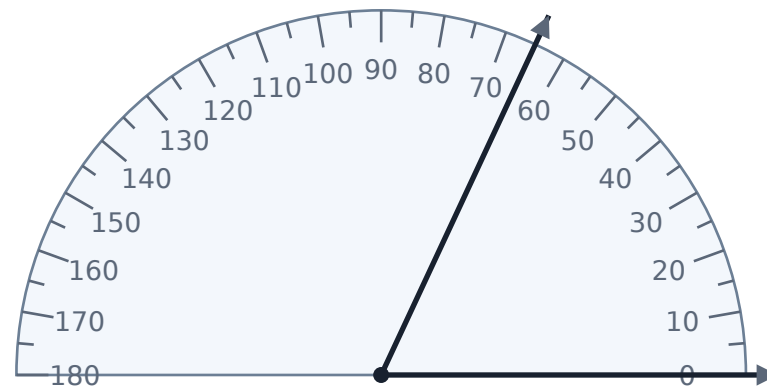
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.6

Use the protractor.

What is the measure of the angle?

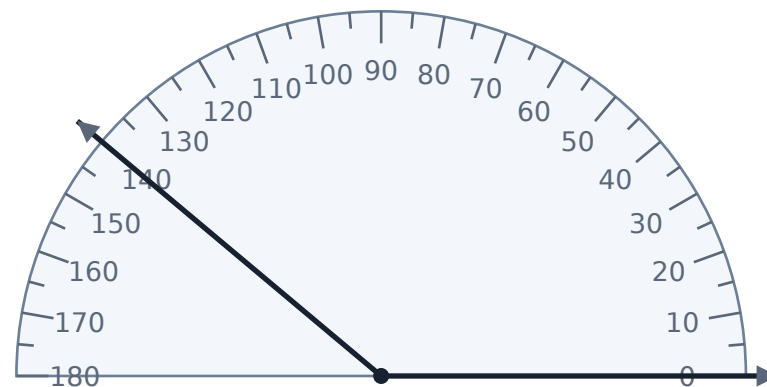
Measure: _____



Use the protractor.

What is the measure of the angle?

Measure: _____



Use the protractor. One ray points to 180 on the outer scale.

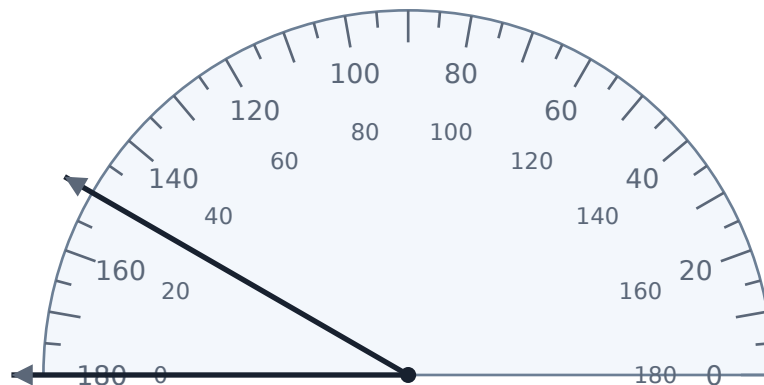
What is the measure of the angle?

A. 150°

B. 120°

C. 180°

D. 30°

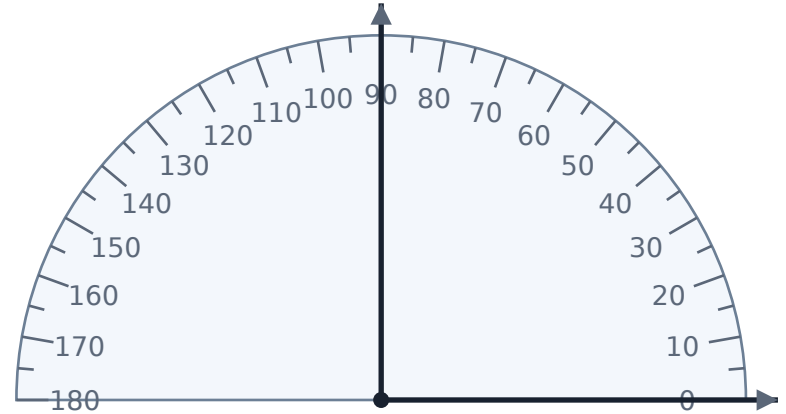


True or false?

The angle measures 90° .

A. True

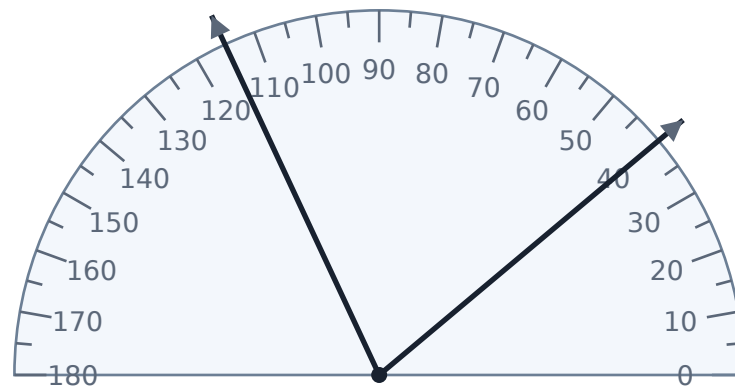
B. False



Use the protractor.

What is the measure of the angle?

Measure: _____



SET 54 • 4.MD.C.6

BACKWARD 1

Measure length with a ruler • Grade 2 • Measurement & Data • 2.MD.A.1

A pencil reaches from 0 to 7 on an inch ruler.
How long is the pencil?

Length: _____

True or false?

When you measure with a ruler, you can line up one end of the object with 0.

- A.** True
- B.** False

A crayon starts at 2 cm and ends at 9 cm on a ruler.

How long is the crayon?

- A.** 2 cm
- B.** 7 cm
- C.** 9 cm
- D.** 11 cm

A leaf reaches from 0 to 5 on a centimeter ruler.
How long is the leaf?

Length: _____

True or false?

An eraser that reaches from 3 to 8 on a ruler is 8 units long.

- A.** True
- B.** False

SET 54 • 4.MD.C.6

BACKWARD 2

Shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

How many angles does a triangle have?

Angles: _____

True or false?

A square has 4 angles.

A. True

B. False

Which shape has 6 angles?

- A.** Pentagon
- B.** Quadrilateral
- C.** Hexagon
- D.** Triangle

How many angles does a pentagon have?

Angles: _____

True or false?

A circle has 4 angles.

A. True

B. False

SET 54 • 4.MD.C.6

FORWARD 1

Attributes of a category belong to its subcategories • Grade 5 • Geometry •
5.G.B.3

True or false?

Every right triangle has one 90° angle. An isosceles right triangle is a right triangle, so it has a 90° angle.

- A.** True
- B.** False

A shape is a rhombus.

Which statement must be true?

- A.** Its opposite angles are equal.
- B.** All its angles are 90° .
- C.** It has 3 sides.
- D.** It has exactly one pair of parallel sides.

A quadrilateral has four 90° angles and four sides of equal length.

What is its most specific name?

Name: _____

True or false?

Every parallelogram has four 90° angles.

A. True

B. False

A triangle has angles of 45° , 45° , and 90° .

Name two categories of triangles it belongs to.

Categories: _____

FORWARD 2

Write and solve $x + p = q$ to find an unknown angle • Grade 6 • Expressions & Equations • 6.EE.B.7

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Two angles together make a 90° angle. One of them is 35° .

Write and solve an equation for the other angle, x .

Equation: _____

$x =$ _____

True or false?

If $x + 120 = 180$, then $x = 60$.

A. True

B. False

Solve for a.

$$a + 48 = 90$$

A. 42

B. 138

C. 48

D. 52

Solve for y .

$$y + 75 = 180$$

$$y = \underline{\hspace{10cm}}$$

True or false?

If $2x = 90$, then $x = 180$.

A. True

B. False

Sketch angles of a given measure using a protractor

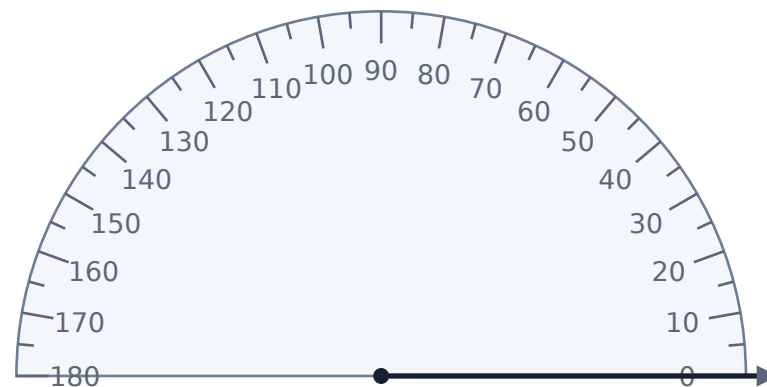
MAIN	Grade 4 main question	4.MD.C.6
BACKWARD 1	Draw shapes with given attributes	2.G.A.1
BACKWARD 2	Defining attributes of shapes	1.G.A.1
FORWARD 1	Graph points in the first quadrant to solve problems	5.G.A.2
FORWARD 2	Draw polygons in the coordinate plane	6.G.A.3

SET 55 • 4.MD.C.6

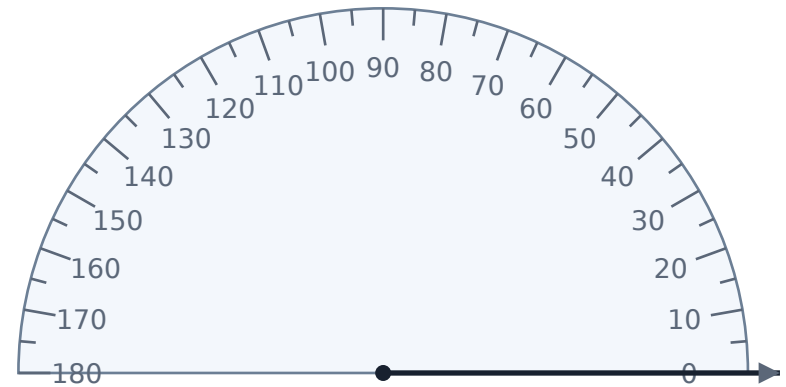
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.6

Use the protractor to draw an angle of 50° . One ray is drawn for you.

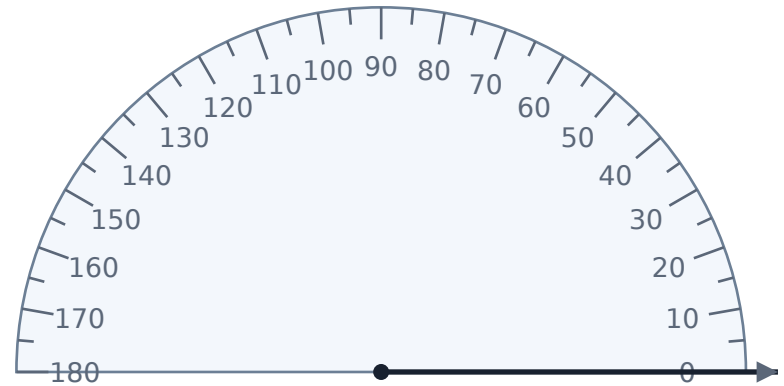


Use the protractor to draw an angle of 120° . One ray is drawn for you.



Use the protractor to draw an angle of 90° . One ray is drawn for you.

What kind of angle is it?

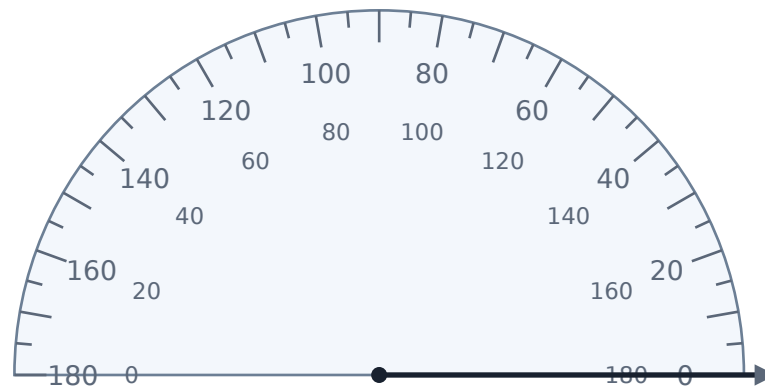


Kind of angle: _____

Kim wants to draw a 70° angle. Her first ray points to 0 on the outer scale.

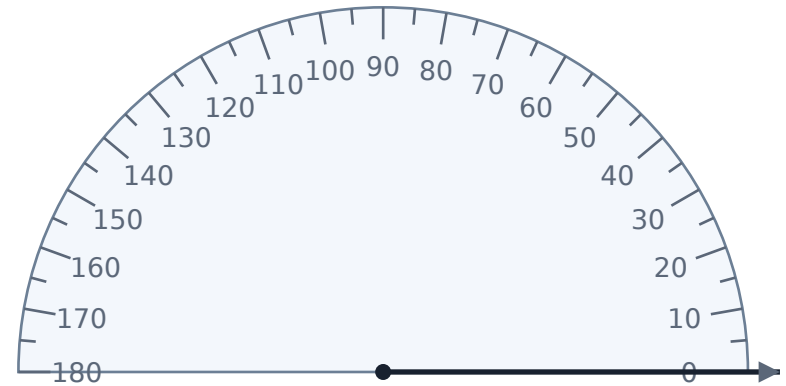
Where should she draw the second ray?

- A.** Through 70 on the inner scale
- B.** Through 0 on the inner scale
- C.** Through 110 on the outer scale
- D.** Through 70 on the outer scale



Use the protractor to draw an angle of 35° . One ray is drawn for you.

Is the angle acute or obtuse?



Answer: _____

SET 55 • 4.MD.C.6

BACKWARD 1

Draw shapes with given attributes • Grade 2 • Geometry • 2.G.A.1

A shape has 4 sides and 4 angles.

Name a shape it could be.

Shape: _____

True or false?

A triangle has 3 sides and 3 angles.

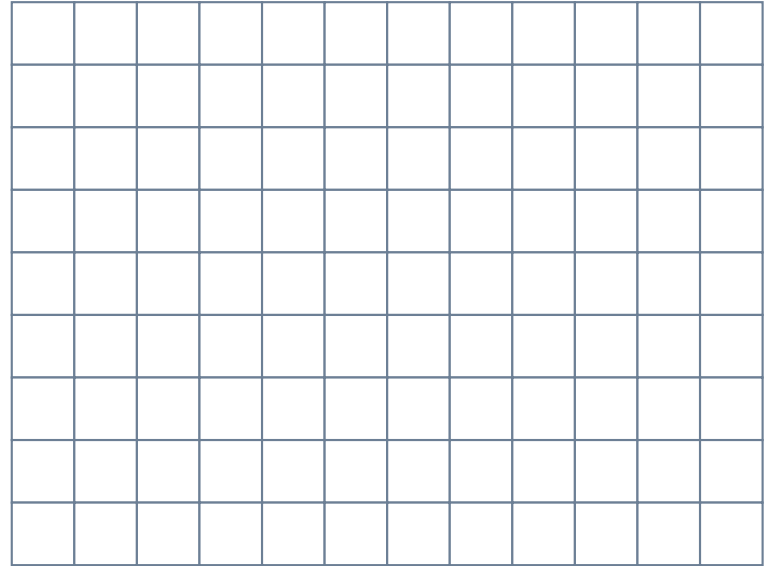
A. True

B. False

Which shape has 5 sides?

- A.** Triangle
- B.** Hexagon
- C.** Pentagon
- D.** Square

Draw a shape that has 3 sides.



True or false?

A shape with 6 angles has 4 sides.

A. True

B. False

SET 55 • 4.MD.C.6

BACKWARD 2

Defining attributes of shapes • Grade 1 • Geometry • 1.G.A.1

True or false?

A triangle can be any color or size and still be a triangle.

A. True

B. False

Which is a defining attribute of a square?

- A.** It is turned on its corner.
- B.** It is blue.
- C.** It is big.
- D.** It has 4 equal sides.

How many corners (vertices) does a rectangle have?

Corners: _____

True or false?

A shape must be large to be a hexagon.

A. True

B. False

A closed shape has 3 straight sides.

What is it called?

Name: _____

SET 55 • 4.MD.C.6

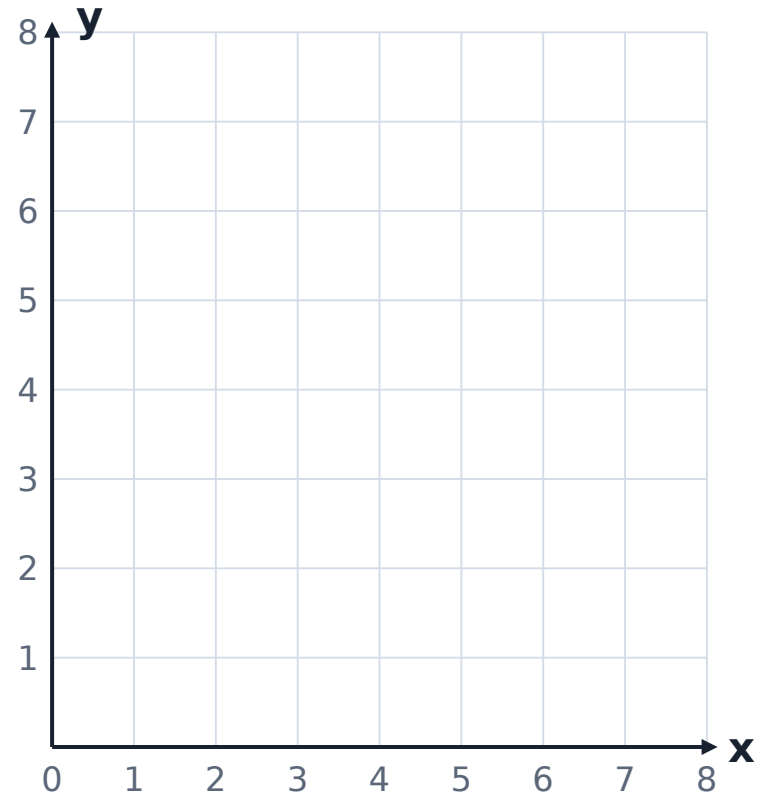
FORWARD 1

Graph points in the first quadrant to solve problems • Grade 5 • Geometry •
5.G.A.2

On a map grid, Ana's house is at (1, 2) and the park is at (1, 7). Each unit is 1 block.

Plot and label both points.
How many blocks apart are they?

Blocks: _____



True or false?

A plant is 4 cm tall on day 0 and grows 2 cm each day. The point (3, 10) shows its height on day 3.

A. True

B. False

A tank holds 2 liters at minute 0 and fills 3 liters each minute.

Which point (minutes, liters) shows the amount after 4 minutes?

A. (4, 14)

B. (3, 4)

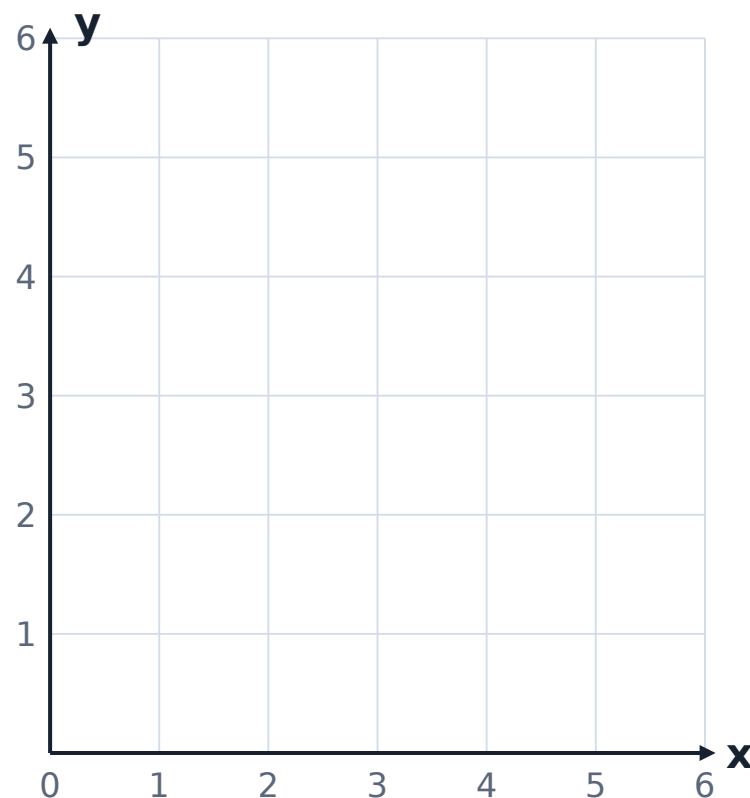
C. (4, 12)

D. (14, 4)

Plot $(1, 1)$, $(5, 1)$, and $(1, 4)$,
and connect them.

What kind of angle is at $(1, 1)$?

Kind of angle: _____



True or false?

On a map grid, a store is at (6, 3) and a bank is at (2, 3). They are 8 units apart.

A. True

B. False

SET 55 • 4.MD.C.6

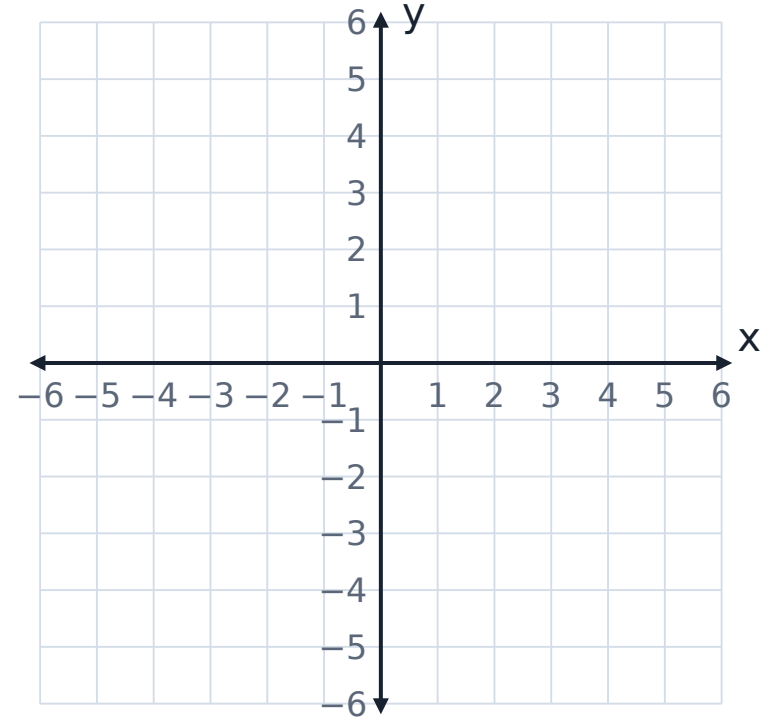
FORWARD 2

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(3, 2)$, $C(3, -2)$, and $D(-3, -2)$. Connect them in order.

What is the most specific name for the polygon?

Name: _____



What is the length of the side from $(-2, 4)$ to $(5, 4)$?

Length: _____

Three vertices of a square are $(0, 0)$, $(4, 0)$, and $(4, 4)$.

What is the fourth vertex?

A. $(8, 4)$

B. $(0, 8)$

C. $(4, 8)$

D. $(0, 4)$

True or false?

The points $(1, 1)$, $(1, 5)$, and $(4, 1)$ are the vertices of a right triangle.

- A. True
- B. False

True or false?

The side from $(-3, -1)$ to $(2, -1)$ is 1 unit long.

A. True

B. False

Angle measure is additive

MAIN	Grade 4 main question	4.MD.C.7
BACKWARD 1	Add within 100	2.NBT.B.5
BACKWARD 2	Area is additive	3.MD.C.7.d
FORWARD 1	Volume is additive — nearest related (no direct Grade 5 step)	5.MD.C.5.c
FORWARD 2	Write expressions and equations for parts of an angle — nearest...	6.EE.B.6

SET 56 • 4.MD.C.7

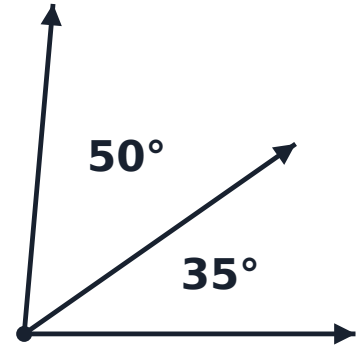
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles of 35° and 50° .

What is the measure of angle ABC?

Measure: _____



True or false?

A 90° angle is split into two angles. One is 40° , so the other is 50° .

A. True

B. False

Angles of 65° and 25° are side by side with no overlap.
What angle do they make together?

- A.** 90°
- B.** 100°
- C.** 80°
- D.** 40°

Angles of 30° , 45° , and 60° are side by side with no overlap.

What angle do they make together?

Measure: _____

True or false?

An angle of 120° can be split into angles of 70° and 60° .

A. True

B. False

SET 56 • 4.MD.C.7

BACKWARD 1

Add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$35 + 50$$

Sum: _____

True or false?

$$65 + 25 = 90$$

A. True

B. False

$$45 + 38 = ?$$

A. 713

B. 93

C. 73

D. 83

Add.

$$27 + 46$$

Sum: _____

True or false?

$$58 + 34 = 82$$

A. True

B. False

SET 56 • 4.MD.C.7

BACKWARD 2

Area is additive • Grade 3 • Measurement & Data • 3.MD.C.7.d

A figure is made of two rectangles that do not overlap.
Their areas are 12 and 15 square units.

What is the area of the figure?

Area: _____

True or false?

The area of a figure made of parts that do not overlap is the sum of the areas of the parts.

A. True

B. False

A figure is made of two rectangles that do not overlap.
Their areas are 12 and 6 square units.

What is the area of the figure?

- A.** 6 square units
- B.** 72 square units
- C.** 12 square units
- D.** 18 square units

Two squares that do not overlap have areas of 9 and 16 square units.

What is their total area?

Area: _____

True or false?

Two rectangles of 10 and 6 square units that do not overlap have a total area of 60 square units.

- A. True
- B. False

FORWARD 1

Volume is additive • Grade 5 • Measurement & Data • 5.MD.C.5.c

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Two prisms that do not overlap have volumes of 24 and 36 cubic units.

What is their total volume?

Volume: _____

True or false?

A figure made of a 2-by-3-by-4 prism and a 1-by-3-by-4 prism has a volume of 36 cubic units.

- A.** True
- B.** False

A figure is made of a 5-by-2-by-2 prism and a 3-by-2-by-1 prism that do not overlap.

What is its volume?

- A.** 26 cubic units
- B.** 20 cubic units
- C.** 120 cubic units
- D.** 6 cubic units

A building is a 9-by-6-by-5 prism with a 3-by-6-by-2 prism attached.

What is its total volume?

Volume: _____

True or false?

The volume of two prisms that touch is the product of their volumes.

- A. True
- B. False

FORWARD 2

Write expressions and equations for parts of an angle • Grade 6 • Expressions & Equations • 6.EE.B.6

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

An angle is split into two parts, x° and 40° .

Write an expression for the measure of the whole angle.

Expression: _____

True or false?

A 90° angle is split into x° and 25° , so
 $x + 25 = 90$.

A. True

B. False

Two equal angles of n° make a straight angle (180°).

Which equation is true?

A. $n + 180 = 2$

B. $2 + n = 180$

C. $n = 180$

D. $2n = 180$

Three equal angles of a° make a full turn (360°).

Write an equation.

Equation: _____

True or false?

An angle made of parts of y° and 30° measures $30y^\circ$.

A. True

B. False

Solve for unknown angles in diagrams and real-world problems with an equation

MAIN	Grade 4 main question	4.MD.C.7
BACKWARD 1	Solve one-step word problems with an unknown in any position	2.OA.A.1
BACKWARD 2	Find an unknown in an equation within 20	1.OA.D.8
FORWARD 1	Evaluate expressions with grouping symbols	5.OA.A.1
FORWARD 2	Write and solve one-step equations for unknown angles — neares...	6.EE.B.7

SET 57 • 4.MD.C.7

MAIN

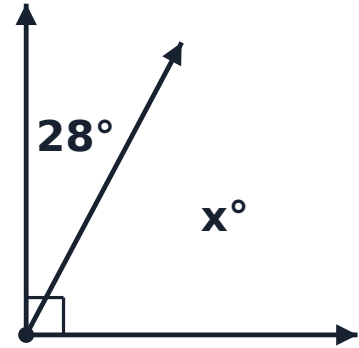
Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.7

The whole angle is a right angle (90°).

Write an equation with a letter for the unknown part.
Then solve it.

Equation: _____

Unknown angle: _____

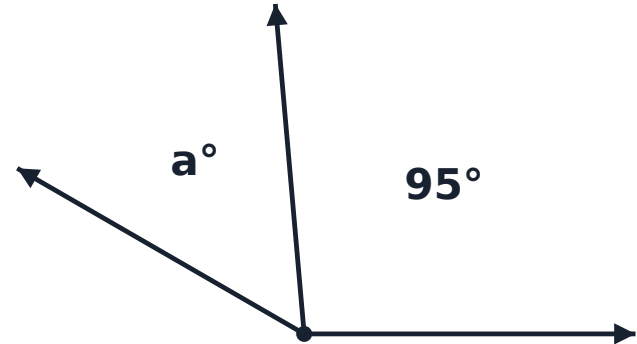


The whole angle is 150° .

Write an equation with a letter for the unknown part.
Then solve it.

Equation: _____

Unknown angle: _____



A straight angle (180°) is split into angles of 110° and m° .

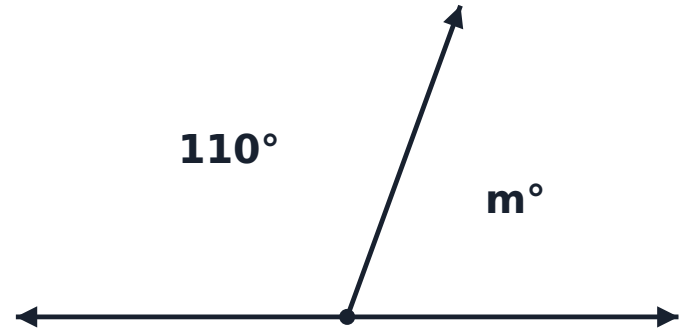
Which equation can be used to find m ?

A. $m - 110$
 $= 180$

B. $110 + m$
 $= 180$

C. $110 + m$
 $= 90$

D. $m = 110$
 $+ 180$



True or false?

A door opens 35° and then 40° more. It has turned through 75° in all.

- A. True
- B. False

A skater turns 90° and then 135° in the same direction.
How many degrees did she turn in all? How many more degrees would make a full turn (360°)?

Total turn: _____

More to a full turn: _____

SET 57 • 4.MD.C.7

BACKWARD 1

Solve one-step word problems with an unknown in any position • Grade 2 •
Operations & Algebraic Thinking • 2.OA.A.1

A box had some crayons. After 28 more crayons were put in, it had 90 crayons.

How many crayons were in the box at first?

Crayons: _____

True or false?

Kim had 45 stickers. She got some more, and now she has 80. She got 35 stickers.

A. True

B. False

Some birds were on a fence. 37 more birds came, and then there were 60 birds.

How many birds were on the fence at first?

A. 97

B. 27

C. 33

D. 23

Ty had 100 cards. He gave some away and has 64 left.
How many cards did he give away?

Cards: _____

True or false?

A shelf had some books. 19 more books were added, making 50 books. There were 41 books at first.

A. True

B. False

SET 57 • 4.MD.C.7

BACKWARD 2

Find an unknown in an equation within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$8 + ? = 15$$

Unknown: _____

True or false?

If $6 + n = 13$, then $n = 7$.

A. True

B. False

$$? - 4 = 9$$

A. 5

B. 13

C. 12

D. 4

Find the unknown number.

$$14 - ? = 6$$

Unknown: _____

True or false?

If $9 + n = 17$, then $n = 9$.

A. True

B. False

SET 57 • 4.MD.C.7

FORWARD 1

Evaluate expressions with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$180 - (45 + 60)$$

Value: _____

True or false?

$$360 - (90 + 90 + 120) = 60$$

A. True

B. False

Evaluate.

$$90 - (12 + 33)$$

A. 45

B. 111

C. 35

D. 69

Evaluate.

$$[360 - (100 + 80)] \div 2$$

Value: _____

True or false?

$$180 - 50 + 30 = 100$$

A. True

B. False

FORWARD 2

Write and solve one-step equations for unknown angles • Grade 6 •
Expressions & Equations • 6.EE.B.7

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Two equal angles together make 130° .

Write and solve an equation for each angle, x .

Equation: _____

$x =$ _____

True or false?

If 4 equal angles make a full turn of 360° , each angle is 90° .

A. True

B. False

Solve for x.

$$x + 72 = 180$$

A. 98

B. 108

C. 118

D. 252

Five equal angles meet at a point and make 360° .

Write and solve an equation for each angle, a .

Equation: _____

$a =$ _____

True or false?

If $x + 45 = 90$, then $x = 135$.

A. True

B. False

Identify points, lines, line segments, and rays

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Shapes with straight sides	1.G.A.1
BACKWARD 2	Measure the length of a segment with a ruler	2.MD.A.1
FORWARD 1	Understand the coordinate plane: perpendicular axes and the orig...	5.G.A.1
FORWARD 2	Name and plot points in all four quadrants	6.NS.C.6.c

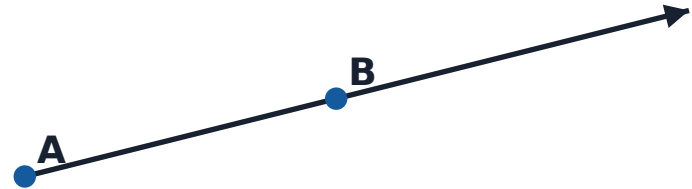
SET 58 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

What is the figure called?

- A.** Point
- B.** Line segment
- C.** Line
- D.** Ray



What is the figure called?

- A.** Angle
- B.** Ray
- C.** Line segment
- D.** Line



True or false?

A line goes on without end in both directions.

A. True

B. False

What is the figure called?

- A.** Line
- B.** Line segment
- C.** Ray
- D.** Point



True or false?

A line segment has an arrow at each end.

A. True

B. False

SET 58 • 4.G.A.1

BACKWARD 1

Shapes with straight sides • Grade 1 • Geometry • 1.G.A.1

True or false?

A triangle has 3 straight sides.

A. True

B. False

Which shape has no straight sides?

- A.** Square
- B.** Triangle
- C.** Circle
- D.** Rectangle

How many straight sides does a rectangle have?

Sides: _____

True or false?

A closed shape with 4 straight sides is a circle.

A. True

B. False

A closed shape has 3 straight sides.

What is it called?

Name: _____

SET 58 • 4.G.A.1

BACKWARD 2

Measure the length of a segment with a ruler • Grade 2 • Measurement & Data
• 2.MD.A.1

A segment starts at 0 cm and ends at 8 cm on a ruler.

How long is it?

Length: _____

True or false?

You can measure the length of a line segment with a ruler.

- A.** True
- B.** False

Which tool is best for measuring the length of a segment?

- A.** A measuring cup
- B.** A clock
- C.** A scale
- D.** A ruler

A segment goes from 3 inches to 7 inches on a ruler.
How long is it?

Length: _____

True or false?

A segment from 2 cm to 6 cm on a ruler is 6 cm long.

- A.** True
- B.** False

SET 58 • 4.G.A.1

FORWARD 1

Understand the coordinate plane: perpendicular axes and the origin • Grade 5

• Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

Answer: _____

True or false?

The x-axis and the y-axis are perpendicular lines.

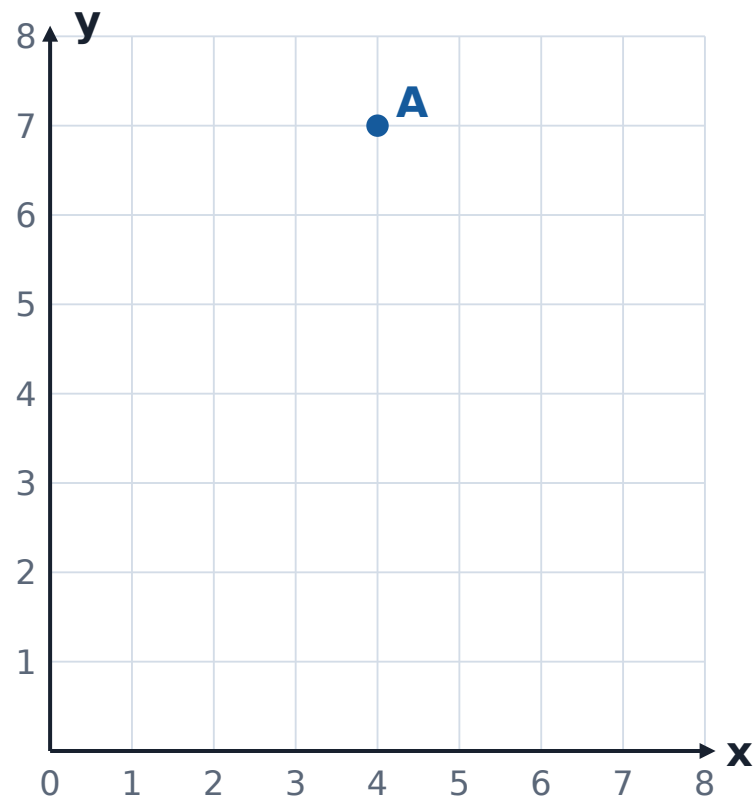
- A. True
- B. False

In the ordered pair (6, 2), what does the 6 tell you?

- A.** Move 6 units along the y-axis
- B.** The point is on the y-axis
- C.** Move 2 units to the right
- D.** Move 6 units along the x-axis

What are the coordinates of point A?

A = _____



True or false?

$(3, 5)$ and $(5, 3)$ name the same point.

A. True

B. False

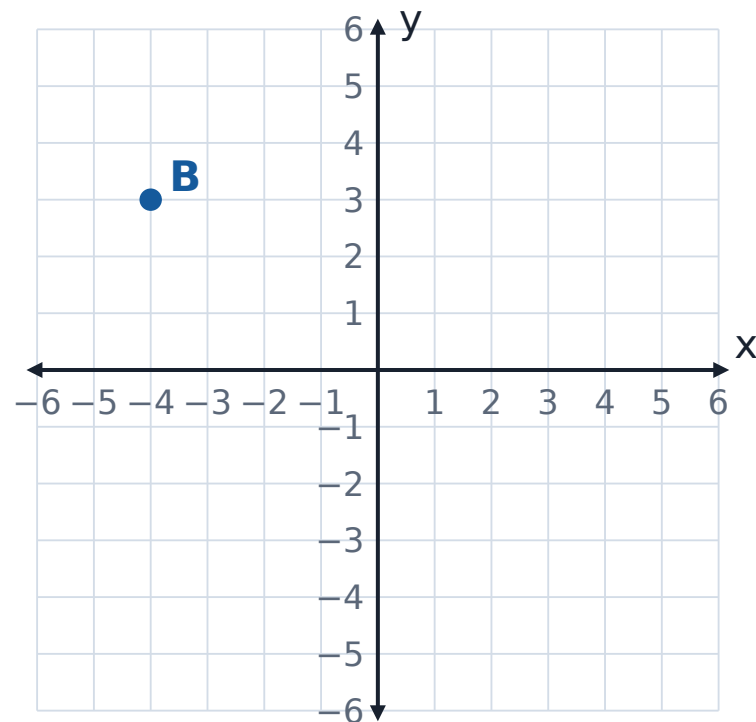
SET 58 • 4.G.A.1

FORWARD 2

Name and plot points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point B?

B = _____



True or false?

The point $(0, -5)$ is on the y -axis.

A. True

B. False

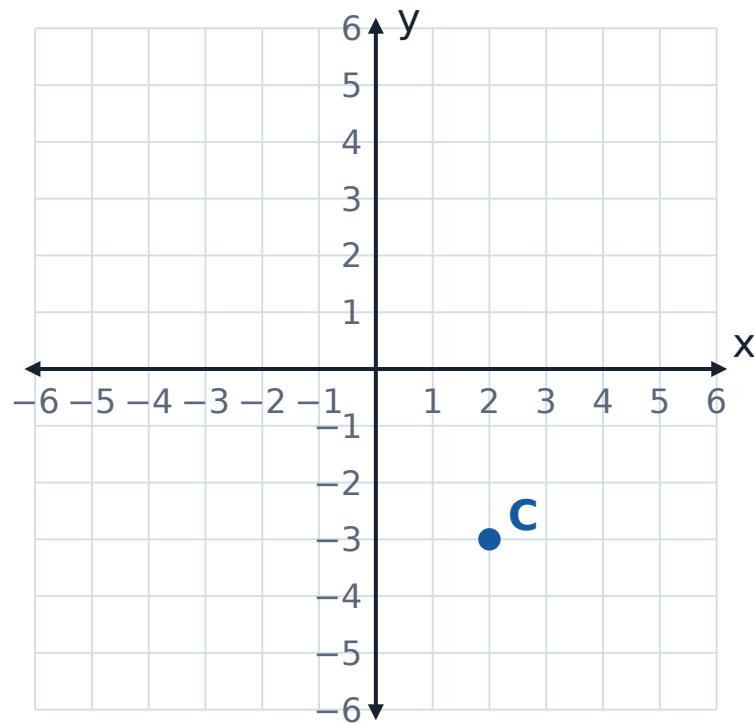
Which ordered pair names point C?

A. $(-3, 2)$

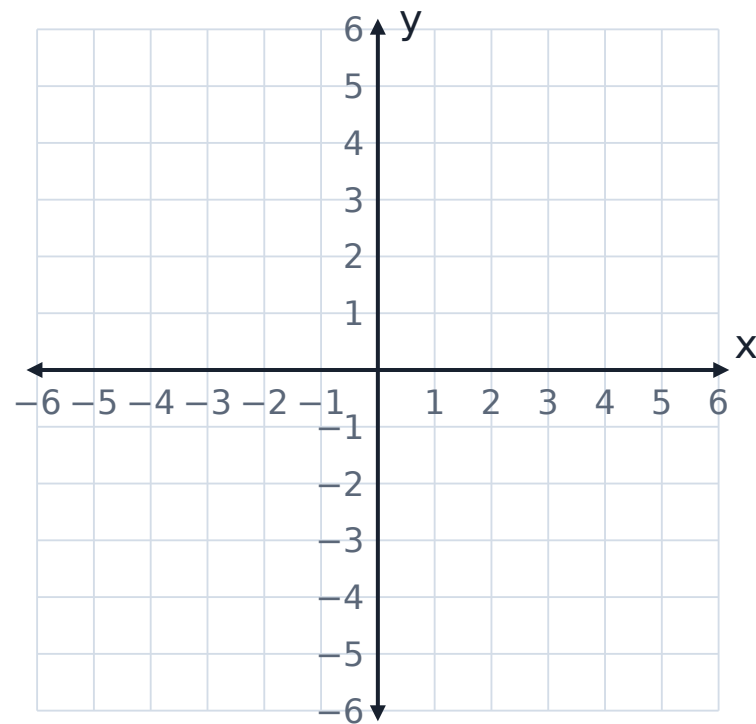
B. $(-2, 3)$

C. $(3, -2)$

D. $(2, -3)$



Plot and label the points
 $D(-2, -4)$ and $E(5, 1)$.



True or false?

The point $(-1, -1)$ is in Quadrant II.

A. True

B. False

Identify right, acute, and obtuse angles

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Shapes with a given number of angles	2.G.A.1
BACKWARD 2	Rectangles and squares have square corners	3.G.A.1
FORWARD 1	Classify triangles by their angles	5.G.B.4
FORWARD 2	Use the right angle of a right triangle to find its area — nearest re...	6.G.A.1

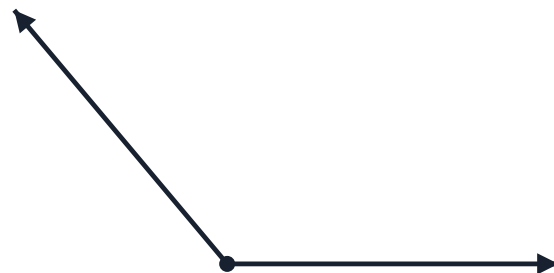
SET 59 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

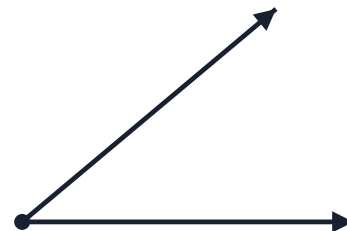
What kind of angle is shown?

- A.** Acute
- B.** Right
- C.** Obtuse
- D.** Straight



What kind of angle is shown?

- A.** Acute
- B.** Obtuse
- C.** Right
- D.** Straight



True or false?

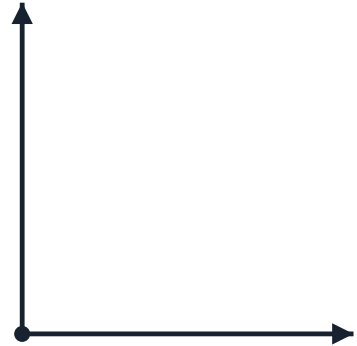
A right angle measures exactly 180° .

A. True

B. False

What kind of angle is shown?

- A.** Straight
- B.** Obtuse
- C.** Acute
- D.** Right



Classify each angle as acute, right, or obtuse.

15° 90° 165°

Answers: _____

SET 59 • 4.G.A.1

BACKWARD 1

Shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

How many angles does a hexagon have?

Angles: _____

True or false?

A quadrilateral has 4 angles.

A. True

B. False

Which shape has 3 angles?

- A.** Triangle
- B.** Pentagon
- C.** Square
- D.** Hexagon

How many angles does a rectangle have?

Angles: _____

True or false?

A pentagon has 6 angles.

A. True

B. False

SET 59 • 4.G.A.1

BACKWARD 2

Rectangles and squares have square corners • Grade 3 • Geometry • 3.G.A.1

True or false?

Every rectangle has 4 square corners.

A. True

B. False

Which shape always has 4 square corners?

- A.** Trapezoid
- B.** Triangle
- C.** Square
- D.** Rhombus

How many square corners does a square have?

Square corners: _____

True or false?

Every quadrilateral has 4 square corners.

A. True

B. False

Name a quadrilateral that has 4 square corners but is not a square.

Shape: _____

SET 59 • 4.G.A.1

FORWARD 1

Classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

A triangle has angles of 100° , 40° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which kind of triangle has three acute angles?

- A.** An obtuse triangle
- B.** An acute triangle
- C.** A straight triangle
- D.** A right triangle

A triangle has angles of 55° , 35° , and 90° .
Classify it by its angles.

Classification: _____

True or false?

An acute triangle can have one obtuse angle.

A. True

B. False

SET 59 • 4.G.A.1

FORWARD 2

Use the right angle of a right triangle to find its area • Grade 6 • Geometry • 6.G.A.1

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A right triangle has legs that are 6 cm and 10 cm long.
What is its area?

Area: _____

True or false?

In a right triangle, the two sides that form the right angle can be used as the base and the height.

- A.** True
- B.** False

An 8-by-5 rectangle is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 40 square units
- B.** 13 square units
- C.** 20 square units
- D.** 10 square units

A triangle has a base of 12 m and a height of 7 m.
What is its area?

Area: _____

True or false?

An obtuse triangle does not have a height.

A. True

B. False

Identify perpendicular and parallel lines

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 2	Quadrilaterals share attributes	3.G.A.1
BACKWARD 3	Square corners and sides that run the same way in a rectangle	3.G.A.1
FORWARD 1	Attributes of a category belong to all its subcategories	5.G.B.3
FORWARD 2	Use coordinates to find parallel and perpendicular sides of polygo...	6.G.A.3

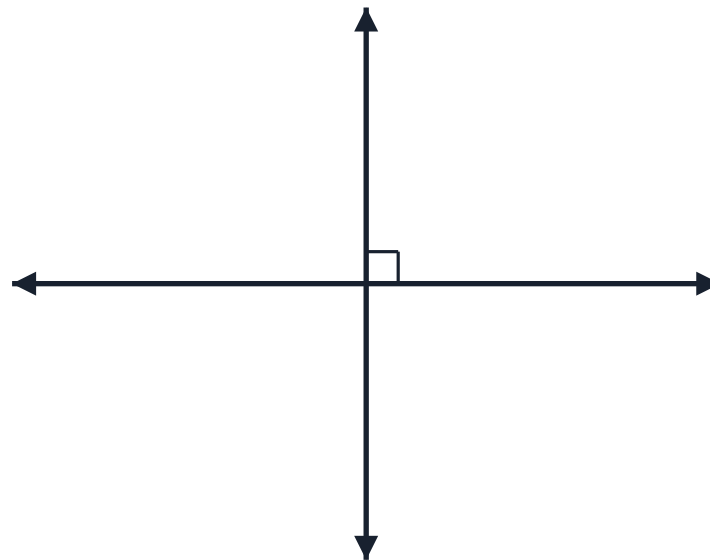
SET 60 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

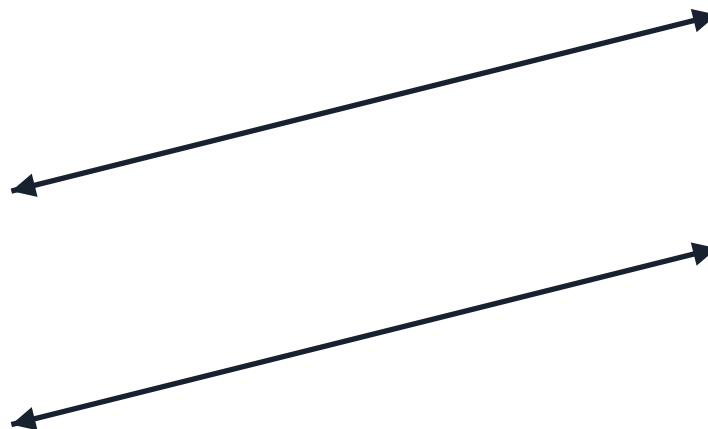
How are the two lines related?

- A.** Parallel
- B.** Perpendicular
- C.** Neither
- D.** They are the same line



How are the two lines related?

- A.** Parallel
- B.** Perpendicular
- C.** Neither
- D.** They are the same line



True or false?

Parallel lines never meet.

A. True

B. False

Lines a and b meet to form a right angle.
Are they parallel or perpendicular?

Answer: _____

True or false?

Perpendicular lines meet to form four obtuse angles.

A. True

B. False

SET 60 • 4.G.A.1

BACKWARD 2

Quadrilaterals share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

A rhombus, a rectangle, and a square are all quadrilaterals.

A. True

B. False

Which shape is a quadrilateral?

A. Pentagon

B. Triangle

C. Circle

D. Rhombus

How many sides does every quadrilateral have?

Sides: _____

True or false?

A triangle is a quadrilateral.

A. True

B. False

Name a quadrilateral that is not a rectangle.

Shape: _____

SET 60 • 4.G.A.1

BACKWARD 3

Square corners and sides that run the same way in a rectangle • Grade 3 •
Geometry • 3.G.A.1

True or false?

In a rectangle, the two sides that meet at a corner make a square corner.

A. True

B. False

In a rectangle, which sides run the same way and never meet?

- A.** All four sides
- B.** The two opposite sides
- C.** Two sides that meet at a corner
- D.** No sides

How many pairs of opposite sides that run the same way does a rectangle have?

Pairs: _____

True or false?

Two sides of a rectangle that meet at a corner run the same way.

A. True

B. False

Which shape has 4 square corners and opposite sides that run the same way?

- A.** Circle
- B.** Rectangle
- C.** Triangle
- D.** Hexagon

SET 60 • 4.G.A.1

FORWARD 1

Attributes of a category belong to all its subcategories • Grade 5 • Geometry •
5.G.B.3

True or false?

All parallelograms have two pairs of parallel sides.
A rectangle is a parallelogram, so every rectangle has two pairs of parallel sides.

A. True

B. False

Which shape must have two pairs of parallel sides?

- A.** Triangle
- B.** Kite
- C.** Trapezoid
- D.** Rhombus

Do the sides of every square meet at right angles?
Explain.

Answer: _____

Explanation: _____

True or false?

Every quadrilateral has at least one pair of parallel sides.

A. True

B. False

Which shape has exactly one pair of parallel sides?

- A.** A square
- B.** A trapezoid with only one pair of parallel sides
- C.** A rectangle
- D.** A rhombus

SET 60 • 4.G.A.1

FORWARD 2

Use coordinates to find parallel and perpendicular sides of polygons • Grade 6
• Geometry • 6.G.A.3

True or false?

The segment from $(1, 2)$ to $(5, 2)$ is parallel to the segment from $(1, -3)$ to $(5, -3)$.

A. True

B. False

What is the length of the side from $(-3, -1)$ to $(-3, 4)$?

Length: _____

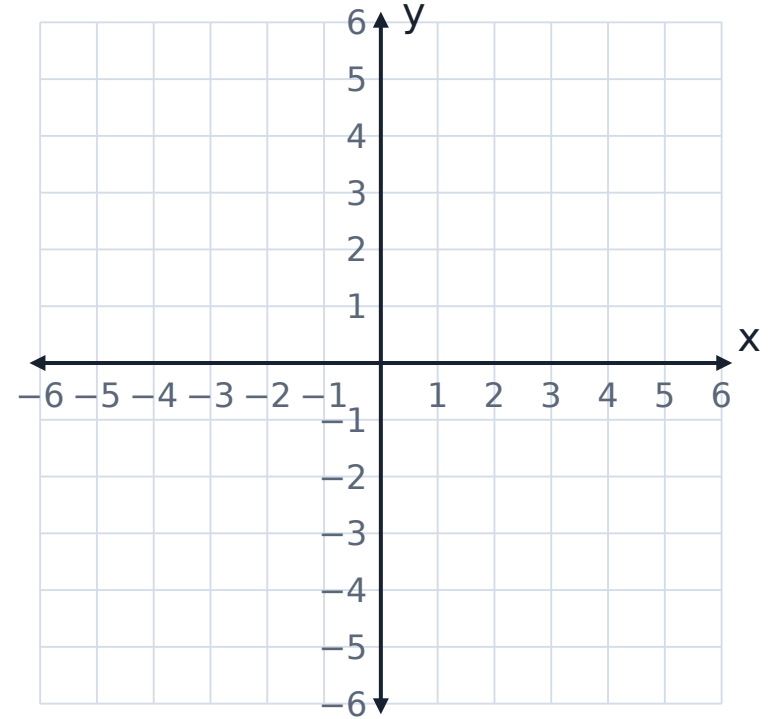
Which segment is perpendicular to the segment from $(0, 0)$ to $(6, 0)$?

- A.** From $(0, 1)$ to $(6, 1)$
- B.** From $(2, 0)$ to $(2, 5)$
- C.** From $(1, 1)$ to $(4, 4)$
- D.** From $(0, 0)$ to $(6, 3)$

Plot $(-2, 1)$, $(4, 1)$, $(4, -3)$,
and $(-2, -3)$. Connect them
in order.

How many pairs of parallel
sides does the polygon have?

Pairs: _____



True or false?

The segment from $(0, 1)$ to $(3, 1)$ is perpendicular to the segment from $(0, 4)$ to $(3, 4)$.

A. True

B. False

Draw points, lines, line segments, and rays

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Draw shapes with straight sides	2.G.A.1
BACKWARD 2	Measure the length of a drawn segment	2.MD.A.1
FORWARD 1	Locate and draw points and segments on a coordinate grid	5.G.A.1
FORWARD 2	Draw points and segments in all four quadrants	6.NS.C.6.c

SET 70 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

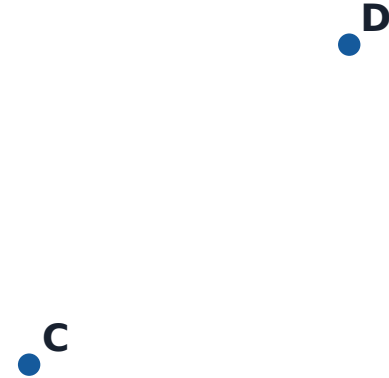
Draw ray PQ. It starts at point P and goes through point Q.



Draw line segment AB.



Draw line CD through points
C and D.



Draw a point that is not on line m , and label it E .



Draw ray ST. It starts at point S and goes through point T.

How is a ray different from a line segment?



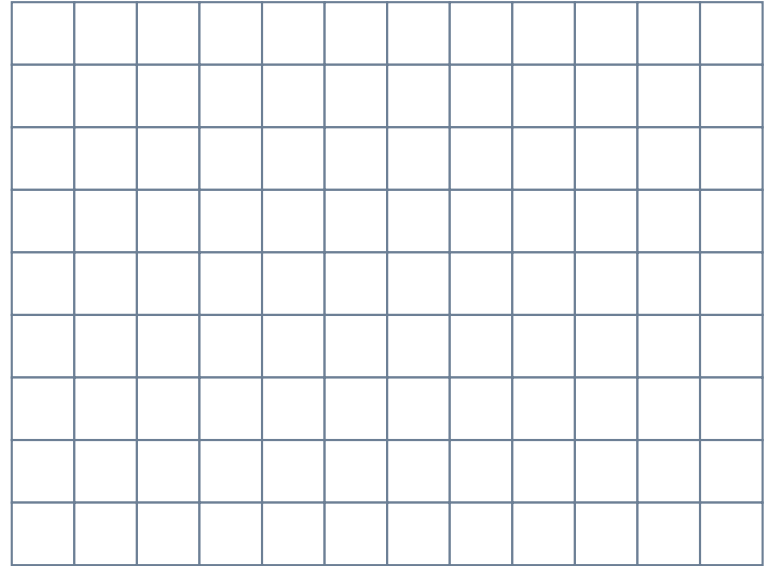
Difference: _____

SET 70 • 4.G.A.1

BACKWARD 1

Draw shapes with straight sides • Grade 2 • Geometry • 2.G.A.1

Draw a closed shape with 4 straight sides.



True or false?

A closed shape drawn with 3 straight sides that meet at 3 corners is a triangle.

A. True

B. False

Which shape has only straight sides?

- A.** Pentagon
- B.** Oval
- C.** Circle
- D.** Half-circle

How many straight sides do you draw to make a hexagon?

Sides: _____

True or false?

A circle is drawn with straight sides.

A. True

B. False

SET 70 • 4.G.A.1

BACKWARD 2

Measure the length of a drawn segment • Grade 2 • Measurement & Data •
2.MD.A.1

A segment starts at 0 and ends at 5 on a centimeter ruler.

How long is it?

Length: _____

True or false?

To draw a segment 6 cm long, you can start at 0 on the ruler and stop at 6.

- A.** True
- B.** False

A segment is drawn from 2 to 9 on an inch ruler.

How long is it?

- A.** 9 inches
- B.** 11 inches
- C.** 7 inches
- D.** 2 inches

Ben draws a segment from 0 to 4 on an inch ruler.
How long is his segment?

Length: _____

True or false?

A segment drawn from 1 cm to 5 cm on a ruler is 5 cm long.

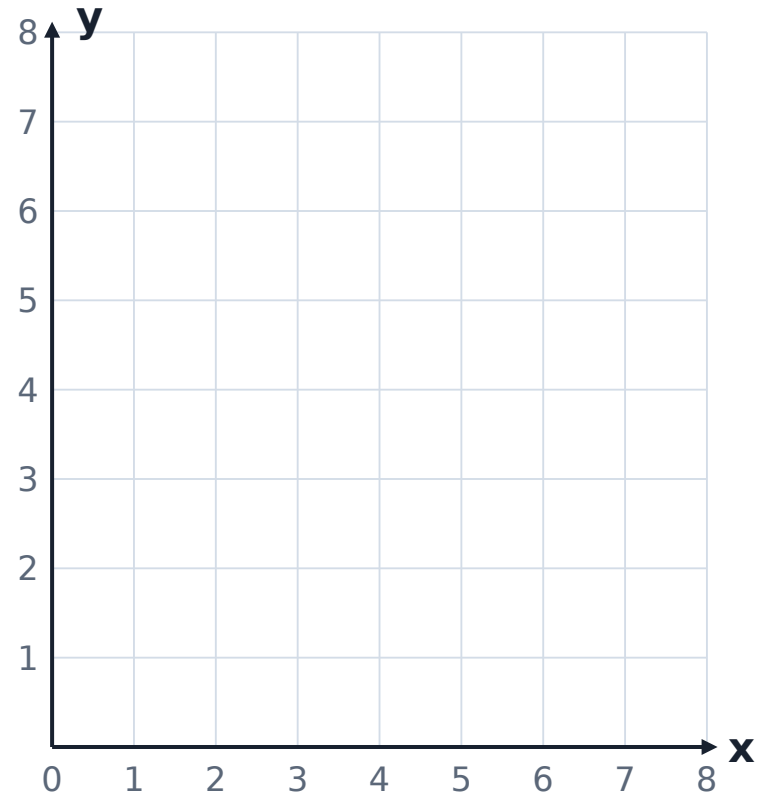
- A. True
- B. False

SET 70 • 4.G.A.1

FORWARD 1

Locate and draw points and segments on a coordinate grid • Grade 5 •
Geometry • 5.G.A.1

Plot the point (3, 5) and label it P.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 70 • FORWARD 1 • Q1/5

True or false?

To locate $(4, 2)$, start at the origin, move 4 units along the x-axis, then move 2 units up.

A. True

B. False

Which point is on the y-axis?

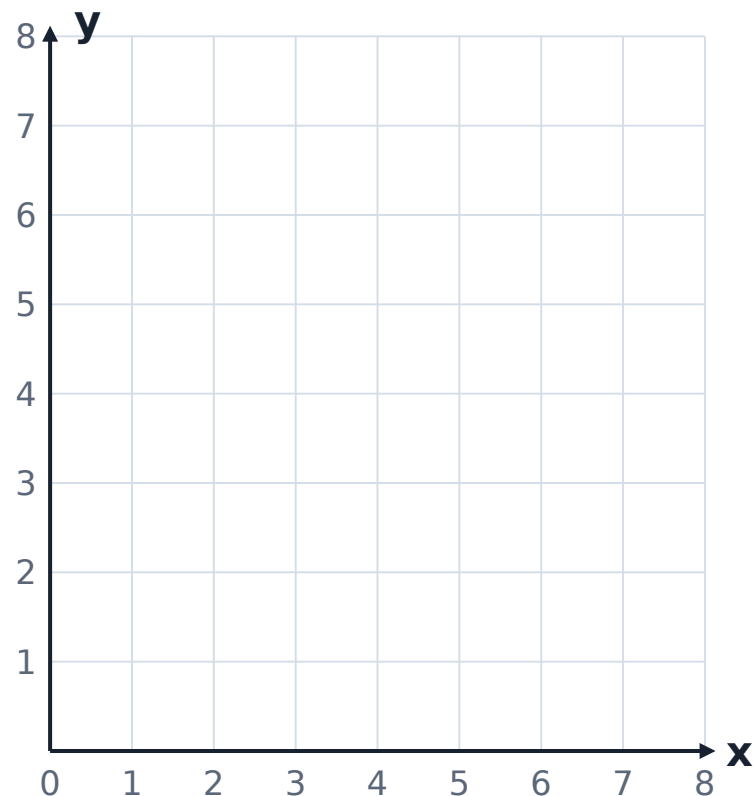
A. (1, 6)

B. (6, 0)

C. (6, 6)

D. (0, 6)

Draw the line segment from $(1, 1)$ to $(6, 1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 70 • FORWARD 1 • Q4/5

True or false?

The origin is the point (1, 1).

A. True

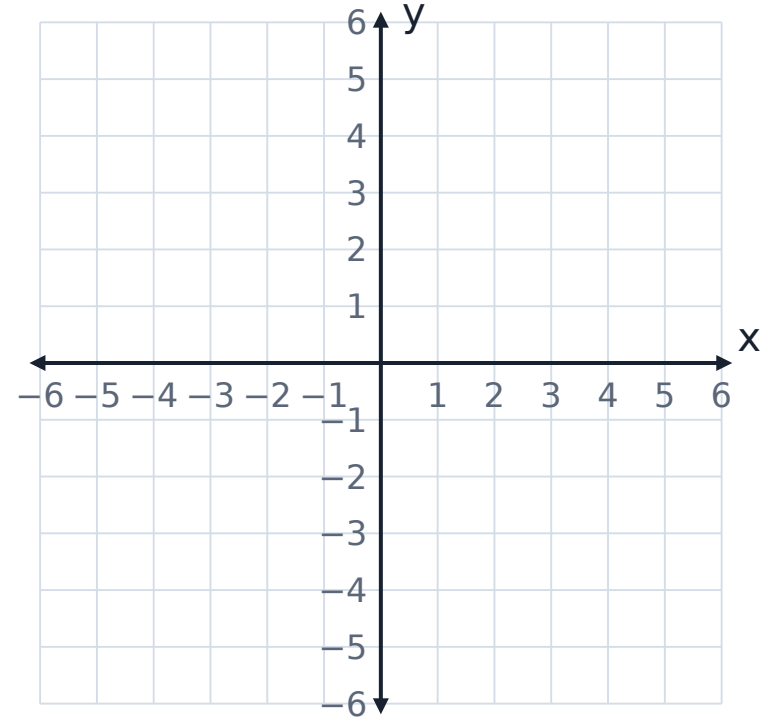
B. False

SET 70 • 4.G.A.1

FORWARD 2

Draw points and segments in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points
 $A(-3, 2)$ and $B(4, -1)$.



True or false?

The point $(-5, 0)$ is on the x-axis.

A. True

B. False

Which point is in Quadrant III?

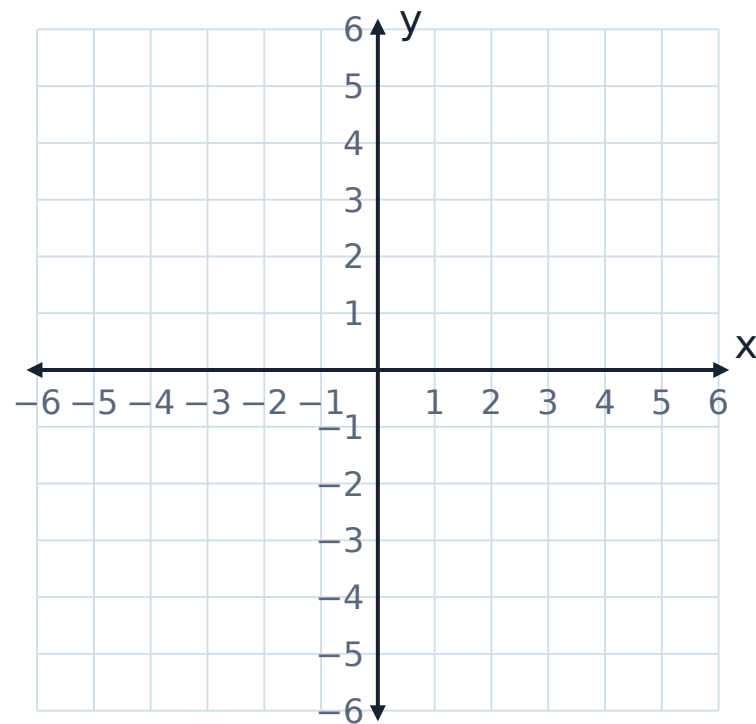
A. $(-2, 4)$

B. $(2, -4)$

C. $(-2, -4)$

D. $(2, 4)$

Draw the segment from $(-4, -3)$ to $(2, -3)$.



True or false?

The point $(3, -2)$ is in Quadrant II.

A. True

B. False

Draw right, acute, and obtuse angles

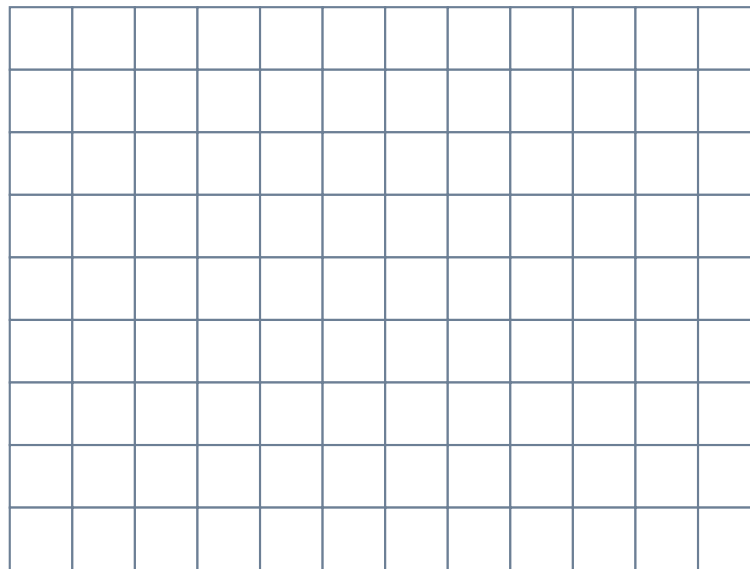
MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Square corners	3.G.A.1
BACKWARD 2	Draw shapes with a given number of angles	2.G.A.1
FORWARD 1	Draw and classify triangles by their angles	5.G.B.4
FORWARD 2	Draw polygons with right angles in the coordinate plane — neares...	6.G.A.3

SET 71 • 4.G.A.1

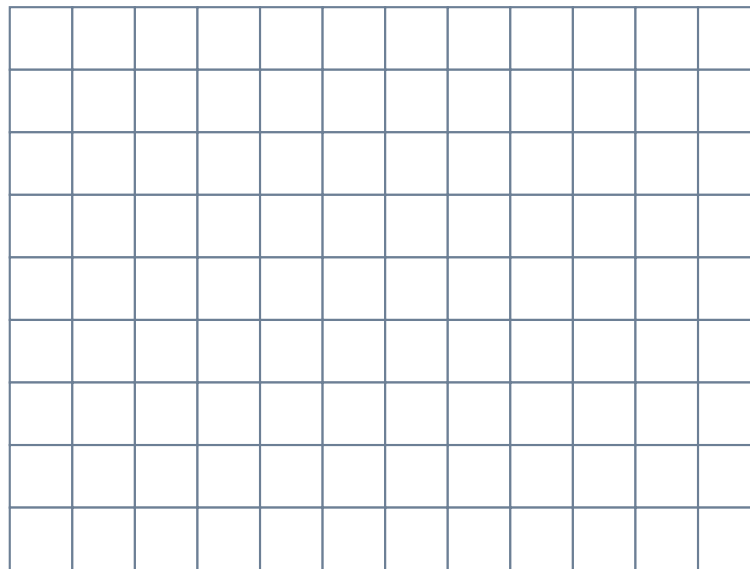
MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

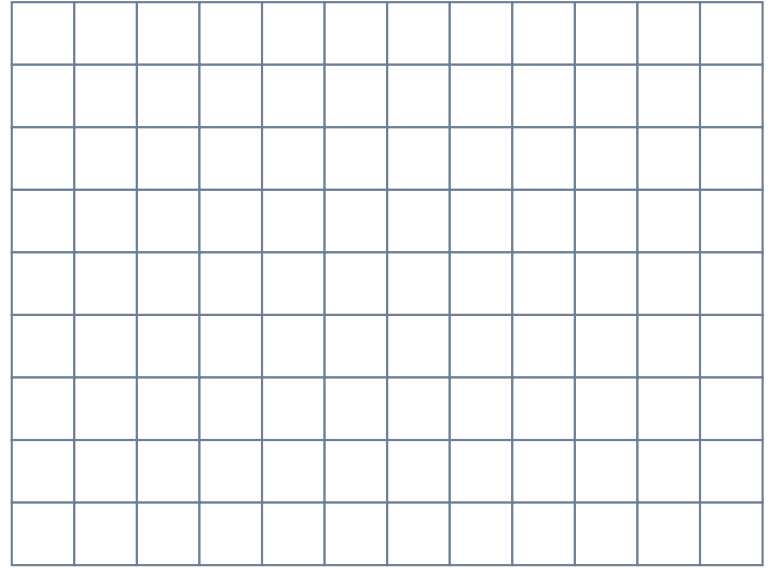
Draw an obtuse angle.



Draw an acute angle.



Draw a right angle. Mark it with a right-angle mark.



Point A is the vertex. Draw ray AB and ray AC so that angle BAC is obtuse.

Is angle BAC greater or less than a right angle?



Answer: _____

Draw an acute angle with its vertex at point V.

How do you know your angle is acute?



Explanation: _____

SET 71 • 4.G.A.1

BACKWARD 1

Square corners • Grade 3 • Geometry • 3.G.A.1

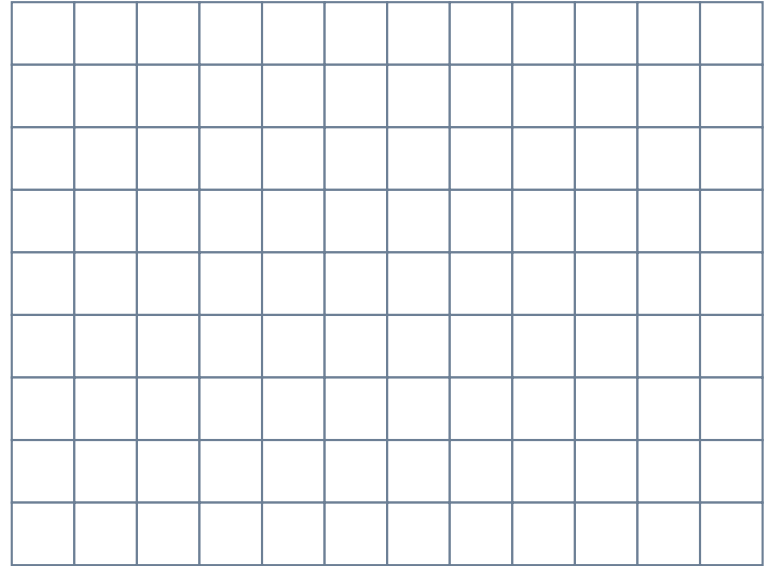
True or false?

The corner of a sheet of paper is a square corner.

A. True

B. False

Draw a shape that has 4 square corners.



Which shape always has 4 square corners?

- A.** Rectangle
- B.** Pentagon
- C.** Trapezoid
- D.** Rhombus

A rectangle is cut into 2 smaller rectangles by one straight cut from one side to the opposite side.

How many square corners does each smaller rectangle have?

Square corners: _____

True or false?

Every triangle has a square corner.

A. True

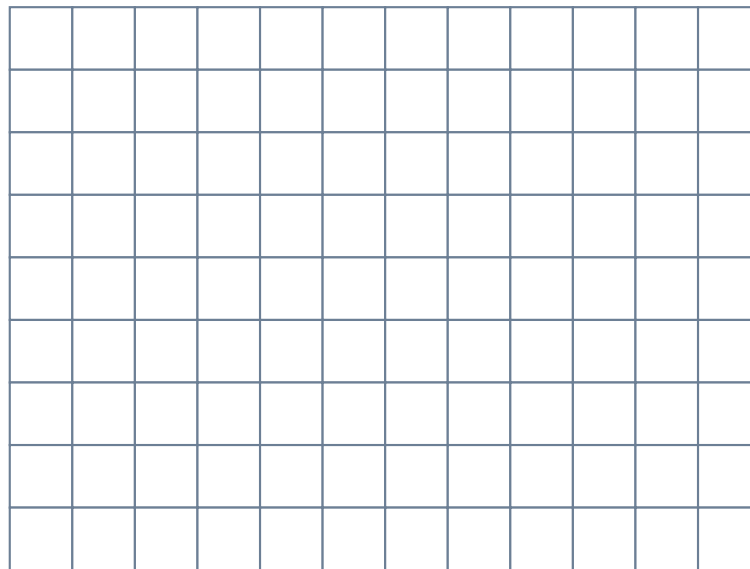
B. False

SET 71 • 4.G.A.1

BACKWARD 2

Draw shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

Draw a shape with 3 angles.



True or false?

A closed shape with 5 angles is a pentagon.

A. True

B. False

How many angles does a hexagon have?

A. 8

B. 4

C. 5

D. 6

How many angles do you draw when you draw a quadrilateral?

Angles: _____

True or false?

A closed shape with 4 angles has 3 sides.

A. True

B. False

SET 71 • 4.G.A.1

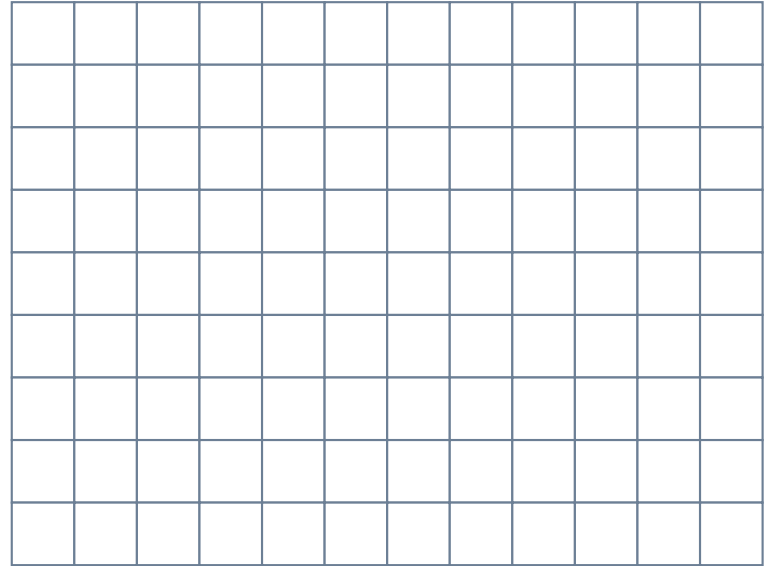
FORWARD 1

Draw and classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

Draw a triangle with one obtuse angle.

What kind of triangle is it?

Kind of triangle: _____



True or false?

A triangle can have one right angle and two acute angles.

A. True

B. False

A triangle has angles of 30° , 60° , and 90° .

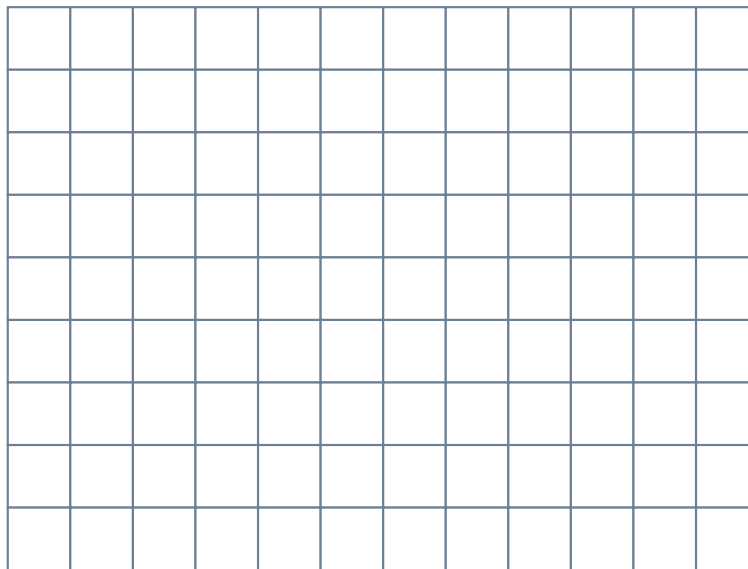
What kind of triangle is it?

- A.** Acute
- B.** Right
- C.** Equilateral
- D.** Obtuse

Draw a triangle with three acute angles.

What kind of triangle is it?

Kind of triangle: _____



True or false?

A triangle can have two obtuse angles.

A. True

B. False

FORWARD 2

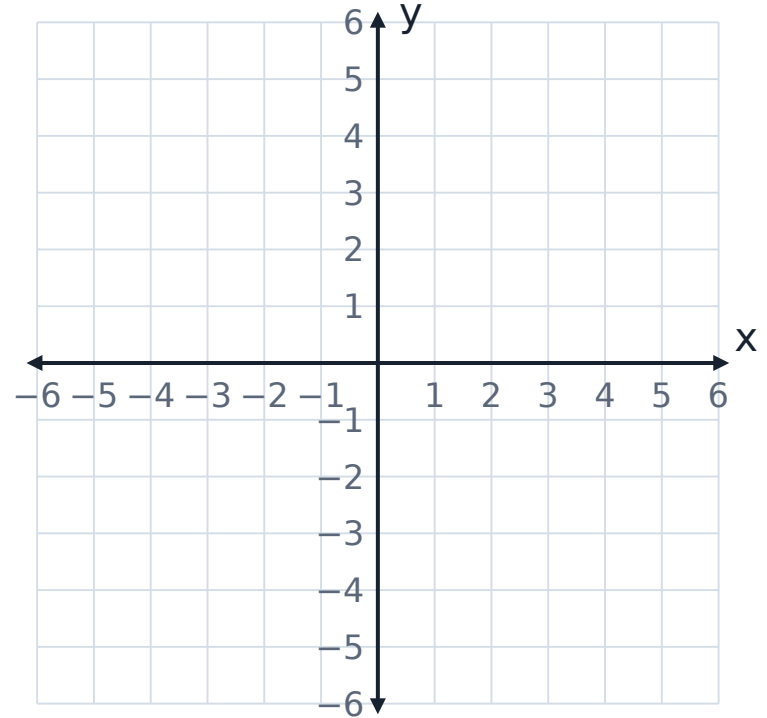
Draw polygons with right angles in the coordinate plane • Grade 6 • Geometry
• 6.G.A.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Plot $(-3, -2)$, $(-3, 3)$, and $(2, -2)$, and connect them.

What kind of angle is at $(-3, -2)$?

Kind of angle: _____



True or false?

The segments from $(0, 0)$ to $(0, 4)$ and from $(0, 0)$ to $(5, 0)$ form a right angle.

- A. True
- B. False

The segment from $(1, 1)$ to $(1, 5)$ is drawn.

Which point makes a right angle at $(1, 1)$ when it is joined to $(1, 1)$?

A. $(4, 4)$

B. $(4, 5)$

C. $(4, 1)$

D. $(0, 5)$

A rectangle has vertices $(-2, 1)$, $(3, 1)$, and $(3, -4)$.

What is the fourth vertex?

Vertex: _____

True or false?

The segments from $(0, 0)$ to $(3, 3)$ and from $(0, 0)$ to $(4, 0)$ form a right angle.

- A. True
- B. False

Draw perpendicular and parallel lines

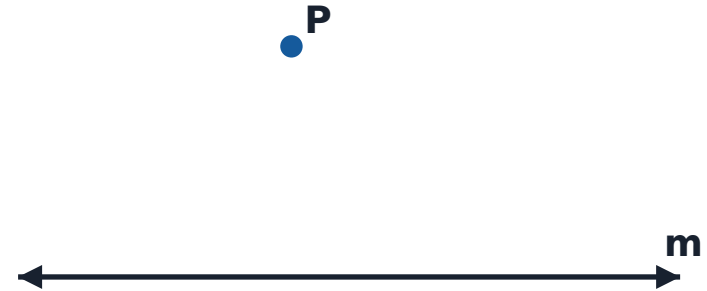
MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Draw shapes with square corners and opposite sides that run the...	3.G.A.1
BACKWARD 2	Draw lines to split a rectangle into rows and columns of squares	2.G.A.2
FORWARD 1	Draw quadrilaterals with given parallel and perpendicular sides	5.G.B.3
FORWARD 2	Draw parallel and perpendicular segments in the coordinate plane	6.G.A.3

SET 72 • 4.G.A.1

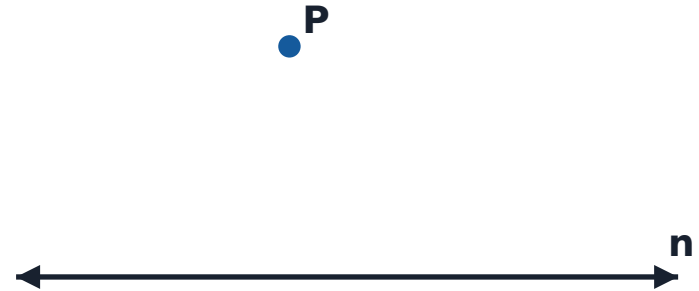
MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

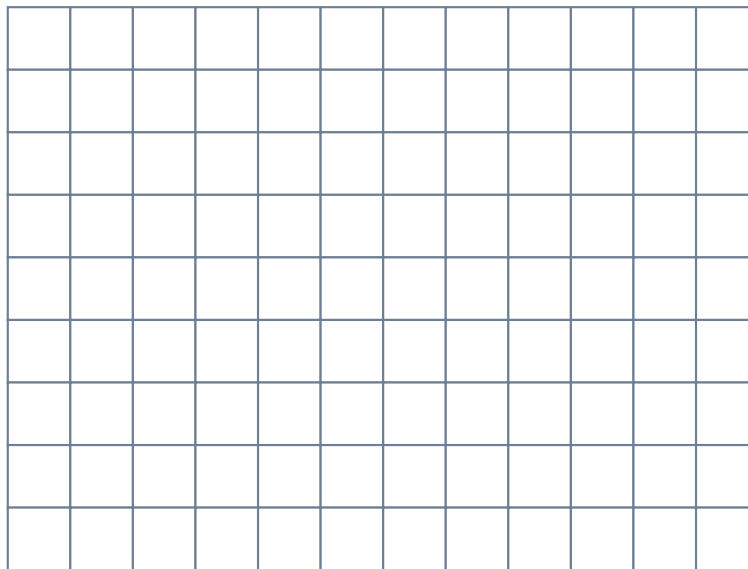
Draw a line through point P
that is parallel to line m.



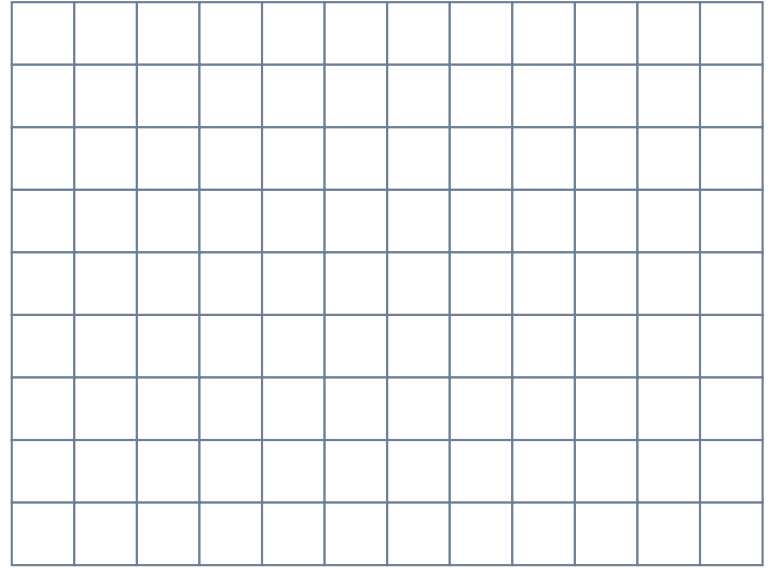
Draw a line through point P
that is perpendicular to line
n.



Draw two lines that are perpendicular to each other. Mark one of the right angles.

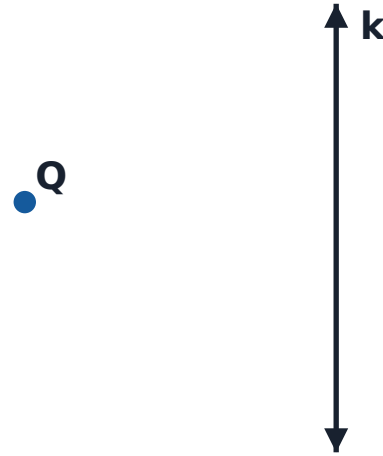


Draw two parallel line segments that are each 4 units long.



Draw a line through point Q
that is perpendicular to line k.

How many right angles do
the two lines make where
they cross?



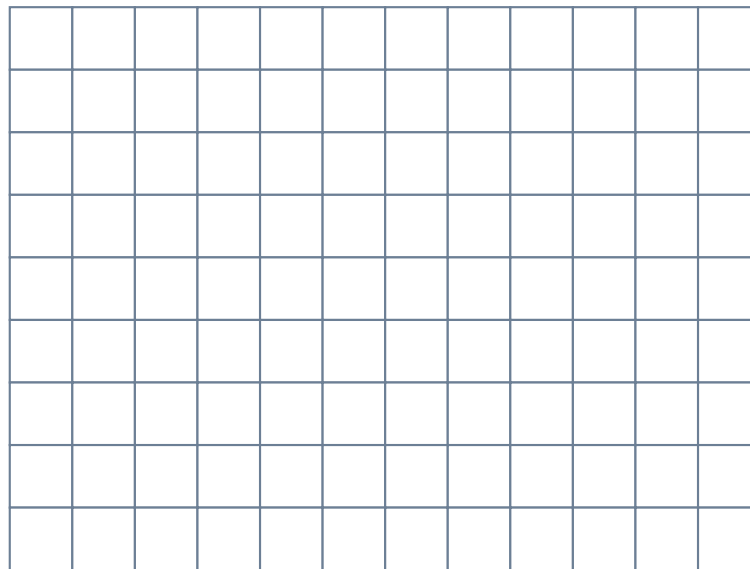
Right angles: _____

SET 72 • 4.G.A.1

BACKWARD 1

Draw shapes with square corners and opposite sides that run the same way •
Grade 3 • Geometry • 3.G.A.1

Draw a rectangle on the grid.



True or false?

Every rectangle has 2 pairs of opposite sides that run the same way.

A. True

B. False

Ann draws a shape with 4 square corners and 4 equal sides.

What shape did she draw?

A. Triangle

B. Trapezoid

C. Hexagon

D. Square

How many square corners must you draw to make a rectangle?

Square corners: _____

True or false?

A shape drawn with 4 square corners can have 5 sides.

- A.** True
- B.** False

SET 72 • 4.G.A.1

BACKWARD 2

Draw lines to split a rectangle into rows and columns of squares • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the rectangle into 2 rows of 3 same-size squares.



True or false?

To split a rectangle into 3 rows, you can draw 2 lines across it that run the same way.

- A.** True
- B.** False

A rectangle is split into 2 rows of 4 same-size squares.
How many squares are there?

- A.** 4
- B.** 8
- C.** 2
- D.** 6

How many lines across do you draw to split a rectangle into 4 rows?

Lines: _____

True or false?

A rectangle split into 2 rows and 2 columns of squares has 2 squares.

- A. True
- B. False

SET 72 • 4.G.A.1

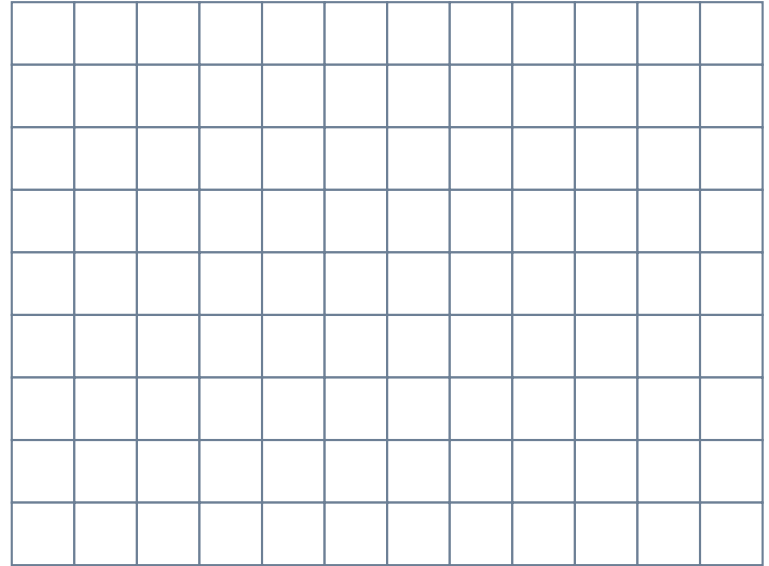
FORWARD 1

Draw quadrilaterals with given parallel and perpendicular sides • Grade 5 •
Geometry • 5.G.B.3

Draw a quadrilateral with two pairs of parallel sides and no right angles.

What is it called?

Name: _____



True or false?

Every square has two pairs of parallel sides and four right angles.

A. True

B. False

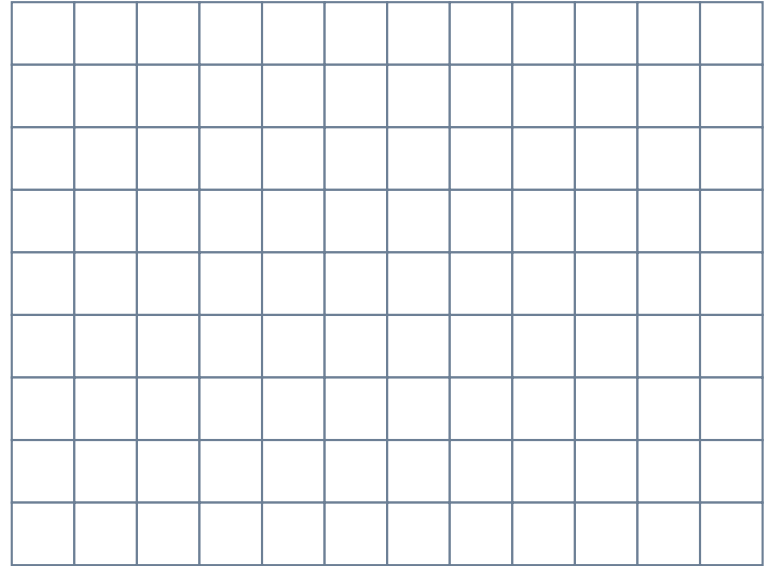
Which shape must have perpendicular sides?

- A.** Rectangle
- B.** Trapezoid
- C.** Parallelogram
- D.** Rhombus

Draw a trapezoid with exactly one pair of parallel sides and two right angles.

How many pairs of perpendicular sides does it have?

Pairs: _____



True or false?

Every parallelogram has perpendicular sides.

A. True

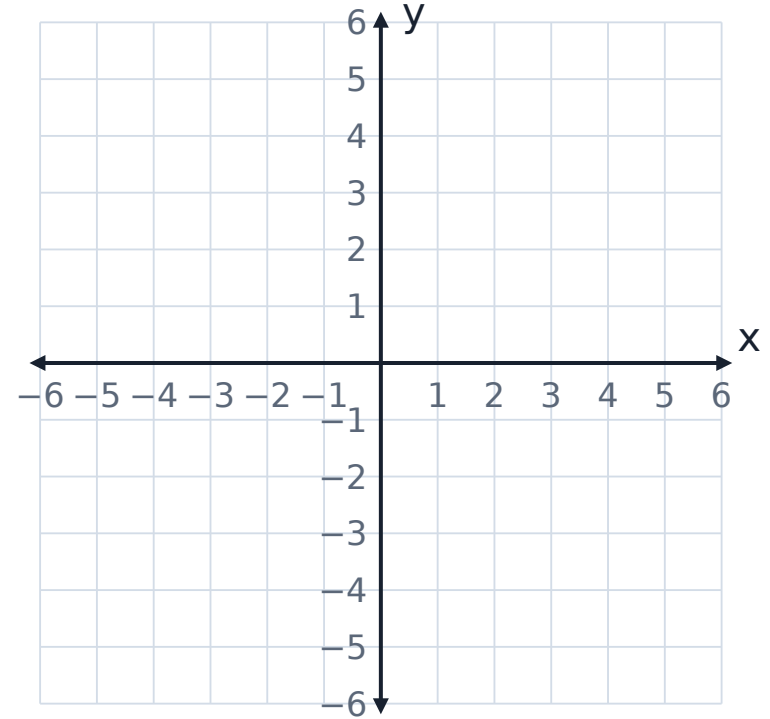
B. False

SET 72 • 4.G.A.1

FORWARD 2

Draw parallel and perpendicular segments in the coordinate plane • Grade 6 •
Geometry • 6.G.A.3

Draw the segment from $(-3, 2)$ to $(3, 2)$. Then draw a segment parallel to it that starts at $(-3, -2)$.



True or false?

The segment from $(2, -4)$ to $(2, 3)$ is perpendicular to the x-axis.

- A. True
- B. False

Which segment is parallel to the segment from $(-4, 1)$ to $(4, 1)$?

- A.** From $(-4, 1)$ to $(4, 5)$
- B.** From $(1, -4)$ to $(1, 4)$
- C.** From $(0, 0)$ to $(4, 4)$
- D.** From $(-4, -3)$ to $(4, -3)$

A segment goes from $(-1, -2)$ to $(-1, 5)$. A segment perpendicular to it starts at $(-1, 0)$.

Give one possible other endpoint.

Endpoint: _____

True or false?

The segment from $(0, 2)$ to $(5, 2)$ is perpendicular to the segment from $(0, -1)$ to $(5, -1)$.

- A. True
- B. False

Identify segments, angles, and perpendicular and parallel sides in two-dimensional figures

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Count sides and angles of shapes	2.G.A.1
BACKWARD 2	Find square corners in shapes	3.G.A.1
FORWARD 1	Use the attributes of a category to find sides and angles in figures	5.G.B.3
FORWARD 2	Identify parallel and perpendicular sides of polygons in the coordi...	6.G.A.3

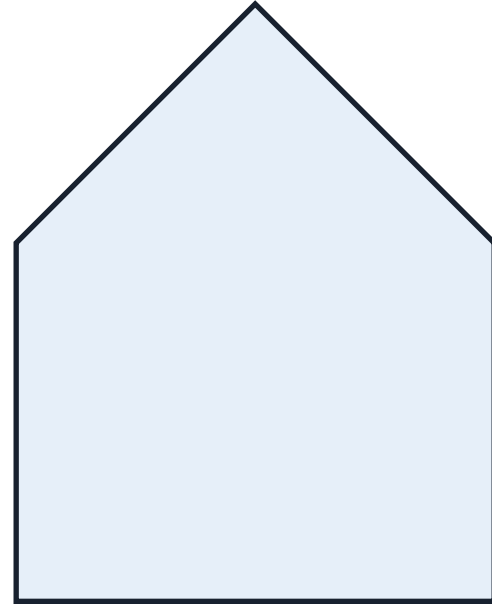
SET 73 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

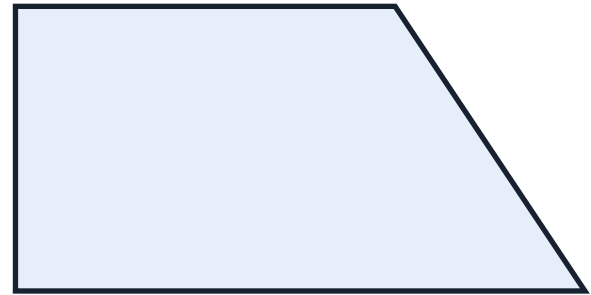
How many right angles does the figure have?

- A.** 1
- B.** 3
- C.** 2
- D.** 5



How many pairs of parallel sides does the figure have?

Pairs: _____



Which sides of rectangle ABCD are perpendicular to side AB?

- A.** CD only
- B.** AD only
- C.** AD and BC
- D.** CD and AD

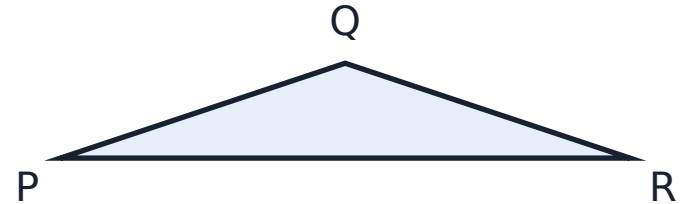


True or false?

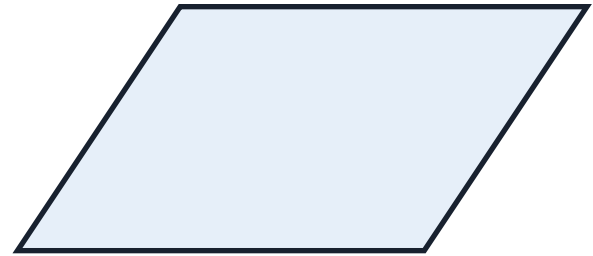
In triangle PQR, angle Q is obtuse.

A. True

B. False



How many line segments make up the sides of the figure? How many of its angles are acute?



Segments: _____

Acute angles: _____

SET 73 • 4.G.A.1

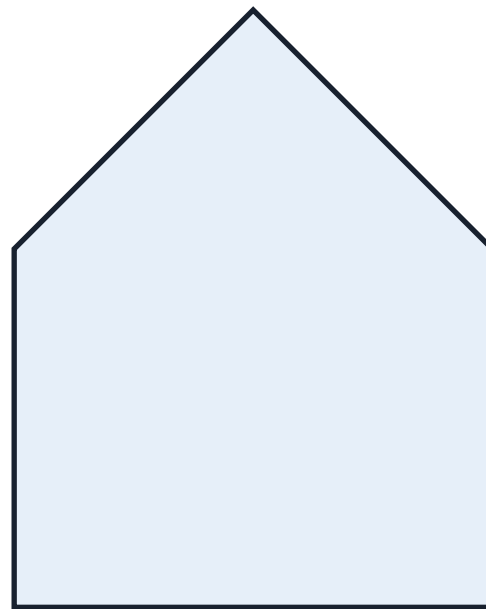
BACKWARD 1

Count sides and angles of shapes • Grade 2 • Geometry • 2.G.A.1

How many sides and how many angles does the shape have?

Sides: _____

Angles: _____



True or false?

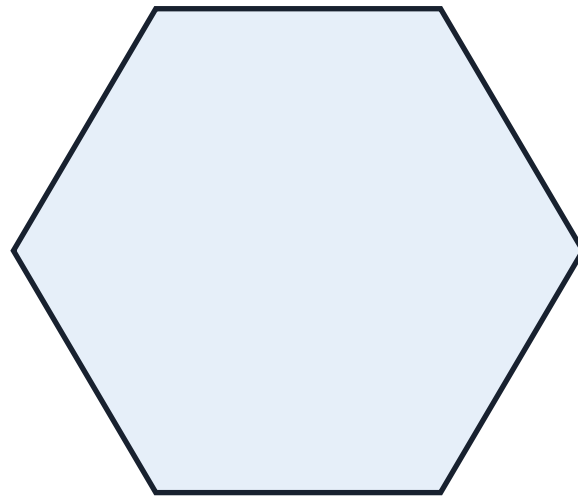
A quadrilateral has 4 sides and 4 angles.

A. True

B. False

What is the shape called?

- A.** Hexagon
- B.** Quadrilateral
- C.** Pentagon
- D.** Triangle



How many corners does a triangle have?

Corners: _____

True or false?

A pentagon has 4 angles.

A. True

B. False

SET 73 • 4.G.A.1

BACKWARD 2

Find square corners in shapes • Grade 3 • Geometry • 3.G.A.1

How many square corners
does the shape have?

Square corners: _____



True or false?

A square has 4 square corners.

A. True

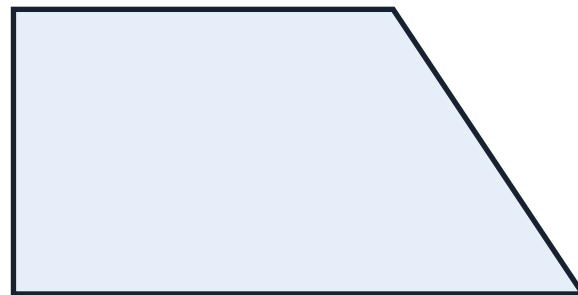
B. False

Which shape can have exactly one square corner?

- A.** Circle
- B.** Rectangle
- C.** Square
- D.** Triangle

How many square corners
does the shape have?

Square corners: _____



True or false?

A triangle can have 2 square corners.

A. True

B. False

SET 73 • 4.G.A.1

FORWARD 1

Use the attributes of a category to find sides and angles in figures • Grade 5 •
Geometry • 5.G.B.3

True or false?

Every rectangle has 4 right angles. A square is a rectangle, so every square has 4 right angles.

A. True

B. False

A figure is a rhombus.

How many pairs of parallel sides must it have?

Pairs: _____

A parallelogram has one right angle.
How many right angles must it have?

- A.** 4
- B.** 2
- C.** 3
- D.** 1

ABCD is a rectangle.

Which side is parallel to side AB?

Side: _____



True or false?

Every trapezoid has a pair of perpendicular sides.

A. True

B. False

SET 73 • 4.G.A.1

FORWARD 2

Identify parallel and perpendicular sides of polygons in the coordinate plane •

Grade 6 • Geometry • 6.G.A.3

True or false?

In the polygon with vertices $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$, the side from $(0, 0)$ to $(6, 0)$ is parallel to the side from $(0, 4)$ to $(6, 4)$.

A. True

B. False

A triangle has vertices $(1, 1)$, $(1, 6)$, and $(5, 1)$.

At which vertex is the right angle?

Vertex: _____

A quadrilateral has vertices $(-3, 0)$, $(3, 0)$, $(1, 4)$, and $(-1, 4)$.

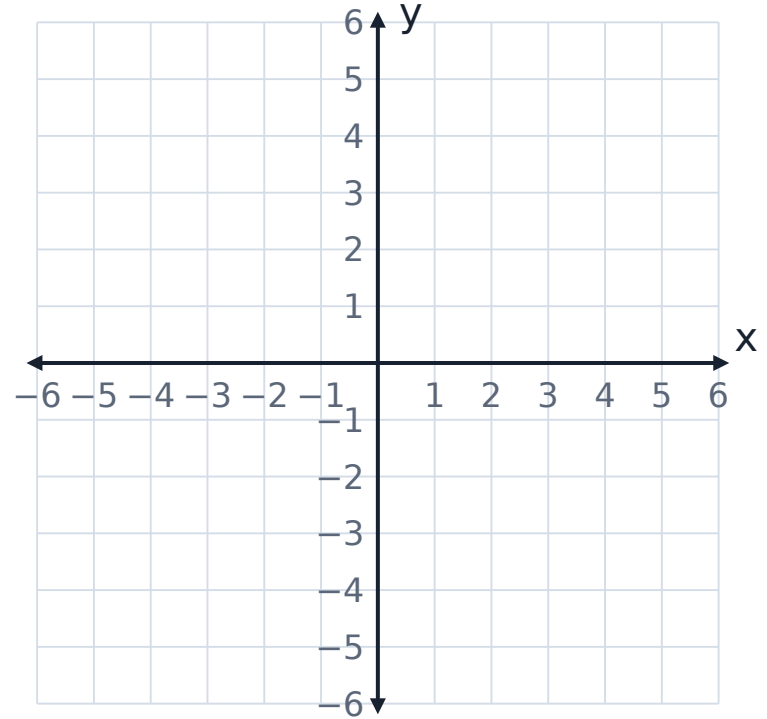
Which two sides are parallel?

- A.** No two sides are parallel **B.** From $(-3, 0)$ to $(3, 0)$ and from $(-1, 4)$ to $(1, 4)$
- C.** From $(3, 0)$ to $(1, 4)$ and from $(-1, 4)$ to $(-3, 0)$ **D.** Every pair of sides is parallel

Plot $(-4, -1)$, $(2, -1)$, $(2, 3)$, and $(-4, 3)$, and connect them in order.

How many right angles does the polygon have?

Right angles: _____



True or false?

The triangle with vertices $(0, 0)$, $(4, 0)$, and $(2, 3)$ has a right angle.

A. True

B. False

Classify two-dimensional figures by parallel or perpendicular sides and by angles

MAIN	Grade 4 main question	4.G.A.2
BACKWARD 1	Quadrilaterals as a category	3.G.A.1
BACKWARD 2	Triangles, quadrilaterals, pentagons, and hexagons	2.G.A.1
FORWARD 1	Classify quadrilaterals in a hierarchy	5.G.B.4
FORWARD 2	Find the area of parallelograms, rhombuses, and trapezoids	6.G.A.1

SET 61 • 4.G.A.2

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.2

Which shape has two pairs of parallel sides and no right angles?

- A.** A square
- B.** A rhombus that is not a square
- C.** A rectangle
- D.** A right triangle

A quadrilateral has exactly one pair of parallel sides.

What is it called?

Name: _____

True or false?

A rectangle has two pairs of parallel sides and four right angles.

A. True

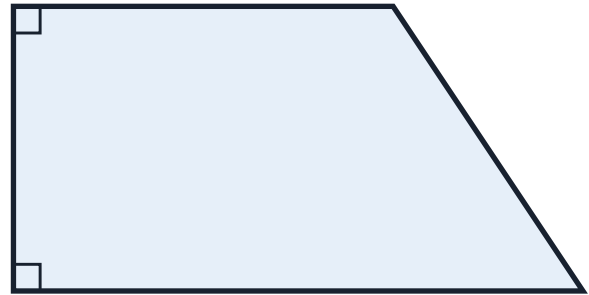
B. False

Which shape always has perpendicular sides?

- A.** Rhombus
- B.** Parallelogram
- C.** Square
- D.** Trapezoid

The shape has one pair of parallel sides and two right angles.

What kind of quadrilateral is it?



Name: _____

SET 61 • 4.G.A.2

BACKWARD 1

Quadrilaterals as a category • Grade 3 • Geometry • 3.G.A.1

True or false?

Squares and rectangles are both quadrilaterals.

- A. True
- B. False

What is a closed shape with 4 straight sides called?

Name: _____

Which shape is NOT a quadrilateral?

- A.** Square
- B.** Rectangle
- C.** Pentagon
- D.** Rhombus

True or false?

All quadrilaterals are squares.

A. True

B. False

Name two different quadrilaterals.

Shapes: _____

SET 61 • 4.G.A.2

BACKWARD 2

Triangles, quadrilaterals, pentagons, and hexagons • Grade 2 • Geometry •
2.G.A.1

What is a shape with 5 sides called?

Name: _____

True or false?

A hexagon has 6 sides.

A. True

B. False

Which shape has 4 sides?

A. Quadrilateral

B. Triangle

C. Pentagon

D. Hexagon

How many sides does a triangle have?

Sides: _____

True or false?

A pentagon has 8 sides.

A. True

B. False

SET 61 • 4.G.A.2

FORWARD 1

Classify quadrilaterals in a hierarchy • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Parallelogram, quadrilateral, square, rhombus
- B.** Quadrilateral, parallelogram, rhombus, square
- C.** Rhombus, square, quadrilateral, parallelogram
- D.** Square, rhombus, parallelogram, quadrilateral

True or false?

Every square is a rhombus.

A. True

B. False

Which of these categories does every rectangle belong to?

quadrilateral, parallelogram, rhombus, square

Categories: _____

True or false?

Every rhombus is a rectangle.

A. True

B. False

A quadrilateral has 4 sides of equal length and no right angles.

What is its most specific name?

A. Rhombus

B. Quadrilateral

C. Square

D. Rectangle

SET 61 • 4.G.A.2

FORWARD 2

Find the area of parallelograms, rhombuses, and trapezoids • Grade 6 •
Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

True or false?

A rhombus with a base of 6 m and a height of 5 m has an area of 30 square meters.

- A. True
- B. False

A trapezoid has bases of 4 ft and 8 ft and a height of 5 ft.

What is its area?

A. 40 square feet

B. 17 square feet

C. 30 square feet

D. 60 square feet

A trapezoid has bases of 6 units and 10 units and a height of 3 units.

What is its area?

Area: _____

True or false?

A parallelogram with a base of 10 units, slanted sides of 6 units, and a height of 5 units has an area of 60 square units.

A. True

B. False

Recognize right triangles as a category, and identify them

MAIN	Grade 4 main question	4.G.A.2
BACKWARD 1	Triangles	2.G.A.1
BACKWARD 2	Square corners	3.G.A.1
FORWARD 1	Right triangles share the attributes of all triangles	5.G.B.3
FORWARD 2	Find the area of right triangles	6.G.A.1

SET 62 • 4.G.A.2

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 90° , 50° , and 40° .
Is it a right triangle? Explain.

Answer: _____

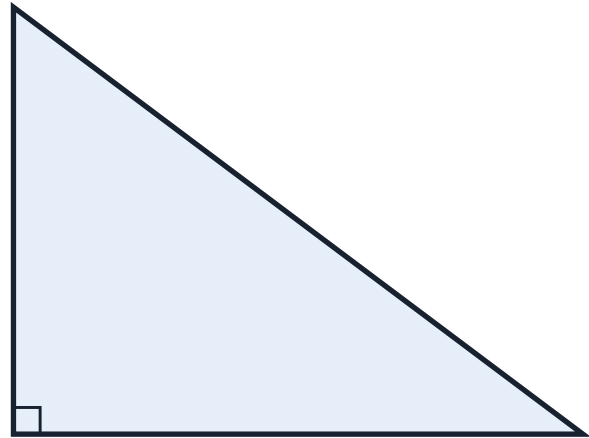
Explanation: _____

True or false?

The triangle shown is a right triangle.

A. True

B. False



Which could be the angles of a right triangle?

- A.** $70^\circ, 70^\circ, 40^\circ$
- B.** $100^\circ, 50^\circ, 30^\circ$
- C.** $90^\circ, 60^\circ, 30^\circ$
- D.** $80^\circ, 60^\circ, 40^\circ$

True or false?

A right triangle can have two right angles.

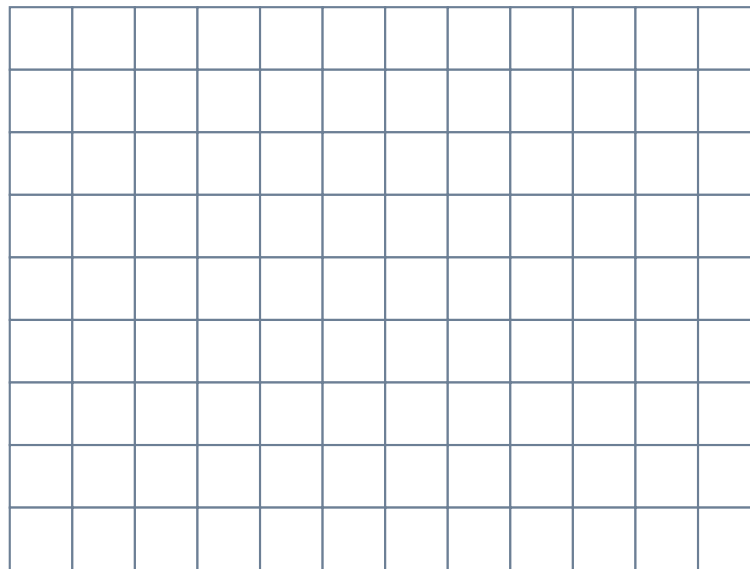
A. True

B. False

Draw a right triangle on the grid and mark its right angle.

How many right angles does your triangle have?

Right angles: _____



SET 62 • 4.G.A.2

BACKWARD 1

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does every triangle have?

Sides: _____

True or false?

A triangle has 3 corners.

A. True

B. False

Which describes a triangle?

- A.** A closed shape with 4 straight sides
- B.** A closed shape with 3 straight sides
- C.** A round shape
- D.** A closed shape with 5 corners

How many angles does a triangle have?

Angles: _____

True or false?

All triangles are the same size.

A. True

B. False

SET 62 • 4.G.A.2

BACKWARD 2

Square corners • Grade 3 • Geometry • 3.G.A.1

True or false?

A square corner is a right angle.

A. True

B. False

Which shape always has a square corner?

- A.** Oval
- B.** Hexagon
- C.** Rectangle
- D.** Circle

How many square corners does a rectangle have?

Square corners: _____

True or false?

A triangle can never have a square corner.

A. True

B. False

Which can you use to check for a square corner: the corner of a sheet of paper or a pencil?

Answer: _____

SET 62 • 4.G.A.2

FORWARD 1

Right triangles share the attributes of all triangles • Grade 5 • Geometry •
5.G.B.3

True or false?

All triangles have 3 sides. A right triangle is a triangle, so it has 3 sides.

A. True

B. False

An isosceles right triangle belongs to which categories?

- A.** Right triangles and isosceles triangles
- B.** Only isosceles triangles
- C.** Only right triangles
- D.** Equilateral triangles

Can a right triangle be scalene? Write yes or no.

Answer: _____

True or false?

An equilateral triangle can be a right triangle.

- A. True
- B. False

A right triangle has sides of 3, 4, and 5 units.
Is it isosceles, equilateral, or scalene?

Answer: _____

SET 62 • 4.G.A.2

FORWARD 2

Find the area of right triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 8 m and 5 m long.

What is its area?

Area: _____

True or false?

Two copies of the same right triangle can be put together to make a rectangle.

- A. True
- B. False

A right triangle has legs that are 12 in. and 6 in. long.

What is its area?

- A.** 9 square inches
- B.** 36 square inches
- C.** 18 square inches
- D.** 72 square inches

A 10-by-4 rectangle is cut along a diagonal into two right triangles.

What is the area of each triangle?

Area: _____

True or false?

A right triangle with legs of 7 and 4 units has an area of 28 square units.

- A. True
- B. False

Recognize lines of symmetry and line-symmetric figures

MAIN	Grade 4 main question	4.G.A.3
BACKWARD 1	Split circles and rectangles into equal shares	2.G.A.3
BACKWARD 2	Halves and fourths	1.G.A.3
FORWARD 1	Graph points that match across a line of symmetry	5.G.A.2
FORWARD 2	Reflect points across the axes	6.NS.C.6.b

SET 63 • 4.G.A.3

MAIN

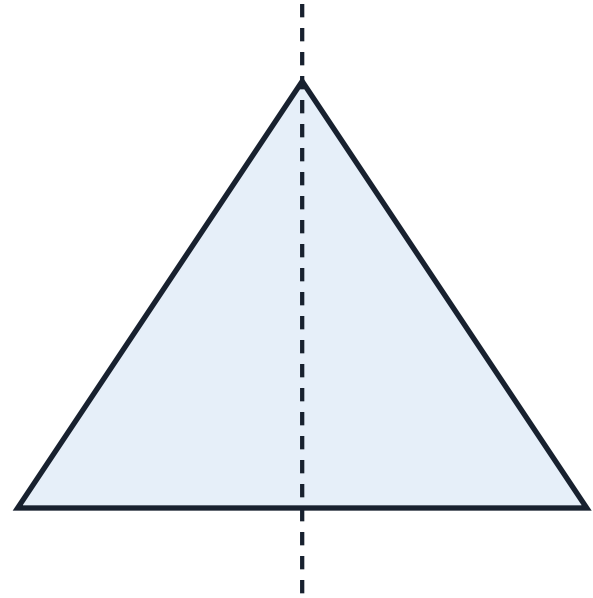
Grade 4 main question • Grade 4 • Geometry • 4.G.A.3

True or false?

The dashed line is a line of symmetry of the triangle.

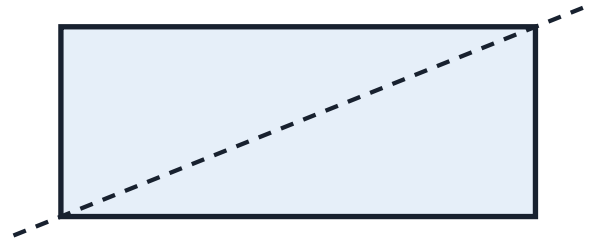
A. True

B. False



True or false?

The dashed line is a line of symmetry of the rectangle.



A. True

B. False

How many lines of symmetry does a square have?

A. 1

B. 4

C. 0

D. 2

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

Which capital letter has a vertical line of symmetry?

A. R

B. J

C. A

D. F

SET 63 • 4.G.A.3

BACKWARD 1

Split circles and rectangles into equal shares • Grade 2 • Geometry • 2.G.A.3

True or false?

A rectangle cut into 2 equal shares is cut into halves.

A. True

B. False

A circle is cut into 4 equal shares.

What is one share called?

Name: _____

A rectangle is cut into 3 equal shares.

What are the shares called?

- A.** Fifths
- B.** Halves
- C.** Thirds
- D.** Fourths

How many halves make a whole?

Halves: _____

True or false?

Equal shares of the same whole must have the same shape.

A. True

B. False

SET 63 • 4.G.A.3

BACKWARD 2

Halves and fourths • Grade 1 • Geometry • 1.G.A.3

A square is cut into 2 equal parts.

What is each part called?

Name: _____

True or false?

Cutting a circle into 4 equal parts makes smaller parts than cutting it into 2 equal parts.

A. True

B. False

How many fourths make a whole?

A. 2

B. 8

C. 4

D. 1

A pizza is cut into 4 equal parts.

What is one part called?

Name: _____

True or false?

Halves are smaller than fourths of the same whole.

A. True

B. False

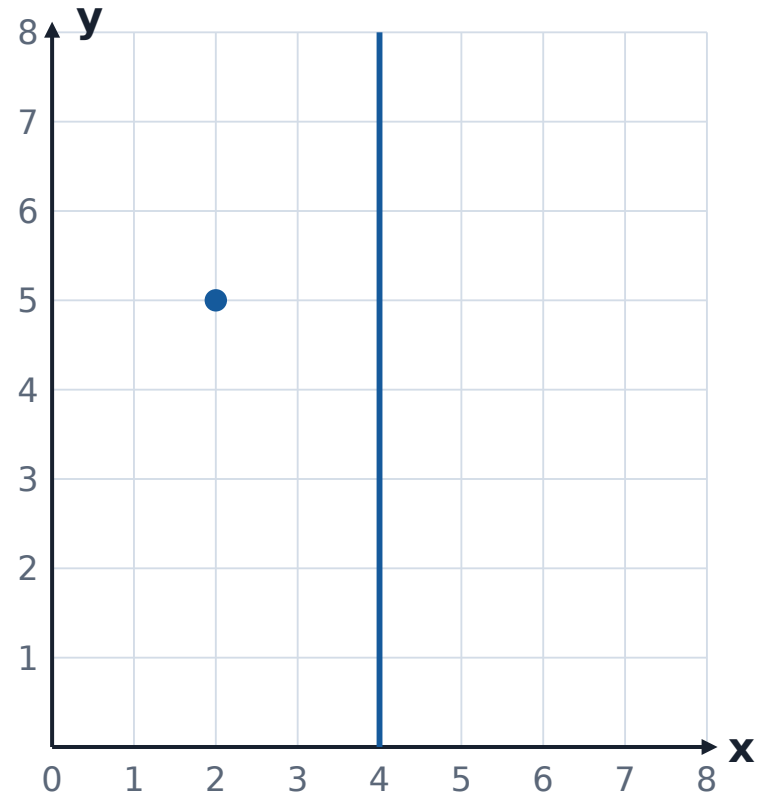
SET 63 • 4.G.A.3

FORWARD 1

Graph points that match across a line of symmetry • Grade 5 • Geometry •
5.G.A.2

The line through $(4, 0)$ and $(4, 8)$ is a line of symmetry.

Plot the point that matches $(2, 5)$ on the other side of the line.



True or false?

Across the line through $(3, 0)$ and $(3, 5)$, the point $(1, 2)$ matches the point $(5, 2)$.

A. True

B. False

Across the line through $(5, 0)$ and $(5, 8)$, which point matches $(7, 4)$?

A. $(5, 4)$

B. $(2, 4)$

C. $(7, 6)$

D. $(3, 4)$

Across the line through $(4, 0)$ and $(4, 8)$, which point matches $(1, 6)$?

Point: _____

True or false?

Across the line through $(2, 0)$ and $(2, 5)$, the point $(0, 3)$ matches the point $(3, 3)$.

A. True

B. False

SET 63 • 4.G.A.3

FORWARD 2

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Reflect $(3, 4)$ across the y -axis.

What are the coordinates of the image?

Image: _____

True or false?

Reflecting $(2, -5)$ across the x-axis gives $(2, 5)$.

- A. True
- B. False

Which point is the reflection of $(-4, 1)$ across the x-axis?

- A.** $(4, 1)$
- B.** $(-4, -1)$
- C.** $(1, -4)$
- D.** $(4, -1)$

Reflect $(-6, -2)$ across the y -axis.

What are the coordinates of the image?

Image: _____

True or false?

Reflecting $(5, 0)$ across the x-axis gives $(-5, 0)$.

A. True

B. False

Draw lines of symmetry

MAIN	Grade 4 main question	4.G.A.3
BACKWARD 1	Split circles and rectangles into equal shares	2.G.A.3
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SET 64 • 4.G.A.3

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.3

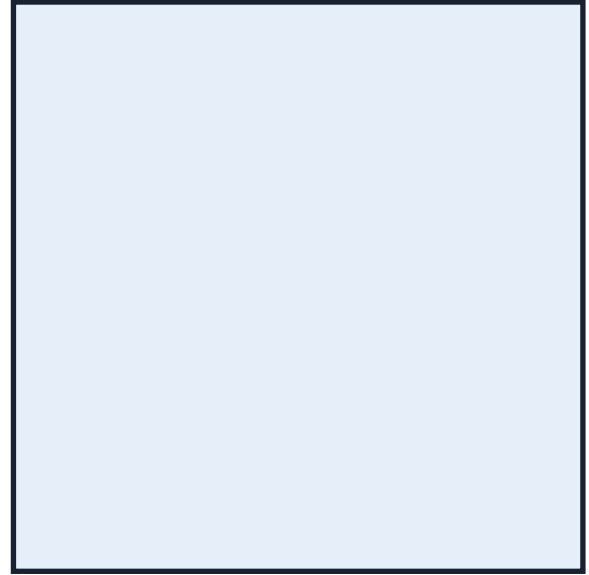
Draw all the lines of symmetry of the rectangle.



Draw all the lines of
symmetry of the square.

How many are there?

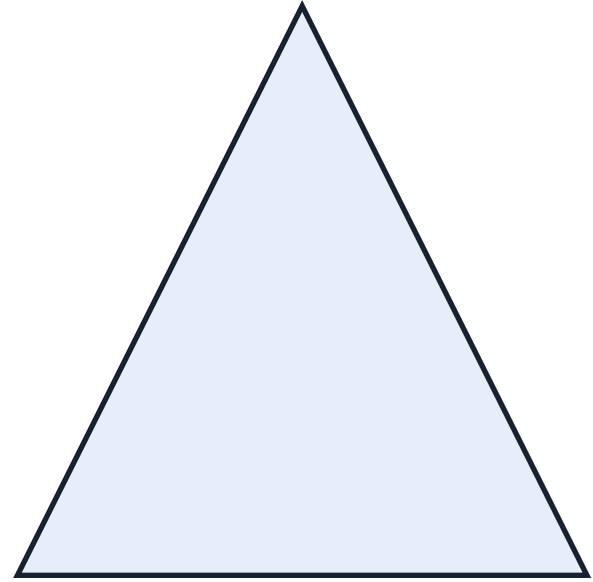
Lines: _____



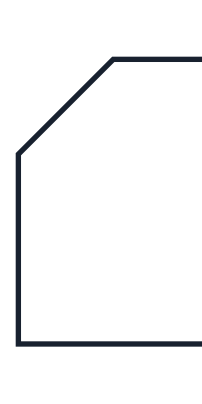
Draw the line of symmetry of the triangle. The triangle has exactly two sides of equal length; its base has a different length.

How many lines of symmetry does it have?

Lines: _____



Complete the figure so that the dashed line is a line of symmetry.



A regular hexagon has 6 equal sides and 6 equal angles.
How many lines of symmetry does it have?

Lines: _____

SET 64 • 4.G.A.3

BACKWARD 1

Split circles and rectangles into equal shares • Grade 2 • Geometry • 2.G.A.3

A square is cut into 4 equal shares.

How many shares make the whole?

Shares: _____

True or false?

A circle can be cut into 3 equal shares called thirds.

A. True

B. False

Which one shows halves?

- A.** A circle cut into 4 equal parts
- B.** A rectangle cut into 2 unequal parts
- C.** A rectangle cut into 2 equal parts
- D.** A circle cut into 3 equal parts

A rectangle is cut into 3 equal shares.

What is one share called?

Name: _____

True or false?

A circle cut into 2 parts of different sizes shows halves.

A. True

B. False

SET 64 • 4.G.A.3

BACKWARD 2

Put shapes together to make new shapes • Grade 1 • Geometry • 1.G.A.2

True or false?

Two matching right triangles can be put together to make a rectangle.

- A. True
- B. False

Which two shapes can be put together to make a square?

- A.** Two matching rectangles
- B.** Two circles
- C.** Three circles
- D.** A circle and a triangle

How many squares do you need to make a rectangle that is 2 squares long and 1 square wide?

Squares: _____

True or false?

Two half-circles can be put together to make a square.

A. True

B. False

Two matching trapezoids are put together to make a shape with 6 sides.

What is the shape called?

Name: _____

SET 64 • 4.G.A.3

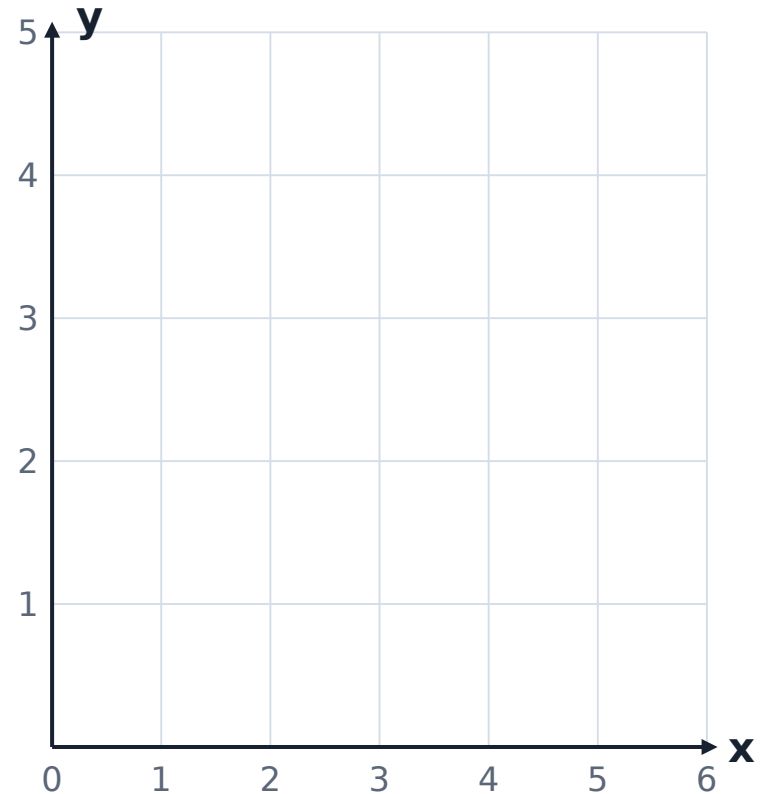
FORWARD 1

Graph points to make and check symmetric figures • Grade 5 • Geometry •
5.G.A.2

Plot $(1, 1)$, $(3, 4)$, and $(5, 1)$ and connect them. Then draw the line of symmetry.

Where does the line of symmetry cross the x-axis?

Crosses at: _____



True or false?

The rectangle with corners $(1, 1)$, $(5, 1)$, $(5, 3)$, and $(1, 3)$ is symmetric about the line through $(3, 0)$ and $(3, 4)$.

A. True

B. False

A shape has a line of symmetry through $(4, 0)$ and $(4, 6)$. The point $(2, 3)$ is on the shape.

Which point must also be on it?

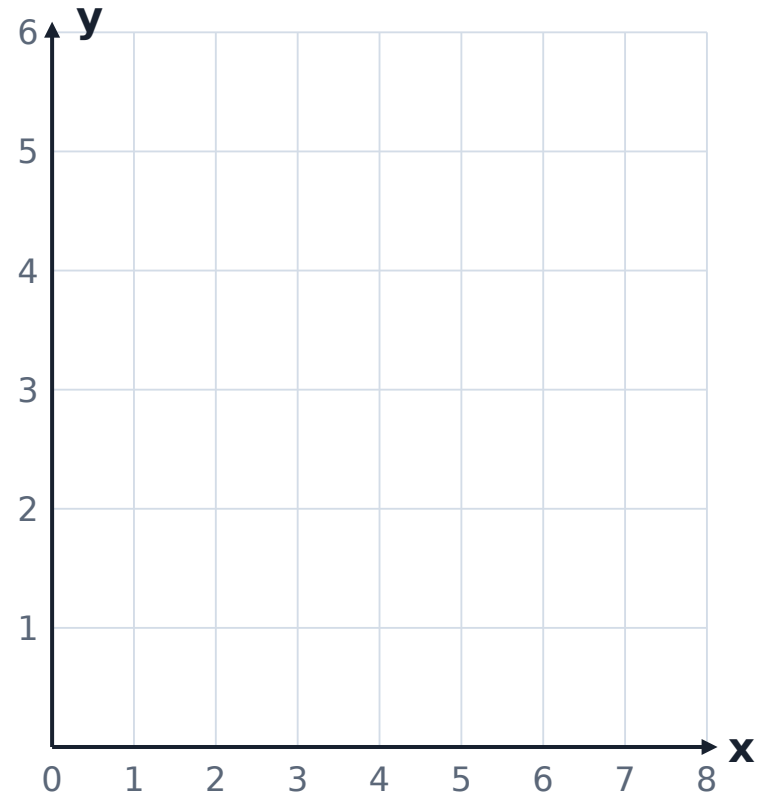
A. $(2, 5)$

B. $(4, 3)$

C. $(8, 3)$

D. $(6, 3)$

Plot $(2, 2)$, $(2, 5)$, and $(6, 2)$.
Then plot one more point so
that the four points are the
corners of a rectangle.



True or false?

The points $(1, 4)$ and $(7, 4)$ match across the line through $(3, 0)$ and $(3, 6)$.

A. True

B. False

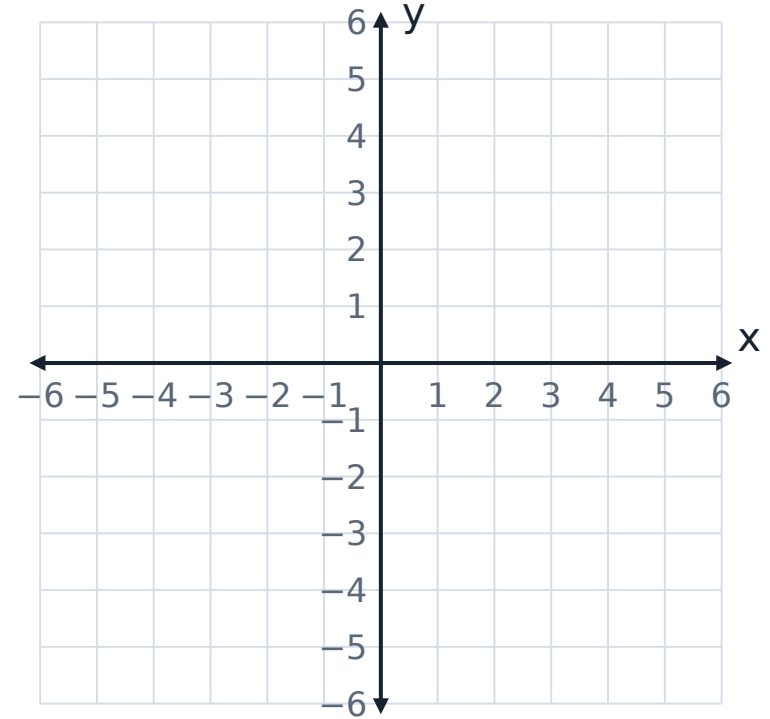
SET 64 • 4.G.A.3

FORWARD 2

Draw polygons in the coordinate plane from their vertices and find side lengths

• Grade 6 • Geometry • 6.G.A.3

Plot $(2, 3)$, $(5, 3)$, $(5, -1)$, and $(2, -1)$. Connect them in order to draw the polygon.



True or false?

The polygon with vertices $(-3, 2)$, $(3, 2)$, $(3, -2)$, and $(-3, -2)$ has a side that is 6 units long.

A. True

B. False

A square has vertices $(-2, -2)$, $(2, -2)$, and $(2, 2)$.

What is the fourth vertex?

A. $(-2, 2)$

B. $(2, -4)$

C. $(-2, 4)$

D. $(0, 2)$

A polygon has vertices $(1, 4)$, $(4, 4)$, and $(4, -2)$.

What is the length of the side from $(4, 4)$ to $(4, -2)$?

Length: _____

True or false?

The segment from $(-4, 1)$ to $(4, 1)$ is 4 units long.

- A. True
- B. False